



Hantro VC9000E Video Encoder Test Bench User Manual

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Preface

This chapter introduces the *Hantro VC9000E Video Encoder Test Bench User Manual*.

About This Document

This document describes the test bench released for Hantro VC9000E multi-format video encoder IP and provides simulation guidelines.

Additional Reading

- Hantro VC9000E Video Encoder Hardware Features
- Hantro VC9000E Video Encoder Memory Buffer and Format Organization
- Hantro VC9000E Video Encoder API User Manual
- Hantro VC9000E Video Encoder Wrapper Layer API User Manual
- **ITU-T Rec. H.265**: High-efficiency video coding
- **ITU-T Rec. H.264**: Advanced video coding for generic audiovisual services
- **ITU-R REC. BT.601**: Studio encoding parameters of digital television for standard 4:3 and wide-screen 16:9 aspect ratios
- **ITU-R REC. BT.709**: Parameter values for the HDTV standards for production and international programme exchange
- **ITU-R REC. BT.2020**: Parameter values for ultra-high-definition television systems for production and international programme exchange

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1 Overview

This chapter gives an overview of the IP core.

1.1 IP Introduction

Designed for easy integration onto the SoC, VeriSilicon's Hantro VC9000E multi-format video encoder is based on silicon-proven designs deployed in millions of smartphones, tablets, set-top boxes, televisions, IP cameras, data centers, server ASIC, and AIoT devices worldwide.

The VC9000E video encoder provides up to 8192x4096 encoding, featured by small silicon footprint, lower power consumption, and negligible load on the host CPU. The integrated reference frame compression (RFC) provides significant savings on system bandwidth, and extensive use of multi-level hierarchical clock gating minimizes the dynamic power consumption.

1.2 Supported Codec Standards and Profiles

The command lines discussed in this version of the document is compatible with the following video codec standards and profiles. Additional levels may be supported via software.

- HEVC - ITU-T Rec. H.265 (04/2013), ISO/IEC 23008-2
 - Main Profile, up to Level 6, High Tier
 - Main 10 Profile, up to Level 6, High Tier
 - Main Still Picture Profile, up to Level 6, High Tier
- AVC - Spec Version 12: ISO/IEC 14496-10 / ITU-T Rec. H.264 (03/2010)
 - Baseline Profile, up to level 6

- Main Profile, up to level 6
- High Profile, up to level 6
- High 10 Profile, level 6
- AV1 - AV1 Bitstream & Decoding Process Specification Version 1.0.0 with Errata 1
 - Main Profile, Level 5.1
- VP9 - VP9 Bitstream & Decoding Process Specification v0.6
 - Profile 0 (8-bit), Level 5.1
 - Profile 1 (10-bit), Level 5.1

2 Options

The chapter describes the options available in the test bench for configuring the video encoder.

- To specify a long option, separate the option and the value with an equal sign (=). For example, `--qpHdr=-1`.
- To specify a short option, separate the option and the value with an equal sign (=) or omit the equal sign. For example, `-q=-1` or `-q-1`.

In this chapter:

- In short options, [s] requires a string value and [n] requires an integer value.
- The value range of an option is provided at the beginning of its description, with the boundaries separated by a couple of periods (...). Both boundaries are inclusive.
- The default value of an option is marked by square brackets [] at the end of the first paragraph of the option description.

For example use cases, see the *Application Examples and Test Utility* chapter.

2.1 Encoder Input and Output

Short	Long Option	Description
-i[s]	--input	The file from which the video encoder reads input video sequence. [input.yuv]
-o[s]	--output	The file to which the video encoder writes output stream. [stream.hevc]
-a[n]	--firstPic	The first picture to be encoded in the input file. [0]

Short	Long Option	Description
-b[n]	--lastPic	The last picture to be encoded in the input file. [100]
	--outReconFrame	Whether to output reconstructed frames. [1] 0 - do not output reconstructed frames. 1 - output reconstructed frames.
-j[n]	--inputRateNumer	1..1048575 The numerator used to calculate the input frame rate. [30]
-J[n]	--inputRateDenom	1..1048575 The denominator used to calculate the input frame rate. [1]
-f[n]	--outputRateNumer	1..1048575 The numerator used to calculate the output frame rate. [Same as input]
-F[n]	--outputRateDenom	1..1048575 The denominator used to calculate the output frame rate. [Same as input]
	--writeReconToDDR	Whether to write reconstructed frames to DDR. [1] 0 - do not write reconstructed frames to DDR. 1 - write reconstructed frames to DDR. The option is valid only for I-frame only encoding.

2.1.1 Input and Encoded Frame Resolutions and Cropping

Short	Long Option	Description
-w[n]	--lumWidthSrc	The width of the input picture, in pixels. [176]
-h[n]	--lumHeightSrc	The height of the input picture, in pixels. [144]
-x[n]	--width	The width of the encoded picture, in pixels. [-lumWidthSrc]
-y[n]	--height	The height of the encoded picture, in pixels. [-lumHeightSrc]
-X[n]	--horOffsetSrc	The horizontal offset, in pixels, of the top-left corner of the encoded picture relative to the input picture. [0] The option value must be an even integer.
-Y[n]	--verOffsetSrc	The vertical offset, in pixels, of the top-left corner of the encoded picture relative to the input picture. [0] The option value must be an even integer.

Short	Long Option	Description
	--inputFileList	The path to a file that contains a list of input files. The option is useful for testing resolution change. Supported sub-options in the file include: -i: the input file path -a: the start picture -b: the end picture -w: the picture width in pixels -h: the picture height in pixels -o: the path to the output file If inputFileList is specified, the setting of the --input option is ignored. The video encoder reads inputs from the listed files. In this case, the --width and --height indicates the allowed maximum width and height used in initialization.

2.2 Pre-Processing



Short Short	Long Option Long Option	Description Description
-l[n]	--inputFormat	The input color format. [0] 0 - YUV420P8b_Raster_A16N 1 - YUV420SP8b_Raster_A16N 2 - YVU420SP8b_Raster_A16N 3 - YUV422YUYV8b_Raster_A16N 4 - YUV422UYVY8b_Raster_A16N 5 - RGB565_Raster_A16N 6 - BGR565_Raster_A16N 7 - XRGB1555_Raster_A16N 8 - XBGR1555_Raster_A16N 9 - XRGB4444_Raster_A16N 10 - XBGR4444_Raster_A16N 11 - XRGB8888_Raster_A16N 12 - XBGR8888_Raster_A16N 13 - XRGB2101010_Raster_A16N 14 - XBGR2101010_Raster_A16N 15 - YUV420P10bWL_Raster_A16N 16 - YUV420SP10bWH_Raster_A16N 17 - YUV420P10b_Raster_A16N 18 - YUV420Y0L210b_Raster_A16N 19 - (Reserved) YUV420P8b_Tile32x32_A16N for HEVC (H.265) 20 - (Reserved) YUV420P8b_Tile16x16_A16N for H.264 (AVC) 21 - YUV420SP8b_YuvSp4x4_A16N 22 - YVU420SP8b_YuvSp4x4_A16N 23 - YUV420SP10bWH_YuvSp4x4_A32N 24 - YUV420SP10bDWL_Raster_A16N 26 - (Reserved) YUV420YVU8b_Tile64x4_A16N 27 - (Reserved) YUV420YUV8b_Tile64x4_A16N 28 - (Reserved) YUV420YUV10b_Tile32x4_A16N 29 - (Reserved) YUV420YUV10b_Tile48x4_A16N 30 - (Reserved) YUV420YVU10b_Tile48x4_A16N 31 - (Reserved) YUV420YVU8b_Tile128x2_A16N 32 - (Reserved) YUV420YUV8b_Tile128x2_A16N 33 - (Reserved) YUV420YUV10b_Tile96x2_A16N 34 - (Reserved) YUV420YVU10b_Tile96x2_A16N 35 - YUV420SP8b_YuvSp8x8_A64N 36 - YUV420SP10bWH_YuvSp8x8_A128N 37 - YUV420PYVU8b_Raster_A16N 38 - (Reserved) YVU420SP8b_Tile64x2_A16N 39 - (Reserved) YVU420SP10b_Tile128x2_A16N 40 - RGB888_Raster_A16N 41 - BGR888_Raster_A16N 42 - BRG888_Raster_A16N

Short	Long Option	Description
		occupies 2B [9 : 0] 57 - Y10bWH_Raster_A16N (YCbCr 4:0:0) Monochrome 10-bit occupies 2B [15 : 6] 58 - YUV420SP10b_Raster_A16N 59 - YUV422P10bWL_Raster_A16N 60 - Y8b_Tile8x8_A16N (YCbCr 4:0:0) Monochrome 8-bit 61 - Y10bWH_Tile8x8_A16N (YCbCr 4:0:0) Monochrome 10-bit occupies 2B [15 : 6] 62 - YUV422YVYU8b_Raster_A16N 63 - YUV422VYUY8b_Raster_A16N 64 - YVU420SP8b_YuvSp8x8_A64N 65 - Y10b_Raster_A16N For details about the color formats, see Hantro VC9000E Series Memory Buffer and Format Organization.
-O[n]	--colorConversion	The RGB-to-YUV color conversion type. [0] 0 - conversion to YUV values without any range change according to Rec. ITU-R BT.601. 1 - conversion to YUV values without any range change according to Rec. ITU-R BT.709. 2 - conversion using custom coefficients. 3 - conversion according to Rec. ITU-R BT.2020. 4 - conversion from full range RGB to limited range YUV according to Rec. ITU-R BT.601. 5 - (just for test) conversion from RGB in range of (0~219) to limited range YUV according to Rec. ITU-R BT.601. 6 - conversion from full range RGB to limited range YUV according to Rec. ITU-R BT.709.
	--mirror	The mirroring mode to pre-process the input image. [0] 0 - do not mirror 1 - mirror
-r[n]	--rotation	The rotation mode to pre-process the input image. [0] 0 - do not rotate 1 - rotate 90 degrees clockwise 2 - rotate 90 degrees counterclockwise 3 - rotate 180 degrees clockwise

Short	Long Option	Description
-Z[n]	--videoStab	Whether to enable video stabilization. [0] 0 - disable 1 - enable When this option is set to 1, the video stabilization module also performs scene change detection if the input picture resolution is the same as the encoded picture resolution.
	--scaledWidth	0..width The width of the down-scaled picture. [0] The value must a multiple of 4. If either this option or --scaledHeight is set to 0, down-scaling is disabled. NOTE: The down-scaled pictures is stored in the scaled.yuv file.
	--scaledHeight	0..height The height of the down-scaled picture. [0] The value must a multiple of 4. If either this option or --scaledWidth is set to 0, down-scaling is disabled. NOTE: The down-scaled pictures is stored in the scaled.yuv file.
	--scaledOutputFormat	The color format of the output down-scaled picture. [0] 0 - YUV422 interleaved 1 - YUV420 semiplanar (NV12)
	--interlacedFrame	Whether the input frames are progressive or interlaced. [0] 0 - progressive frame input 1 - interlaced framed input
	--fieldOrder	The interlaced field order. [0] 0 - bottom field first 1 - top field first
	--codedChromaIdc	The chroma sampling modes relative to luma sampling. [1] 0 - 4:0:0 for HEVC (H.265) and H.264 (AVC) only 1 - 4:2:0 2 - 4:2:2 (under development) 3 - 4:4:4 for HEVC (H.265) only This field is used as the SPS syntax element chroma_format_idc specified in ITU-T Rec. H.265 and H.264.

Short	Long Option	Description
	--scanType	The scan type of input image. [0] 0 - raster 1 - supertileX

2.2.1 Input Conversion to Customized Formats

Short	Long Option	Description
	--formatCustomizedType	-1..13 (For test use) Converts the input image to the specified custom format. The data after format conversion is fed to the video encoder as its input. [-1] An option value in range from 0 to 13 indicates a pre-defined custom format. If this option is set to -1, format conversion before input to the encoder is disabled. If this option is specified, dividing the input image into multiple tile columns and rows is not supported.

2.2.2 DEC400 Compression Table (Tile Status)

Short	Long Option	Description
	--dec400TableInput	The file from which the video encoder reads the DEC400 compression table for video layer input. [dec400CompTableinput.bin]
	--osdDec400TableInputN	The file from which the video encoder reads the DEC400 compression table for OSD input. [NULL]
	--dec400TileSize	0 - default 1 - 128byte. 2 - 512byte. 3 - 256byte.

Short	Long Option	Description
	--dec400TileMode	0 - default.[0] 1 - TILE_8x8_x. 2 - TILE_8x8_y. 3 - TILE_16x4. 4 - TILE_8x4. 5 - TILE_4x8. 6 - RASTER_16x4. 7 - TILE_64x4. 8 - TILE_32x4. 9 - RASTER256x1. 10 - RASTER128x1. 11 - RASTER64x1. 12 - TILE_16x8. 13 - RASTER32x4. 14 - RASTER32x1. 15 - RASTER16x1. 16 - TILE_8x4_s. 17 - TILE_16x4_s. 18 - TILE_32x4_s. 19 - TILE_32x8.
	--dec400DataAlignment	0 - default 1 - 32byte. 2 - 64byte. 3 - 1byte. 4 - 16byte.
	--dec400RGBAX	0 - ARGB.[0] 1 - XRGB.
	--dec400RGBFormat	0 - default 1 - RGB8. 2 - RGB10. 3 - RGB4. 4 - RGB1555. 5 - RGB565.
	--dec400TSHeaderEnable	Whether to process dec400 tile status header buffer.(128byte) 0 - disable 1 - enable

2.2.3 UFBC Parameters

Short	Long Option	Description
	--ufbcMode	The core mode of UFBC 0 - disable 1 - afbc (only support 32x8 pixels) 2 - afbc (support 32x8 and 16x16 pixels) 3 - dec400 4 - pvrhc
	--ufbcYuvTrans	Whether to disable interal YUV transformation. [0] 0 - enable 1 - disable
	--ufbcBlockType	The size of the superblock. Version 1.x 0 - 32x8 pixels 1 - 16x16 pixels Version 2.x 0 - 8x8 pixels 1 - 16x4 pixels 2 - 32x2 pixels
	--ufbcBlockSplit	Whether to enable the superblock split mode for version 1.x. 0 - disable 1 - enable
	--ufbcConstantVal	The constant color for version 2.x.

2.3 Stream Coding Tools

2.3.1 Output Stream Format

Short	Long Option	Description
	--codecFormat	The video codec standard. [0] 0 or hevc - HEVC (H.265) 1 or h264 - H.264 (AVC) 2 or av1 - AV1 3 or vp9 - VP9

Short	Long Option	Description
-P[n]	--profile	The stream profile. [-1] -1 - Adaptive profile HEVC (H.265) profiles: 0 - Main profile 1 - Main Still Picture profile 2 - Main 10 profile 3 - Monochrome profile 4 - Monochrome 10 profile 5 - Main 10 Still Picture profile 6 - Main 422 10 profile 7 - Main 444 10 profile 8 - SCC Main profile 9 - SCC 444 10 profile H.264 (AVC) profiles: 9 - Baseline profile 10 - Main profile 11 - High profile 12 - High 10 profile AV1 profiles: 0 - Main profile VP9 profiles: 0 - Main profile 2 - High profile

Short	Long Option	Description
-L[n]	--level	The stream level. If the option is set to 0xFFFF, the encoder software automatically selects the level. HEVC (H.265) levels [180]: Levels up to level 5.1 High tier are supported in real-time. Levels 5.0 and 5.1 with resolutions up to 4096x2048 are supported with 4K@60FPS performance. Levels greater than level 5.1 are supported in non-realtime mode. Value - Level - Resolution - Main tier max bit rate - High tier max bit rate 30 - 1.0 - QCIF - 128 kbps - N/A 60 - 2.0 - CIF - 1.5 Mbps - N/A 63 - 2.1 - Q720p - 3.0 Mbps - N/A 90 - 3.0 - QHD - 6.0 Mbps - N/A 93 - 3.1 - 1280x720 - 10.0 Mbps - N/A 120 - 4.0 - 2Kx1080 - 12.0 Mbps - 30 Mbps 123 - 4.1 - 2Kx1080 - 20.0 Mbps - 50 Mbps 150 - 5.0 - 4096x2160 - 25.0 Mbps - 100 Mbps 153 - 5.1 - 4096x2160 - 40.0 Mbps - 160 Mbps 156 - 5.2 - 4096x2160 - 60.0 Mbps - 240 Mbps 180 - 6.0 - 8192x4320 - 60.0 Mbps - 240 Mbps 183 - 6.1 - 8192x4320 - 120.0 Mbps - 480 Mbps 186 - 6.2 - 8192x4320 - 240.0 Mbps - 800 Mbps H.264 levels [51]: 10 - H264_LEVEL_1 99 - H264_LEVEL_1_b 11 - H264_LEVEL_1_1 12 - H264_LEVEL_1_2 13 - H264_LEVEL_1_3 20 - H264_LEVEL_2 21 - H264_LEVEL_2_1 22 - H264_LEVEL_2_2 30 - H264_LEVEL_3 31 - H264_LEVEL_3_1 32 - H264_LEVEL_3_2 40 - H264_LEVEL_4 41 - H264_LEVEL_4_1 42 - H264_LEVEL_4_2 50 - H264_LEVEL_5 51 - H264_LEVEL_5_1 52 - H264_LEVEL_5_2 60 - H264_LEVEL_6 61 - H264_LEVEL_6_1

Short	Long Option	Description
		10 - AV1_LEVEL_4.2 11 - AV1_LEVEL_4.3 12 - AV1_LEVEL_5.0 13 - AV1_LEVEL_5.1
	--tier	The stream tier for HEVC (H.265) or AV1. [0] 0 - Main tier 1 - High tier
	--bitDepthLuma	The bit depth of luma samples in the encoded stream. [8] 8 - 8-bit luma samples 10 - 10-bit luma samples
	--bitDepthChroma	The bit depth of chroma samples in the encoded stream. [8] 8 - 8-bit Chroma samples 10 - 10-bit Chroma samples The value of bitDepthChroma must be the same as that of bitDepthLuma.
-N[n]	--byteStream	(Only for HEVC/H.264) The stream type. [1] 0 - NAL unit stream. NAL sizes are stored in the nal_sizes.txt file. 1 - Byte stream. For AV1, only the byte stream type is supported.
	--ivf	(For AV1/VP9 only) The IVF encapsulation. [1] 0 - OBU raw output 1 - IVF output
	--sendAUD	Whether to enable the access unit delimiter (AUD). [0] 0 - do not send 1 - send
	--smoothingIntra	(For HEVC only) Whether to enable normal or strong smoothing. [1] 0 - normal 1 - strong
-p[n]	--cabaclnitFlag	(Obsoleted) The initialization value for CABAC. [0] The option value is fixed to 0.

Short	Long Option	Description
-K[n]	--enableCabac	(Only for H.264) The entropy coding mode. [1] 0 - content-adaptive variable-length coding (CAVLC) 1 - content-based adaptive binary arithmetic coding (CABAC)
-e[n]	--sliceSize	(For HEVC/H.264/only) 0..height/ctu_size The number of CTB or MB rows in each slice. [0] If the option is set to 0, the encoder encodes a picture in each slice.
-D[n]	--disableDeblocking	Whether to disable inloop de-blocking filters. [0] 0 - enable. This value provides better quality. 1 - disable.
	--enableTS	Disable/Enable HEVC Transform Skip [0] 0 - disable hevc transform skip 1 - enable hevc transform skip
	--log2MaxTSBlockSizeMinus2	0..2 Max transform skip size minus 2 [0]
-M[n]	--enableSao	Whether to enable sample adaptive offset (SAO) for HEVC (H.265) or CDEF filter for AV1. [1] 0 - disable 1 - enable
-W[n]	--tc_Offset	-6..6 The de-blocking parameter tc_offset for HEVC (H.265) or alpha_c0_offset for H.264 (AVC). [0]
-E[n]	--beta_Offset	-6..6 The de-blocking parameter beta_offset for HEVC (H.265) and H.264 (AVC). [0]
	--enableDeblockOverride	(Only for HEVC) Whether to enable de-blocking override between frames. [0] 0 - disable 1 - enable
	--deblockOverride	(Only for HEVC) Whether the frame has de-blocking override information in the slice header. [0] 0 - does not have 1 - has

Short	Long Option	Description
	--tile=a:b:c	(For HEVC only) The tile settings. [1 : 1 : 1] - a indicates the number of tile columns to divide the picture. - b indicates the number of tile rows to divide the picture. - c indicates whether to enable the loop filter across the tile boundary. Value 0 indicates to disable and value 1 indicates to enable. Tiling is enabled when $a * b > 1$.
	--enableScalingList	(For HEVC only) Whether to use the average or default scaling list. [0] 0 - average scaling list 1 - default scaling list
	--RPSInSliceHeader	(For HEVC only) Whether to encode RPS in the slice header. [0] 0 - do not encode 1 - encode
	--resendParamSet	Whether to re-send VPS, PPS, and SPS before each IDR frame. [0] 0 - do not re-send 1 - re-send
	--POCConfig=a:b:c	The POC settings. [0 : 16 : 12] - a indicates the method to encode POC in the slice header. - b indicates the number of bits used to encode POC in the slice header. - c indicates the number of bits used to encode frame_num in the slice header.
	--psyFactor	0..4.0 The strength of psycho-visual encoding. <float> [0] If the option is set to 0, psycho-visual encoding is disabled.

Short	Long Option	Description
	--layerInRefIdc	(Only for H.264) Whether to enable layer information in the nal_ref_idc syntax element. [0] 0 - disable. In this case, the value of nal_ref_idc can be 0 or 1. 1 - enable. In this case, the value of nal_ref_idc can be 0 to 3.
	--prefixNalSvcFlag	(Only for H.264) Whether to add prefix NAL units between slices for temporal scalable video coding (SVC-T). [0] 0 - do not add 1 - add
	--svctEnable	(Only for H.264) Whether to enable SVC-T. [0] 0 - disable 1 - enable
	--enableTMVP	Whether to enable temporal motion vector prediction (TMVP) in inter prediction. 0 - disable 1 - enable The default value is 1 for HEVC, AV1, and VP9 if TMVP is supported, and 0 in other cases.
	--Heifcfg	The path to the HEIF configuration file. (Only support codec format HEVC and AV1)

Constant Chroma

Short	Long Option	Description
	--enableConstChroma	Whether to encode the picture with constant U and V values. [0] 0 - do not encode with a constant U or V value 1 - encode with a constant U or V value
	--constCb	The constant U value. The option value range for 8-bit encoding color space is from 0 to 255, inclusive. The default value is 128. The option value range for 10-bit encoding color space is from 0 to 1023, inclusive. The default value is 512. This option is valid only if enableConstChroma is set to 1.

Short	Long Option	Description
	--constCr	The constant V value. The option value range for 8-bit encoding color space is from 0 to 255, inclusive. The default value is 128. The option value range for 10-bit encoding color space is from 0 to 1023, inclusive. The default value is 512. This option is valid only if enableConstChroma is set to 1.

2.3.2 GOP Structure

Short	Long Option	Description
	--smoothPsnrInGOP	(Obsoleted) Whether to enable smooth PSNR for frames in a GOP. [0] 0 - disable 1 - enable This option is valid only if gopSize is set to 1.
	--gopSize	0..8 The size of the GOP structure. [0] If the option is set to 0, the encoder adaptively selects a GOP structure size. If the option is set to a value from 1 to 8, the encoder uses a fixed GOP structure size with a pre-defined GOP structure.
	--gopMaxBSize	0..7[default -255, disable] The GOP max B frame number for AGOP

Short	Long Option	Description
	--gopConfig	The path to a custom GOP configuration file, which defines the GOP structure table. If the gopConfig option is not specified, the default configurations take effect. The GOP configuration file must include gopSize normal configuration lines. Each normal configuration line must contain the following fields in order: - FrmN: the ID of the frame, where N is an integer ranging from 1 to gopSize. In the configuration file, frames must be listed in decoding order. - Type: the slice type, which can be P, B, or nrefB. - POC: the display order of the frame within the GOP, ranging from 1 to gopSize. - QPoffset: the offset to be added to the QP parameter to obtain the final QP value used for the frame. - QPfactor: the weight used during rate distortion optimization. - TemporalId: the temporal layer ID. - num_ref_pics: the number of reference frames kept for the current frame. The reference frames can be used by either the current frame or a subsequent frame. - ref_pics: a list of num_ref_pics integers, each indicating the POC offset of the reference frame relative to the current frame or the LTR index. - used_by_cur: a list of num_ref_pics binaries, each indicating whether the corresponding frame in the ref_pics list is used as a reference for encoding of the current frame. Each special configuration line must contain the following fields in order: - Frame0: indicates that the current line contains special GOP configurations (for example, for long-term reference). - Type: the slice type, which can be either P, B, or I. [-255] If the value is not -255, it overrides the value specified in the normal configurations. - QPoffset: the offset to be added to the QP parameter to obtain the final QP value used for this frame. [-255] If the value is not -255, it overrides the value specified in the normal configurations. - QPfactor: the weight used during rate distortion optimization. [-255] If the value is not -255, it overrides the value specified in the normal configurations. - TemporalId: temporal layer ID. [-255] If the value is not -255, it overrides the value specified in the

Short	Long Option	Description
		indicating whether the frame is used as a reference for encoding of the current frame. - LTR: the LTR index of the frame. Value 0 indicates a common frame that uses LTR. A value in range from 1 to VCENC_MAX_LT_REF_FRAMES indicates an LTR frame. - Offset: If LTR equals 0, the Offset field indicates the POC delta between the LTR frame and the first subsequent frame that uses the LTR frame as reference. If LTR falls in the range from 1 to VCENC_MAX_LT_REF_FRAMES, the Offset field indicates the POC delta between the first LTR frame and the first encoded frame. - Interval: If LTR equals 0, the Interval field indicates the POC delta between two adjacent frames that use the same LTR frame as reference. If LTR falls in the range from 1 to VCENC_MAX_LT_REF_FRAMES, the Interval field indicates the POC delta between two adjacent frames with the same LTR index.
	--gopLowdelay	Whether to use the default low-delay GOP configuration. [0] 0 - do not use 1 - use The option is valid only if --gopConfig is not specified and the value of gopSize is less than or equal to 4.
	--numRefP	1..2 The number of reference frames kept for each P frame. [1] Change the number of reference frames for predefined GOP structure P frame to use
	--lowdelayB	0..1 Use lowdelay B frame to replace regular P frame 0 - turn off 1 - to replace predefined GOP structure P frame Default 0 for GOP1, default 1 for GOP size > 1 and 2pass
	--flexRefs	The path to a custom reference description file. For details, see Section Flexible Referenc Configuration in Hantro VC9000E Video Encoder Test Bench User Manual.
-V[n]	--bFrameQpDelta	-1..51 The QP delta of B-frames to the target QP. [-1] If the option is set to -1, it indicates that no QP delta is set. The option is valid only if --gopConfig is not specified.

Short	Long Option	Description
	--refRingBufEnable	Whether to enable ring buffer for reference frame store. [0] 0 - disable 1 - enable The option is valid for P or I frame encoding only. Since only one reference frame is stored, some coding tools are invalid, such as lookahead, reEncode, B frame, two reference P frame, multi-core and parallel encoding, LTR.

2.3.3 Rate Distortion Optimization

Short	Long Option	Description
	--preset	The pre-defined RDO scheme. A higher value leads to higher quality but worse performance, and vice versa. To use this option, explicitly claim it.
	--rdoLevel	The RDO level. [3] The option value range is from 1 to 3, inclusive. A higher value leads to higher quality but better performance, and vice versa.
	--enableDynamicRdo	Whether to enable dynamic RDO level selection. [0] 0 - disable 1 - enable
	--dynamicRdoCu16Bias	The bias used for 16x16 CU cost calculation in dynamic RDO level selection. [3] The option value range is from 0 to 255, inclusive.
	--dynamicRdoCu16Factor	The factor used for 16x16 CU cost calculation in dynamic RDO level selection. [80] The option value range is from 0 to 255, inclusive.
	--dynamicRdoCu32Bias	The bias used for 32x32 CU cost calculation in dynamic RDO level selection. [2] The option value range is from 0 to 255, inclusive.
	--dynamicRdoCu32Factor	The factor used for 32x32 CU cost calculation in dynamic RDO level selection. [32] The option value range is from 0 to 255, inclusive.

Short	Long Option	Description
	--enableRdoQuant	Whether to enable rate-distortion optimized quantization (RDOQ). 0 - disable 1 - enable The default value is 0 if the hardware does not support RDOQ, and 1 if the hardware supports RDOQ.

2.3.4 VUI and SEI

SEI

Short	Long Option	Description
-S[n]	--sei	(Only for HEVC/H.264) Whether to insert picture timing and buffering period SEI messages into the stream. [0] 0 - do not insert 1 - insert
-z[s]	--userData	The path to an SEI user data file. The file is parsed and inserted as an SEI message before the first frame.
	--extSEI	The path to an external SEI data file. The file is parsed and inserted as SEI messages for each frame.
	--jsonHDR	The file with dynamic HDR10+ and/or Dolby Vision information. If this option is specified, add the HDR_HELPER_SUPPORT=y option when generating the executable binary.
	--t35=a:b	The country code and area code. [0 : 0] - a indicates the country code, in one byte. - b indicates the area code, in one byte. For details, see Annex A and B of ITU-T Rec. T.35.
	--payloadT35File	The path to a file that contains the T.35 payload data (t35_payload_bytes) in binary bytes.

VUI

Short	Long Option	Description
	--vuiAspectRatio [=aspectratioWidth :aspectratioHeight]	The sample aspect ratio in the VUI. [0 : 0] - aspectratioWidth: The sample aspect ratio in horizontal direction, in an arbitrary unit. - The value range is from 0 to 65535, inclusive. - aspectratioHeight: The sample aspect ratio in vertical direction, in the same arbitrary unit as aspectratioWidth. - The value range is from 0 to 65535, inclusive. If either aspectratioWidth or scaledHeight is set to 0, it indicates that the sample aspect ratio is not specified.
	--vuiVideosignalPresent	Whether to present the video signal type in the VUI. [0] 0 - do not present 1 - present
	--vuiVideoFormat	The video format in the VUI. [5] 0 - component 1 - PAL 2 - NTSC 3 - SECAM 4 - MAC 5 - UNDEF
-k[n]	--videoRange	The video signal sample range in the encoded stream. [0] 0 - Y samples range from 16 to 235 and UV samples range in from 16 to 240. 1 - YUV samples range from 0 to 255.

Short	Long Option	Description
	--vuiColordescription [=primary:transfer :matrix]	The color description in the VUI. - primary: The index of chromaticity coordinates. [2] Value 2 indicates unspecified. The value range is from 0 to 255, inclusive. For details, see Table E.3 in ITU-T Rec. H.265. - transfer: The reference of the opto-electronic transfer function of the source picture. [2] Value 1 indicates IT-R BT.709-6 Value 2 indicates unspecified. Value 14 indicates ITU-R Rec. BT.2020-2. Value 18 indicates ARIB STD-B67. For details, see Table E.4 in ITU-T Rec. H.265. - matrix: The index of matrix coefficients used for deriving luma and chroma signals from green, blue, and red or Y, Z, and X primaries. [2] Value 2 indicates unspecified. The value range is from 0 to 255, inclusive. For details, see Table E.5 in ITU-T Rec. H.265.

HDR10

Short	Long Option	Description
	--writeOnceHDR10	Whether to write HDR10 information only before the first IDR frame. [0] 0 - write before each IDR frame. 1 - write only before the first IDR frame.

Short	Long Option	Description
	--HDR10_display [=dx0:dy0:dx1:dy1: dx2:dy2:wx:wy: max:min]	The color volume SEI message of the mastering display. - dx0: the normalized X chromaticity coordinate of component 0. [0] The value range is from 0 to 50000, inclusive. - dy0: the normalized Y chromaticity coordinate of component 0. [0] The value range is from 0 to 50000, inclusive. - dx1: the normalized X chromaticity coordinate of component 1. [0] The value range is from 0 to 50000, inclusive. - dy1: the normalized Y chromaticity coordinate of component 1. [0] The value range is from 0 to 50000, inclusive. - dx2: the normalized X chromaticity coordinate of component 2. [0] The value range is from 0 to 50000, inclusive. - dy2: the normalized Y chromaticity coordinate of component 2. [0] The value range is from 0 to 50000, inclusive. - wx: the normalized X chromaticity coordinate of the white point. [0] The value range is from 0 to 50000, inclusive. - wy: the normalized Y chromaticity coordinate of the white point. [0] The value range is from 0 to 50000, inclusive. - max: the nominal maximum display luminance. [0] - min: the nominal minimum display luminance. [0]
	--HDR10_lightlevel [=maxlevel:avglevel]	The content light level information SEI message. - maxlevel: the maximum content light level. - avglevel: the maximum picture average light level.

2.3.5 (Obsoleted) Noise Reduction

Short	Long Option	Description
	--noiseReductionEnable	(Obsoleted) Whether to enable noise reduction (3DNR). [0] 0 - disable 3DNR 1 - enable 3DNR
	--noiseLow	(Obsoleted) The minimum noise value. [10] The option value range is from 1 to 30, inclusive.
	--noiseFirstFrameSigma	(Obsoleted) The noise estimation for start frames. [11] The option value range is from 1 to 30, inclusive.
	--noiseReductionStrength_IntraY	0..31 denoise strength for intra Y channel [7]
	--noiseReductionStrength_IntraU	0..31 denoise strength for intra U channel [7]
	--noiseReductionStrength_IntraV	0..31 denoise strength for intra V channel [7]
	--noiseReductionStrength_InterY	0..31 denoise strength for inter Y channel [7]
	--noiseReductionStrength_InterU	0..31 denoise strength for inter U channel [7]
	--noiseReductionStrength_InterV	0..31 denoise strength for inter V channel [7]
	--noiseReduction_ChromaMaxMV	0..15 only denoise U/V with MV x/y <= ChromaMaxMV [4]

2.3.6 (Obsoleted) Smart Background Detection

Short	Long Option	Description
	--smartConfig	(Obsoleted) The path to the custom configuration file for the smart algorithm.

2.3.7 End-of-Sequence Disabling

Short	Long Option	Description
	--disableEOS	Whether to write end-of-sequence (EOS) bytes before stream is closed. [0] 0 - do not write 1 - write

2.3.8 Parameters for Enable intraRecon

Short	Long Option	Description
	--intraReconEnable	Enable/disable intra recon when HW support. [1] 0 = Disable intraRecon. 1 = Enable intraRecon.

2.3.9 AV1 Verification for FFmpeg Framework

Short	Long Option	Description
	--modifiedTileGroupSize	Whether to modify the space size of OBU_TILE_GROUP coded by the hardware. [0] 0 - do not modify 1 - modify Use this option for AV1 verification only when the FFmpeg framework is used.

2.3.10 Re-Encoding

Short	Long Option	Description
	--reEncode	Whether to re-encode frames in exceptional scenarios such as output buffer overflow. [0] 0 - do not re-encode frames 1 - re-encode frames For example, when an output buffer overflows, it can be reallocated through a callback function. Then, if this option is set to 1, the encoder re-encodes the current frame with the same settings but the new output buffer. Re-encoding is not supported if the value of --parallelCoreNum is greater than 1.

2.3.11 Motion Estimation and Global Motion

Short	Long Option	Description
	<code>--gmVFile</code>	(Reserved) The path to a file that contains ME search range offsets for reference list 0. In the file, the search offsets of each frame must be listed sequentially line by line. To set a frame-level offset, there should be only one pair of offsets per line for the frame. For example: (MVX, MVY) When <code>SEARCH_RANGE_ROW_OFFSET_TEST</code> is enabled, to set CTU-row-level offsets, there should be multiple offset pairs per line for all CTU rows. For example: (MVX0, MVY0) (MVX1, MVY1) ... Common delimiters such as white spaces, commas (,), and semicolons (;) are allowed between offsets.
	<code>--gmV=MVX:MVY</code>	The frame-level ME search range offsets for reference list 0. [0 : 0] - MVX: the horizontal offset, in pixels. The value must be 64 aligned and within the range from -128 to +128, inclusive. - MVY: the vertical offset, in pixels. The value must be 16 aligned and within the range from -128 to +128, inclusive. If both <code>gmVFile</code> and <code>gmV</code> are specified, the settings of <code>gmVFile</code> take effect.
	<code>--gmVList1=MVX:MVY</code>	(Reserved) The frame level ME search range offsets for reference list 1. [0 : 0] - MVX: the horizontal offset, in pixels. The value must be 64 aligned and within the range from -128 to +128, inclusive. - MVY: the vertical offset, in pixels. The value must be 16 aligned and within the range from -128 to +128, inclusive. If both <code>--gmVList1File</code> and <code>--gmVList1</code> are specified, the settings of <code>--gmVList1File</code> take effect.

Short	Long Option	Description
	--gmVList1File	(Reserved) The path to a file that contains ME search range offsets for reference list 1. In the file, the search offsets of each frame must be listed sequentially line by line. To set a frame-level offset, there should be only one pair of offsets per line for the frame. For example: (MVX, MVY) When SEARCH_RANGE_ROW_OFFSET_TEST is enabled, to set CTU-row-level offsets, there should be multiple offset pairs per line for all CTU rows. For example: (MVX0, MVY0) (MVX1, MVY1) ... Common delimiters such as parentheses (), white spaces, commas (,), and semicolons (;) are allowed between offsets.
	--MEVertRange	The ME vertical search range, in pixels. [0] For HEVC (H.265) and AV1, valid option values include 0, 40, and 64. For H.264 (AVC), valid option values include 0, 24, 48, and 64. If the option is set to 0, the maximum supported search range specified through EWLHwConfig_t.meVertSearchRangeHEVC or EWLHwConfig_t.meVertSearchRangeHEVC is used. This option is valid only if the value of EWLHwConfig_t.meVertRangeProgramable is 1.

2.4 Rate Control

2.4.1 Rate Control Mode



Short	Long Option	Description
Short	Long Option	Description
	--rcMode	The rate control mode. [0] 0 or cvbr - constrained variable bit rate (CVBR) mode The CVBR mode reduces encoded bits in slow-motion and simple scenes, and uses the reduced bits for complex scenes to improve overall quality. If --rcMode is set to 0 with options set as --cpbSize=x, --vbr=1, or --crf=x, the applied rate control mode is adjusted to another mode instead of CVBR. However, if rcMode is set to another value, the rcMode setting has a higher priority than these options. 1 or cbr - constant bit rate (CBR) mode The CBR mode ensures output at a constant bit rate to the transmission path. In CBR mode, make sure that cpbSize is set to an appropriate value. - A greater value of cpbSize helps avoid data overflow. In this case, to avoid underflow, you can set --hrdConformance=1. - A smaller value generates a more stable bit rate, but increases the risk of data overflow. A value of (2 x bitPerSecond) is recommended. Values less than bitPerSecond are not recommended. 2 or vbr - variable bit rate (VBR) The VBR mode allows for bit rate control with limitations on the minimum QP and maximum bit rate. It is recommended that --qpMin be set to an appropriate value. --qpMin=10 is recommended. NOTE: Setting --rcMode=2 is equivalent to setting --rcmode=0 --vbr=1. 3 or abr - average bit rate (ABR) mode The ABR mode tunes AP based on the gap between the actual target bit rates instead of scene complexity. This ensures the average bit rate approximates the target bit rate. 4 or crf - constant rate factor (CRF) mode The CRF mode ensures a certain level of quality with a constant slice QP derived from a given rate factor and frame rate. The CRF mode is available only if look-ahead encoding is enabled. If both CRF and VBR modes are set to be enabled, only the CRF mode is enabled and the VBR mode is disabled. In CRF mode, make sure that --crf is set to an appropriate value. 5 or cqp - constant quantization parameter (CQP) mode In CQP value, a constant QP value is used for all slices with a separate QP delta (--intraQpDelta) for I slices. In CQP mode, the value of --picRc is fixed to 0.

Short	Long Option	Description
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2.4.2 Output Stream Bit Rate

Short	Long Option	Description
-R[n]	--intraPicRate	The interval between two IDR frames. [0] The encoder forces a frame to be encoded as an IDR frame every N frames. If the option is set to 0, the encode does not force any frame to be encoded as an IDR.
	--intraPeriod	The interval between two non-IDR intra frames. [0] Starting from an IDR frame, the encoder forces a frame to be encoded as an intra frame every N frames.
-B[n]	--bitPerSecond	The target bit rate for rate control, in bits per second. [1000000] The option value range is from 10000 to the maximum limited by the stream level, inclusive. The option value must be greater than or equal to 10000. If HRD is enabled, the maximum target bit rate is limited by the stream tier and level selected during encoder initialization. For detailed limitations, see ITU-T Rec. H.265.
	--tolMovingBitRate	The tolerance percentage of the maximum moving bit rate over the target bit rate. [100] The average bit rate of the monitored frames is limited to [bitPerSecond * (1 + tolMovingBitRate/100)]. The option value range is from 0 to 2000, inclusive.
	--monitorFrames	The number of frames to be monitored for calculating the moving bit rate. [outputRateNumer/outputRateDenom] The option value range is from 10 to 120, inclusive.
	--bitVarRangel	(Obsoleted) The permitting variance percentage over average bits per I-frame from the target bit rate. [10000] The option value range is from 10 to 10000, inclusive.
	--bitVarRangeP	The permitting variance percentage over average bits per P-frame from the target bit rate. [10000] The option value range is from 10 to 10000, inclusive.

Short	Long Option	Description
	--bitVarRangeB	The permitting variance percentage over average bits per B-frame from the target bit rate. [10000] The option value range is from 10 to 10000, inclusive.
	--staticScenelbitPercent	The percentage of I-frame bits in static scenes. [80] The option value range is from 0 to 100, inclusive.
	--crf[n]	The constant rate factor. [-1] The option value range is from -1 to 51, inclusive. If the option is set to -1, the constant rate factor mode is disabled.
-U[n]	--picRc	Whether to enable picture-level rate control to adjust QP between pictures. [0] 0 - disable 1 - enable
	--tolRcUnderflow	[0, 99] Tolerance percent of frame RC underflow bitrate [50] Percentage of underflow bitrate tolerance for better quality Save bits in simple scene for complex scene Higher the value, larger underflow may happen Worst underflow bitrate: $\text{TargetBits} * (100 - \text{tolRcUnderflow}) / 100$
	--picQpDeltaRange [=Min:Max]	The QP delta range in picture-level rate control. - Min indicates the minimum QP delta in picture-level rate control. [-4] The value range is from -10 to -1, inclusive. - Max indicates the maximum QP delta in picture-level rate control. [4] The value range is from 1 to 10, inclusive. This QP delta range applies only to two adjacent frames of the same coding type. It does not apply if HRD overflow occurs.
	--fillerData	Whether to fill data when HRD is disabled. [0] 0 - do not fill 1 - fill If this option is set to 1, cpbSize must be specified.

Short	Long Option	Description
-C[n]	--hrdConformance	(Only for HEVC/H.264) Whether to enable HRD conformance checking. [0] 0 - disable 1 - enable If this option is set to 1, the encoder uses the HRD model to restrict the bit rate variance.
-c[n]	--cpbSize	The size of the CPB in bits. [0] When HRD is enabled, each encoded frame cannot be larger than the specified size. By default, the video encoder uses the maximum size allowed by the stream level. Setting this field to 0 restores the default value. A value of (2 * bitPerSecond) is recommended.
	--cpbMaxRate	The maximum bit rate of the CPB, in bits per second. [0] If --cpbSize is specified and --cpbMaxRate is unspecified or less than --bitPerSecond, --cpbMaxRate is fixed to the same value as --bitPerSecond regardless of its setting. If --cpbMaxRate is equal to --bitPerSecond, the rate control algorithm works in CBR mode. If --cpbMaxRate is greater than --bitPerSecond, the rate control algorithm works in VBR mode.
-g[n]	--bitrateWindow	1..300 The number of frames within which the rate control algorithm tries to achieve the target bit rate. [intraPicRate] The rate control algorithm allocates bits for each window and tries to match the target bit rate at the end of the window. Typical windows begin with an intra frame, which is not mandatory.
	--LTR=a:b:c[:d]	The long term reference (LTR) settings. - a: the POC delta between two LTR frames. - b: the POC delta between the LTR frame and the first subsequent frame that uses the LTR frame as reference. - c: the POC delta between two adjacent frames that use the same LTR frame as reference. - d: the QP delta for frames using LTR as reference. [0]

Short	Long Option	Description
-s[n]	--picSkip	Whether to allow skippint pictures for bit rate control. [0] 0 - disallow 1 - allow
-q[n]	--qpHdr	-1..51 The default or initial picture-level QP value. Default 26 for fixed QP method. Default -1 for frameRC method. when qpHdr is -1, the encoder calculates the initialQP according to the target bit rate when enable FrameRc.
-n[n]	--qpMin	0..51 The minimum QP for all slices. [0]
-m[n]	--qpMax	0..51 The maximum QP for all slices. [51]
	--qpMinI	0..51 The minimum QP for I slices. [0] For I slices, if both --qpMin and --qpMinI are specified, --qpMinI takes precedence.
	--qpMaxI	0..51 The maximum QP for I slices. [51] For I slices, if both --qpMax and --qpMaxI are specified, --qpMaxI takes precedence.
-A[n]	--intraQpDelta	-51..51 The delta between the target QP and intra QP values. [-5]
-G[n]	--fixedIntraQp	0..51 The fixed QP value for all intra frames. [0] If this option is set to 0, the QP value for intra frames is not fixed.
-l[n]	--chromaQpOffset	-12..12 The chroma QP offset. [0]
	--vbr	Whether to enable variable bit rate control based on the minimum QP allowed. [0] 0 - disable 1 - enable
	--sceneChange	The frames with scene changes. Set the option value in format Frame1:Frame2:...:Frame20. Separate frame numbers with colons (:). A maximum of 20 scene change frames can be specified.

Short	Long Option	Description
	--gdrDuration	The number of the P frame to be refreshed as an I frame in gradual decoder refresh (GDR). [0] If the option is set to 0, GDR is disabled. NOTE: The starting point of GDR is the frame with type set to VCENC_INTRA_FRAME. intraArea and roi1Area are used to implement the GDR function. The GDR begin to work from the second IDR frame.
	--skipFramePOC	The POC of the frame to be force encoded as a skip frame. [0] If this option is set to 0, the encoder does not force encode any frame as a skip frame.
	--insertIDR	The picture_cnt of the frame to be force encoded as an IDR frame. If this option is set to 0, the encoder does not force encode any frame as an IDR frame.
	--minIprop	0..100 The minimum ratio of I-frame bits to P-frame bits (I/P ratio).
	--maxIprop	minIprop..100 The maximum ratio of I/P ratio. The range specified by the --minIprop and --maxIprop options is used to clamp the I/P ratio. A larger value of --minIprop leads to clear I-frames and blurred P-frames. It is not recommended to restrict it, in order to avoid breathing effects and rate fluctuations.
	--changePos	50..100 The ratio of the bit rate at the initialization of QP adjustment to the maximum bit rate.
	--hieQpDeltaEnable	[0,1] Frame QP delta based on GOP hierarchical layer. [1] Only available in single pass encoding, default 0 for GOP1 0 : Disable. 1 : Enable.
	--aifEnable	Whether to enable anti-intra flicker (AIF). [0] 0 - disable 1 - enable NOTE: AIF works only for 1-pass 8-bit encoding. It is not available for multi-task, GDR, or low-latency encoding or if the input is tiled.
	--aifQpDelta	-51..51 The QP delta of AIF frames to their reference frames. [-5]

2.4.3 Block-level rate control

Short	Long Option	Description
-u[n]	--ctbRc	The block-level rate control mode for adjusting QP inside a frame. [0] 0 - block-level rate control disabled 1 - subjective rate control only 2 - precise rate control only This value is available only if the value of EWLHwConfig_t.CtbRcVersion is greater than or equal to 1. 3 - mixed subjective and precise rate control This value is available only if the value of EWLHwConfig_t.CtbRcVersion is greater than or equal to 1. 4 - (Reserved) precise rate control only 6 - (Reserved) mixed subjective and precise rate control
	--blockRCSIZE	The block size for block-level rate control [0] 0 - 64x64 pixels 1 - 32x32 pixels 2 - 16x16 pixels
	--rcQpDeltaRange	The maximum absolute delta between block-level and picture-level QP values in block-level rate control. [10] If the value of EWLHwConfig_t.CtbRcVersion is less than 1, the option value range is from 0 to 15, inclusive. If the value of EWLHwConfig_t.CtbRcVersion is greater than or equal to 1, the option value range is from 0 to 51, inclusive.
	--rcBaseMBComplexity	The MB complexity threshold for subjective block-level rate control. [15] When the MB complexity equals the threshold, the block-level QP is not adjusted. The QP value increases when the MB complexity exceeds the threshold, and vice versa. The option value range is from 0 to 31, inclusive.

Short	Long Option	Description
	--tolCtbRcInter	The tolerance of block-level rate control for inter frames. $[-1.0]$ The rate control algorithm tries to limit inter frame bits within range from $\text{targetPicSize}/(1 + \text{tolCtbRcInter})$ to $\text{targetPicSize} * (1 + \text{tolCtbRcInter})$. The <code>--tolCtbRcInter</code> option value is a floating-point number. A negative value indicates no bit rate limit in block-level rate control.
	--tolCtbRcIntra	The tolerance of block-level rate control for intra frames. $[-1.0]$ The rate control algorithm tries to limit intra frame bits within range from $\text{targetPicSize}/(1 + \text{tolCtbRcIntra})$ to $\text{targetPicSize} * (1 + \text{tolCtbRcIntra})$. The <code>--tolCtbRcIntra</code> option value is a floating-point number. A negative value indicates no bit rate limit in block-level rate control.
	--ctbRcRowQpStep	The maximum accumulated QP adjustment step per CTB Row allowed by CTB rate control <code>ctbrcv1</code>: default value is [4] <code>ctbrcv2</code>: 0..7 and default [2]
	--ctbRcRowQpDeltaRange	0..31 Qp Delta based on <code>picQP[1]</code>. Only work when <code>ctbrcV2</code>
	--ctbRcDirection	0..15 Subscript of minus direction threshold. [8] It used with <code>ctbRcThreshold</code> to decrease/increase block QP. When the block complexity is between <code>ctbRcThreshold[0]</code> and <code>ctbRcThreshold[Direction]</code> will decrease QP, and the complexity is bigger than <code>ctbRcThreshold[Direction]</code> will increase QP. It is used for subjective block-level rate control.
	--ctbRcThresholdI =a0:a1:a2:a3:a4:a5:a6:a7: a8:a9:a10:a11:a12:a13: a14:a15	0..255 A threshold array, it used with <code>ctbRcDirection</code> to decrease/increase block QP of Intra frame. [0:0:0:0:3:3:5:5:6:6:6:11:11:13:17:17] It is used for subjective block-level rate control.

Short	Long Option	Description
	--ctbRcThresholdP =b0:b1:b2:b3:b4:b5:b6:b7: b8:b9:b10:b11:b12:b13: b14:b15	0..255 A threshold array, it used with ctbRcDirection to decrease/increase block QP of P frame. [0:0:0:0:3:3:5:5:6:6:6:11:11:13:17:17] It is used for subjective block-level rate control.
	--ctbRcThresholdB =c0:c1:c2:c3:c4:c5:c6:c7: c8:c9:c10:c11:c12:c13: c14:c15	0..255 A threshold array, it used with ctbRcDirection to decrease/increase block QP of B frame. [0:0:0:0:3:3:5:5:6:6:6:11:11:13:17:17] It is used for subjective block-level rate control.
	--ctbRcSkinQPDelta	0..7 Negative qp delta of skin area for subjective block-level rate control. [2]
	--ctbRcSkinMinQPDelta	0..15 It subtract ctbRcDirection value is the minimal qp delta of skin area for subjective block-level rate control. [3] Skin area not set too small QPDelta, so use ctbRcSkinMinQPDelta - direction to limit it. ctbRcSkinMinQPDelta - direction always less than 0.
	--ctbRcSkinCbRange=a:b	a:b 0..255 Cb value range to be regarded as skin Area for subjective block-level rate control.[0:0] We use Skin area color range to detect skin Area, otherwise use default algorithm to detect skin area.
	--ctbRcSkinCrRange=a:b	a:b 0..255 Cr value range to be regarded as skin Area for subjective block-level rate control.[0:0]
	--ctbRcSkinLumRange=a:b	a:b 0..255 Lum value range to be regarded as skin Area for subjective block-level rate control.[0:255]

2.4.4 Look-Ahead Encoding

Short	Long Option	Description
	--lookaheadDepth	The number of look-ahead frames. [0] 0 - disables look-ahead encoding. 4..40 - enables look-ahead encoding with the specified number of look-ahead frames.
	--halfDsInput	The path to the file that contains the external provided half-downsampled YUV input.

Short	Long Option	Description
	--aq_mode	The adaptive quantization mode. [0] 0 - none 1 - uniform AQ 2 - auto variance 3 - auto variance with bias to dark scenes
	--aq_strength	0..3.0 The strength of adaptive quantization. [1 . 0] A great value reduces blocking and blurring in flat and textured areas.
	--tune	The type of the target based on which some encoding settings are forced for quality tuing. [0] 0 or psnr - PSNR with forced settings --aq_mode=0 --psyFactor=0 1 or ssim - SSIM with forced settings --aq_mode=2 --psyFactor=0 2 or visual - Visual with forced settings --aq_mode=2 --psyFactor=0.75 3 or sharpness_visual - sharpness-based visual with forced parameters --aq_mode=2 --psyFactor=0.75 --inLoopDSRatio=0 --enableRdoQuant=0 4 or vmaf - (Reserved) VMAF pre-processing enabled
	--inLoopDSRatio	The in-loop down-scaling ratio for the pass-1 encoder. [1] 0 - down-scaling disabled 1 - 1/2 down-scaling ratio

2.5 Visual Quality

2.5.1 Parameters Affecting Visual Tools

Short	Long Option	Description
	--ctbRcTrailStrengthMax	0..3 Trailing area detection max strength. [0] 0 turns off trailing detection. Suggested value is [3] Only available with GOP size 1 and ctbRc Version=2
	--ctbRcTrailDeltaQp	-7..0 Trailing area base qp delta. [-4] QP delta for 16x16 trailing CU, adjusted by CU size
	--IntraBiasChromaStrength	0..10 Chroma error detection strength. Default off; to activate this feature, the suggested value is 7 [0] 0 turns off chroma error detection.

Short	Long Option	Description
	--IntraBiasStrength	0..15 Strength of intra mode is preferred which will guarantee visual quality. Default off; to activate this feature, the suggested value is 10 [0] 0 turns off intra bias.
	--IntraBiasMvThreshold	Int Mv limit for intra bias decision.[5] 0 no mv limit for intra bias decision.
	--bTrailAvoidIntraBias	Whether intraBias is disabled in the trailing area.[1] 0 enable intraBias in the trailing area. 1 disable intraBias in the trailing area.
	--visualBitRateTolerance	-1..1000 Percent of target bitrate tolerance allowed for visual tools to exceed. [50] -1 no limit for visual tools to spend extra bits Trail Reduce and Intra Bias will adjust strength at frame level adaptively to keep bitrate less than $(1 + \text{visualBitRateTolerance}/100) * \text{target bitrate}$ Only available with GOP size 1

2.6 ROI

2.6.1 ROI-Based Coding Control

Short	Long Option	Description
	--cir =[start:interval]	Forces one MB or CTB to be encoded with intra mode at an - start: the order count of the start MB or CTB. - interval: the interval.
	--intraArea =left:top:right:bottom	The ROI for forcing the intra mode. - left: the leftmost MB or CTB column inside the ROI. - top: the top MB or CTB column inside the ROI. - right: the rightmost MB or CTB column inside the ROI. - bottom: the bottom MB or CTB column inside the ROI.

Short	Long Option	Description
	--ipcm1Area =left:top:right:bottom	ROI 1 for forcing the IPCM mode. - left: the leftmost MB or CTB column inside the ROI. - top: the top MB or CTB column inside the ROI. - right: the rightmost MB or CTB column inside the ROI. - bottom: the bottom MB or CTB column inside the ROI.
	--ipcm2Area =left:top:right:bottom	ROI 2 for forcing the IPCM mode.
	--ipcm3Area =left:top:right:bottom	ROI 3 for forcing the IPCM mode.
	--ipcm4Area =left:top:right:bottom	ROI 4 for forcing the IPCM mode.
	--ipcm5Area =left:top:right:bottom	ROI 5 for forcing the IPCM mode.
	--ipcm6Area =left:top:right:bottom	ROI 6 for forcing the IPCM mode.
	--ipcm7Area =left:top:right:bottom	ROI 7 for forcing the IPCM mode.
	--ipcm8Area =left:top:right:bottom	ROI 8 for forcing the IPCM mode.
	--ipcmMapEnable	Whether to enable the ROI map for forcing IPCM. [0] 0 - disable 1 - enable
	--ipcmMapFile	The path to the file that contains the IPCM map. The IPCM map defines the IPCM flag for each block in the frame. For the IPCM map, the block size is 64x64 pixels for HEVC (H.265) and 16x16 pixels for H.264 (AVC).
	--roi1Area =left:top:right:bottom	ROI 1 for forcing a QP delta or an absolute QP value. - left: the leftmost MB or CTB column inside the ROI. - top: the top MB or CTB column inside the ROI. - right: the rightmost MB or CTB column inside the ROI. - bottom: the bottom MB or CTB column inside the ROI.
	--roi2Area =left:top:right:bottom	ROI 2 for forcing a QP delta or an absolute QP value.

Short	Long Option	Description
	--roi3Area =left:top:right:bottom	ROI 3 for forcing a QP delta or an absolute QP value.
	--roi4Area =left:top:right:bottom	ROI 4 for forcing a QP delta or an absolute QP value.
	--roi5Area =left:top:right:bottom	ROI 5 for forcing a QP delta or an absolute QP value.
	--roi6Area =left:top:right:bottom	ROI 6 for forcing a QP delta or an absolute QP value.
	--roi7Area =left:top:right:bottom	ROI 7 for forcing a QP delta or an absolute QP value.
	--roi8Area =left:top:right:bottom	ROI 8 for forcing a QP delta or an absolute QP value.
	--roi1DeltaQp	The QP delta value for blocks in ROI 1. [0] If absolute QP is supported, the option value range is from -51 to 51, inclusive. If absolute QP is not supported, the option value range is from -30 to 0, inclusive. Set either the QP delta or absolute QP value for each ROI.
	--roi2DeltaQp	The QP delta value for blocks in ROI 2. [0]
	--roi3DeltaQp	The QP delta value for blocks in ROI 3. [0]
	--roi4DeltaQp	The QP delta value for blocks in ROI 4. [0]
	--roi5DeltaQp	The QP delta value for blocks in ROI 5. [0]
	--roi6DeltaQp	The QP delta value for blocks in ROI 6. [0]
	--roi7DeltaQp	The QP delta value for blocks in ROI 7. [0]
	--roi8DeltaQp	The QP delta value for blocks in ROI 8. [0]
	--roi1Qp	0..51 The absolute QP value for blocks in ROI 1. [-1] Negative values are invalid. Options --roi1Qp to --roi8Qp are valid only if absolute QP is supported. Set either the QP delta or absolute QP value for each ROI.
	--roi2Qp	0..51 The absolute QP value for blocks in ROI 2. [-1]
	--roi3Qp	0..51 The absolute QP value for blocks in ROI 3. [-1]

Short	Long Option	Description
	--roi4Qp	0..51 The absolute QP value for blocks in ROI 4. [-1]
	--roi5Qp	0..51 The absolute QP value for blocks in ROI 5. [-1]
	--roi6Qp	0..51 The absolute QP value for blocks in ROI 6. [-1]
	--roi7Qp	0..51 The absolute QP value for blocks in ROI 7. [-1]
	--roi8Qp	0..51 The absolute QP value for blocks in ROI 8. [-1]
	--roiMapDeltaQpEnable	Whether to enable the ROI map for forcing QP delta or absolute QP values. [0] 0 - disable 1 - enable The block size is specified by --roiMapDeltaQpBlockUnit. The QP values are specified by --roiMapDeltaQpFile in text format or --roiMapInfoBinFile in binary format. The map size is calculated according to the picture width and height after being aligned to 64 pixels.
	--roiMapDeltaQpBlockUnit	The block size for the QP ROI map. [0] 0 - 64x64 pixels 1 - 32x32 pixels 2 - 16x16 pixels 3 - 8x8 pixels For H.264 (AVC), 8x8 pixel block is not supported.
	--roiMapConfigFile	The path to the ROI definition file.

Short	Long Option	Description
	<code>--roiMapDeltaQpFile</code>	The path to the file that contains the QP map in text format. The QP map defines the delta QP or absolute QP value for each block in the frame. If <code>--RoiQpDeltaVer</code> is set to 1, 2, or 3, both QP delta and absolute QP are supported. - The QP delta value range is from -51 to 51, inclusive. - The absolute QP value range is from 0 to 51, inclusive. Each absolute QP value must be prefixed with 'a', for example, 'a26'. If <code>--RoiQpDeltaVer</code> is set to 4, only QP delta is supported. - The QP delta value range is from -32 to 31, inclusive.
	<code>--roiMapInfoBinFile</code>	The path to the file that contains the QP map in binary format. The QP map in binary format uses one byte to define the delta QP or absolute QP value for each block in the frame. The block size is specified by <code>--roiMapDeltaQpBlockUnit</code>. The byte bitmap of the QP map depends on the setting of <code>--RoiQpDeltaVer</code>. If <code>--RoiQpDeltaVer</code> is set to 1, 2, or 3, both QP delta and absolute QP are supported. - The QP delta value range is from -51 to 51, inclusive. - The absolute QP value range is from 0 to 51, inclusive. If <code>--RoiQpDeltaVer</code> is set to 4, only QP delta is supported. - The QP delta value range is from -32 to 31, inclusive. This option is valid only when the value of <code>--lookaheadDepth</code> is equal to 0.

Short	Long Option	Description
	<code>--RoiQpDeltaVer</code>	1..4 The format version of the ROI map. This option determines the format used in the file specified by <code>--roiMapInfoBinFile</code>. For details about ROI map formats, see Hantro VC9000E Series Memory Buffer and Format Organization.
	<code>--RoimapCuCtrlInfoBinFile</code>	The path to the file that contains the CU control map in binary format. The CU control map defines the CU control information for each block in the frame. The block size is specified by <code>--roiMapDeltaQpBlockUnit</code>. If both a CU control map and a QP map are enabled and specified, the QP map takes precedence and the QP information in the CU control map is ignored. The CU control map size depends on the setting of <code>--RoiCuCtrlVer</code>. The encoded picture is divided into controlled blocks according to the picture width and height after being aligned to 64 pixels.
	<code>--RoiCuCtrlVer</code>	0, 3..7 The format version of the CU control map. This option determines the format used in the file specified by <code>--RoimapCuCtrlInfoBinFile</code>. Setting it to 0 indicates to disable the CU control map. For details about CU control map formats, see Hantro VC9000E Series Memory Buffer and Format Organization.

Short	Long Option	Description
	<code>--roiMapDeltaQpBinFile</code>	The path to the file that contains the QP map in binary format for pass-2 encoding. Each byte in the file indicates the delta QP of a block unit. The block unit size is defined by <code>--roiMapDeltaQpBlockUnit</code>. The encoded picture is divided into controlled blocks according to the picture width and height after being aligned to 64 pixels. In the file, data is arranged in raster-scan order of blocks within each frame and frame data is organized in display order.
	<code>--RoimapCuCtrlIndexBinFile</code>	(Reserved) The path to the CU control map index binary file. The file specifies the ROIs for which the ROI information in the <code>--RoimapCuCtrlInfoBinFile</code> is valid. If <code>--RoimapCuCtrlInfoBinFile</code> is not specified, all ROI information in the <code>--RoimapCuCtrlInfoBinFile</code> is valid.

2.6.2 Skip-Mode Map

Short	Long Option	Description
	<code>--skipMapEnable</code>	Whether to enable the skip-mode map. [0] 0 - disable 1 - enable To enable skip-mode map, you also need to specify <code>--RoiQpDeltaVer=2</code> if one of the following condition is met: - The value of <code>EWLHwConfig_t.ROIMapVersion</code> is 2 with <code>HW_ID_MAJOR >= 0x82</code> or <code>(HW_ID_MAJOR == 0x60 && HW_ID_MINOR >= 0x010)</code>. - The value of <code>EWLHwConfig_t.ROIMapVersion</code> is 3. In other cases, specifying <code>--RoiQpDeltaVer</code> is not required.

Short	Long Option	Description
	--skipMapBlockUnit	The block size for the skip-mode map. [0] 0 - 64x64 pixels 1 - 32x32 pixels 2 - 16x16 pixels For HEVC (H.265), only 64x64 and 32x32 pixel blocks are supported.
	--skipMapFile	The path to the file that contains the skip-mode map. The skip-mode map defines whether each block in the frame is force encoded in skip mode. - Value 0 indicates not to force encode the block in skip mode. - Value 1 indicates to force encode the block in skip mode. The block width and height are calculated based on the picture width and height after alignment to 64 pixels should be used.

2.6.3 RDOQ Map

Short	Long Option	Description
	--rdoqMapEnable	Whether to enable the RDOQ map. [0] 0 - disable 1 - enable

2.7 Coding Statistics Output

Short	Long Option	Description
	--enableOutputCulInfo	Whether to enable CU/MB statistics output to DDR. [0] 0 - disable 1 - enable 2 - enable and write the statistics to the culInfo.txt file.

Short	Long Option	Description
	--cuInfoVersion	The CU/MB statistics format. [-1] -1 - format automatically decided based on software and hardware support by VCE and HW support. 0 - format 0, which is available only if the value of EWLHwConfig_t.cuInforVersion is 0. 1 - format 1, which is available if the value of EWLHwConfig_t.cuInforVersion is 1 or 2. 2 - format 2, which is available only if the value of EWLHwConfig_t.cuInforVersion is 2. If look-ahead encoding is enabled, --cuInfoVersion only affects collected statistics of the pass-2 encoder output. The pass-1 encoder is collected as IM input in format 2. For CuTree, only format 1 is supported.
	--enableOutputCtbBits	Whether to enable bit statistics output to DDR. [0] 0 - disable. 1 - enable with 2-byte data per CTB, in raster scan order of CTBs 2 - enable and write the statistics to the ctbBits.txt file.
	--hashtype	The method to calculate the hash value of the output stream. [0] 0 - hash value calculation disabled 1 - standard CRC32 2 - 32-bit checksum
	--ssim	Whether to enable SSIM calculation. [1] 0 - disable 1 - enable
	--psnr	Whether to enable PSNR calculation. [1] 0 - disable 1 - enable
	--enableVuiTimingInfo	Whether to write VUI timing information in SPS. [1] 0 - disable 1 - enable
	--enableFrameInfoVersion	Whether to enable frame statistics and luma statistics output to DDR. [0] 0 - disable 1 - frame information 2 - luma information and frame information

Short	Long Option	Description
	--sse0Enable	Whether to enable region 0 for SSE statistics. 0 - disable 1 - enable
	--sse1Enable	Whether to enable region 1 for SSE statistics. 0 - disable 1 - enable
	--sse2Enable	Whether to enable region 2 for SSE statistics. 0 - disable 1 - enable
	--sse3Enable	Whether to enable region 3 for SSE statistics. 0 - disable 1 - enable
	--sse4Enable	Whether to enable region 4 for SSE statistics. 0 - disable 1 - enable
	--sse5Enable	Whether to enable region 5 for SSE statistics. 0 - disable 1 - enable
	--sse6Enable	Whether to enable region 6 for SSE statistics. 0 - disable 1 - enable
	--sse7Enable	Whether to enable region 7 for SSE statistics. 0 - disable 1 - enable
	--sse0Rect	Region 0 for SSE statistics. 0 - disable 1 - enable
	--sse1Rect	Region 1 for SSE statistics. 0 - disable 1 - enable
	--sse2Rect	Region 2 for SSE statistics. 0 - disable 1 - enable
	--sse3Rect	Region 3 for SSE statistics. 0 - disable 1 - enable
	--sse4Rect	Region 4 for SSE statistics. 0 - disable 1 - enable

Short	Long Option	Description
	--sse5Rect	Region 5 for SSE statistics. 0 - disable 1 - enable
	--sse6Rect	Region 6 for SSE statistics. 0 - disable 1 - enable
	--sse7Rect	Region 7 for SSE statistics. 0 - disable 1 - enable

2.8 Peripheral and Hardware Setup

2.8.1 Reference Frame Compression

Short	Long Option	Description
	--compressor	Whether to enable embedded reference frame compression. [0] 0 - disable compression 3 - enable compression for both luma and chroma data
	--enableP010Ref	Whether to store reference frames in tiled or raster P010 format in buffers. [0] 0 - tiled 1 - raster

2.8.2 OSD & Mosaic

OSD Control

The encoder supports a maximum of 12 OSD regions. The same CTB cannot be shared across OSD regions.

In the following table, index N can be 01 to 12. For example, the option to configure the input file of OSD region 1 is `--olInput 01`.

Short	Long Option	Description
	--overlayEnables	Each bit indicates whether to enable the corresponding overlay region. [0] For example: - Value 1 indicates region 1 enabled. - Value 2 indicates region 2 enabled. - Value 3 indicates regions 1 and 2 enabled.
	--olInputN	The path to the file that contains the input file to each OSD region. [olInputi.yuv]
	--olFormatN	The input format of each OSD region. [0] 0 - ARGB8888 1 - NV12 2 - bitmap
	--olSuperTileN	Whether the OSD input data is organized in supertile mode. [0] 0 - non-supertile mode. 1 - X-major supertile mode. 2 - Y-major supertile mode. NOTE: super tile is only valid when input format is ARGB8888.
	--olAlphaN	0..255 The global alpha value for the OSD region. [0] This option is invalid for ARGB8888.
	--olWidthN	The width of each OSD input. [0] If a region is enabled, this option must be specified for the region. The option value must be 8 aligned for the bitmap format.
	--olHeightN	The height of each OSD input. [0] If a region is enabled, this option must be specified for the region. The option value must be 8 aligned for the bitmap format.
	--olXoffsetN	The horizontal offset, in pixels, of the top-left corner of each OSD region relative to the encoder picture. [0] The option value must be 2 aligned.
	--olYoffsetN	The vertical offset, in pixels, of the top-left corner of each OSD region relative to the encoder picture. [0] The option value must be 2 aligned.
	--olYStrideN	The Y stride of each OSD input, in bytes. For ARGB8888, the default value is (olWidthi * 4). For NV12, the default value is olWidthi For bitmap, the default value is (olWidthi/8).

Short	Long Option	Description
	--olUVStrideN	The UV stride of each OSD input, in bytes. The default value depends on the Y stride.
	--olCropXoffsetN	The horizontal offset, in pixels, of the top-left corner of the crop area in each OSD input relative to the overlay input picture. [0] The option value must be 8 aligned for the bitmap format, and 2 aligned for other formats.
	--olCropYoffsetN	The vertical offset, in pixels, of the top-left corner of the crop area in each OSD input relative to the overlay input picture. [0] The option value must be 8 aligned for the bitmap format, and 2 aligned for other formats.
	--olCropWidthN	The width of the crop area in each OSD input, which is the width of the final OSD region. [olWidth <i>i</i>] The option value must be 8 aligned for the bitmap format.
	--olCropHeightN	The height of the crop area in each OSD input, which is the height of the final OSD region. [olHeight <i>i</i>] The option value must be 8 aligned for the bitmap format.
	--olBitmapYN	The Y value for the bitmap format. [0]
	--olBitmapUN	The U value for the bitmap format. [0]
	--olBitmapVN	The V value for the bitmap format. [0]

OSD Map

The encoder supports a maximum of 12 colors for the OSD map.

In the following table, index N can be 01 to 12. For example, the option to specify the alpha value for OSD color 1 is `--osdMapAlpha01`.

Compared with overlay and mosaic regions, the OSD map has the lowest priority. The OSD map and OSD regions with super tile input cannot be enabled at the same time.

Short	Long Option	Description
	--osdMapEnable	The OSD map enable. [0] 0:disable 1:enable
	--osdMapInput	The input file for the OSD map. [NULL]
	--osdMapStride	The OSD map stride in bytes. [0] The stride must be greater than or equal to $(\text{codingWidth} + 15[\text{or } 7]) \gg 4[\text{or } 3]$ where codingWidth equals the value of <code>--width</code> , because the basic unit of the 4-bit OSD map is a 8x8 or 4x4 pixel block.

Short	Long Option	Description
	--osdMapBlockSize	OSD Map different block size in pixel. [0] 8:8x8 4:4x4
	--osdMapAlphaN	0..255 The alpha value for color N. [0]
	--osdMapYN	The Y value of the OSD map color N. [0]
	--osdMapUN	The U value of the OSD map color N. [0]
	--osdMapVN	The V value of the OSD map color N. [0]

Mosaic Control

The encoder supports a maximum of 12 mosaic regions. Each mosaic region must be aligned to the CTB size.

In the following table, index N can be 01 to 12. For example, the option to specify mosaic region 1 is `-mosArea01`.

Mosaic and OSD regions of the same index cannot be enabled at the same time. If a mosaic region and an OSD region of the same index are enabled, the OSD region is invalid. For example: OSD region 1 cannot be enabled if mosaic region 1 is enabled. OSD region 1 can be enabled if any of mosaic regions 2 to 12 is enabled.

Short	Long Option	Description
	--mosaicEnables	Each bit indicates whether to enable the corresponding mosaic region. [0] For example: - Value 1 indicates region 1 enabled. - Value 2 indicates region 2 enabled. - Value 3 indicates regions 1 and 2 enabled.
	--mosSizeIndex	different Mosaic size.[0] 0:8x8 1:16x16 2:32x32 3:64x64 4:128x128
	--mosAreaN =left:top:right:bottom	Mosaic region N. - left: the leftmost pixel inside the region. - top: the top pixel inside the region. - right: the rightmost pixel inside the region. - bottom: the bottom pixel inside the region. Make sure that all mosaic regions are aligned to CTB.

2.8.3 Low-Latency Encoding

Short	Long Option	Description
	<code>--inputLineBufferMode</code>	The low-latency encoding mode. [0] 0 - disable low-latency encoding 1 - enable low-latency encoding with software handshaking and loopback enabled. 2 - enable low-latency encoding with hardware handshaking and loopback enabled.(effective only when the upstream IP is used, only VCE IP cannot be tested) 3 - enable low-latency encoding with software handshaking enabled and loopback disabled. 4 - enable low-latency encoding with hardware handshaking enabled and loopback disabled.(effective only when the upstream IP is used, oingle VCE IP cannot be tested)
	<code>--inputLineBufferDepth</code>	0..511 The number of CTB/MB rows to control loopback and handshaking. [1] If the loopback mode is enabled, there are two continuous ping-pong input line buffers. Each contains <code>inputLineBufferDepth</code> CTB or MB rows. If hardware handshaking is enabled, handshaking signals are processed per <code>inputLineBufferDepth</code> CTB/MB rows. If software handshaking is enabled, IRQ is issued and the read count register is updated each time <code>inputLineBufferDepth</code> CTB/MB rows are read. Value 0 is supported only with <code>--inputLineBufferMode=3</code>. In this case, IRQ is not sent and the read count register is not updated.

Short	Long Option	Description
	--inputLineBufferAmountPerLoopback	0..1023 The number of handshake synchronizations every loopback. [0] If the option is set to 0, input line buffer is disabled.
	--segmentUnitHeight	8, 16 The height of each FLEXA SBI segment unit. [16] This option is valid only if the value of EWLHwConfig_t.prpSbiSupport is 1.
	--sliceIrqEnable	Whether to enable the hardware to issue an interrupt each time it finishes encoding a slice. [0] 0 - disable 1 - enable
	--inputSliceInfoEn	Whether to enable low-latency encoding in DDR mode. 0 - disable 1 - enable If low-latency encoding in DDR mode is enabled, the hardware polls a 64-byte DDR space for slice information.
	--lowlatGatingDisable	0..1 Lowlatency handshake auto gating disable[1] 0: enable lowlatency handshake auto gating. 1: disable lowlatency handshake auto gating.
	--lowlatGatingType	0..1 Lowlatency handshake auto gating type.[0] 0: check bus idle counter. 1: check emc, recon idle.
	--lowlatGatingCyc	0..255 Bus idle counter max value. It would be (sw_enc_lowlat_gating_cyc + 1) * 256 cycles when counting. default: hevc - [31], h264 - [7]

2.8.4 Input Frame and Reference Frame Buffer Alignment

Short	Long Option	Description
	--inputAlignmentExp	The input frame buffer alignment. [6] 0 - alignment disabled 4..12 - The base address and each line of the input frame buffer are aligned to two to the power of inputAlignmentExp.
	--refAlignmentExp	The reference frame buffer alignment. [6] 0 - alignment disabled 4..12 - The base address and each line of the reference frame buffer are aligned to two to the power of refAlignmentExp.
	--refChromaAlignmentExp	The alignment of the chroma reference frame buffer. [6] 0 - alignment disabled 4..12 - The base address and each line of the chroma reference frame buffer are aligned to two to the power of refChromaAlignmentExp.
	--aqInfoAlignmentExp	The alignment of the adaptive quantization output buffer. [0] 0 - alignment disabled 4..12 - The base address and each line of the adaptive quantization output buffer are aligned to two to the power of aqInfoAlignmentExp.
	--tileStreamAlignmentExp	0..15 The tile stream buffer alignment. [0] The base address and size of the tile stream buffer are aligned to two to the power of --tileStreamAlignmentExp.

2.8.5 Output Stream Buffer

Short	Long Option	Description
	--streamBufChain	Whether to enable one or two output stream buffers. [0] 0 - one output stream buffer 1 - two chained output stream buffers NOTE: The minimum allowed size of the first stream buffer is 11 KB.

2.8.6 (Reserved) Multi-Segment Stream Output

Short	Long Option	Description
	--streamMultiSegmentMode	(Reserved) The multi-segment mode of output stream buffers. [0] 0 - single-segment mode 1 - multi-segment mode with software handshaking disabled and loopback enabled 2 - multi-segment mode with software handshaking and loopback enabled.
	--streamMultiSegmentAmount	(Reserved) 2..16 The number of segments to control loopback, software handshaking, and IRQ. [4]

2.8.7 Parallel Flow Control

Short	Long Option	Description
	--parallelCoreNum	1..4 The number of cores that run in parallel at the frame level. [1] If the number of tile columns for picture division is set to 1 through --tile, set the --parallelCoreNum option to 1.
	--batchEnable	0..1 enable or disable multi-frame aggregation. [0] 0 - disable batch mode. 1 - enable batch mode.
	--batchCount	1..[parallelCoreNum-1] frame numbers for aggregation encoding. It should be smaller than parallelCoreNum. It can be modified during the encoding process by setting after @frame_number

2.8.8 AXI Settings

Short	Long Option	Description
	--AXIAlignment	The AXI alignment in hexadecimal format. [data from fuse for each bit] - Bits 35:32: AXI_burst_align_wr_cuinfo - Bits 31:28: AXI_burst_align_wr_common - Bits 27:24: AXI_burst_align_wr_stream - Bits 23:20: AXI_burst_align_wr_chroma_ref - Bits 19:16: AXI_burst_align_wr_luma_ref - Bits 15:12: AXI_burst_align_rd_common - Bits 11:8: AXI_burst_align_rd_prp - Bits 7:4: AXI_burst_align_rd_ch_ref_prefetch - Bits 3:0: AXI_burst_align_rd_lu_ref_prefetch
	--burstMaxLength	The maximum AXI burst length, in the unit of AXI bus width. [1 6]

2.8.9 IRQ Control

IRQ Type Mask of Encoder

Short	Long Option	Description
	--irqTypeMask	The IRQ type mask setting in binary format. [111110000] - Bit 8: irq_type_sw_reset_mask - Bit 7: irq_type_fuse_error_mask - Bit 6: irq_type_buffer_full_mask - Bit 5: irq_type_bus_error_mask - Bit 4: irq_type_timeout_mask - Bit 3: irq_type_strm_segment_mask - Bit 2: irq_type_line_buffer_mask - Bit 1: irq_type_slice_rdy_mask - Bit 0: irq_type_frame_rdy_mask

IRQ Type Mask of CuTree

Short	Long Option	Description
	--irqTypeCutreeMask	The IRQ type mask setting of CuTree, in binary format. [111110000] - irq_type_sw_reset_mask - irq_type_fuse_error_mask - irq_type_buffer_full_mask - irq_type_bus_error_mask - irq_type_timeout_mask - irq_type_strm_segment_mask - irq_type_line_buffer_mask - irq_type_slice_rdy_mask - irq_type_frame_rdy_mask The option supports 9 bits for consistency with --irqTypeMask. However, only irq_type_bus_error_mask, irq_type_timeout_mask and irq_type_frame_rdy_mask are valid.

2.8.10 (Only for C-Model) Peripheral Control

Short	Long Option	Description
	--useVcmd	(Only valid fo CModel) Whether to enable VCMD. 0 - disable 1 - enable
	--useDec400	(Reserved)Whether to enable DEC400. 0 - disable 1 - enable
	--useL2Cache	(Reserved)Whether to enable L2Cache. 0 - disable 1 - enable

2.8.11 External SRAM

Short	Long Option	Description
	--extSramLumHeightBwd	The capacity of external SRAM for luma backward reference. 0 - no external SRAM 1..16 - the number of lines = 4 * extSramLumHeightBwd The default value is 16 for HEVC (H.265) and 12 for H.264 (AVC).
	--extSramChrHeightBwd	The capacity of external SRAM for chroma backward reference. 0 - no external SRAM 1..16 - the number of lines = 4 * extSramChrHeightBwd The default value is 8 for HEVC (H.265) and 6 for H.264 (AVC)
	--extSramLumHeightFwd	The capacity of external SRAM for luma forward reference. 0 - no external SRAM 1..16 - the number of lines = 4 * extSramLumHeightFwd The default value is 16 for HEVC (H.265) and 12 for H.264 (AVC).
	--extSramChrHeightFwd	The capacity of external SRAM for chroma forward reference. 0 - no external SRAM 1..16 - the number of lines = 4 * extSramChrHeightFwd The default value is 8 for HEVC (H.265) and 6 for H.264 (AVC).

2.9 Debugging and Testing

2.9.1 Debugging

Short	Long Option	Description
	--dumpRegister	Whether to dump register values. [0] 0 - do not dump 1 - dump

Short	Long Option	Description
	--rasterscan	Whether to dump reconstructed YUV frames in tiled or raster format when the encoder runs on FPGA and hardware. [0] 0 - tiled format 1 - raster format

2.9.2 Testing

NOTE: Options listed in this section are designed for testing multiple encoder instances in multi-thread or multi-process mode.

Short	Long Option	Description
	--multimode	The parallel running mode. [0] 0 - disable 1 - multi-thread mode 2 - multi-process mode
	--streamcfg	The path to the file that contains encoder options.

2.9.3 vcmd priority and core bit mask

Short	Long Option	Description
	--priority	The priority of current instance in vcmd mode. [0] 0 - normal priority 1 - high priority
	--core_mask	The core bit mask of current instance. [0] 0 - not specify, anycore. 1 - bit[0]=1: specify core 0. 2 - bit[1]=1: specify core 1.

2.9.4 Log Output Control

Short	Long Option	Description
	--logOutDir	The log output mode. [0] 0 - outputs all logs to stdout 1 - outputs all logs to the same file 2 - outputs logs of each thread to a separate log file

Short	Long Option	Description
	--logOutLevel	The log level. [3] 0 - no logs output 1 - the fatal level 2 - the error level 3 - the warning level 4 - the information level 5 - the debugging level: 6 - all logs output
	--logTraceMap	Whether to output each type of trace logs. [63]=b'0111111 Bit 0 - encoder API call logs Bit 1 - register configuration logs Bit 2 - EWL API call logs Bit 3 - memory usage logs Bit 4 - rate control status logs Bit 5 - command line logs Bit 6 - performance logs Value 0 indicates not to output the type of logs. Value 1 indicates to output the type of logs.
	--logCheckMap	(Reserved) Whether to output each type of check logs. [1]=b`00001 Bit 0 - reconstructed YUV data Bit 1 - PSNR/SSIM for each frame Bit 2 - VBV information for rate control checking Bit 3 - rate control information for rate control profiling Bit 4 - feature information for coverage checking Value 0 indicates not to output the type of logs. Value 1 indicates to output the type of logs.

2.9.5 (Obsoleted) Runtime Log Output Control

Short	Long Option	Description
	--verbose	(Obsoleted) The log printing mode. [0] 0 - prints brief information 1 - prints detailed information

2.10 Internal Use

NOTE: Options listed in this section are for internal use only. The features may be under development. Specifying these options is not recommended.

Short	Long Option	Description
	--testId	The internal test ID. [0]
	--rdLog	Whether to output rate distortion logs to the profile.log file. 0 - do not output 1 - output
	--TxTypeSearchEnable	(For AV1 only) Whether to enable TX type search. [0] 0 - disable 1 - enable
	--av1InterFiltSwitch	(For AV1 only) Whether to enable interpolation filter switch. [1] 0 - disable 1 - enable
	--replaceMvFile	The path to the file that contains ME1N MVs. Data in the file is organized in raster scan of 32x32 pixel blocks. For each block, the file contains 21 MVs at the following positions: 32x32(0,0), 16x16(0,0), 16x16(16,0), 16x16(0,16), 16x16(16,16), 8x8(0,0), 8x8(8,0), 8x8(16,0), 8x8(24,0), 8x8(0,8), 8x8(8,8), 8x8(16,8), 8x8(24,8), 8x8(0,16), 8x8(8,16), 8x8(16,16), 8x8(24,16), 8x8(0,24), 8x8(8,24), 8x8(16,24), and 8x8(24,24). For each MV, (Hor + Ver) * sizeof(i16) * (L0 + L1) = 8 bytes
	--SRDEnable	Detect static region and reduce number of bits spent in these regions. [0] 0 = Disable SRD. 1 = Enable SRD.
	--SRDThreshold	0..10 Threshold for static region detection [4] Bigger threshold value cause more regions detected as static.
	--disableTU32	Whether to disable TU32 based on tu32Enable fuse value.[0] HW to do, only for test. 1 = Disable TU32.

3 Input Color Formats Mappings to FFmpeg and Microsoft-defined Formats

Table 3.1 lists typical input formats and their mappings to FFmpeg formats and formats defined by Microsoft. For details about the color formats, see *Hantro VC9000E Series Memory Buffer and Format Organization*.

Table 3.1 Mappings of Input Color Formats to FFmpeg and Microsoft-defined Formats

--inputFormat	Description	FFmpeg Format	Format Defined by Microsoft
0	YUV420P8b_Raster_A16N (IYUV/I420)	yuv420p	IMC4
1	YUV420SP8b_Raster_A16N (NV12)	nv12	NV12
2	YVU420SP8b_Raster_A16N (NV21)	nv21	-
3	YUV422YUYV8b_Raster_A16N (YUYV/YUY2)	yuyv422	YUY2
4	YUV422UYVY8b_Raster_A16N (UYVY/Y422)	uyvy422	UYVY
5	RGB565_Raster_A16N	rgb565le	-
6	BGR565_Raster_A16N	bgr565le	-
7	XRGB1555_Raster_A16N	rgb555le	-
8	XBGR1555_Raster_A16N	bgr555le	-
9	XRGB4444_Raster_A16N	rgb444le	-
10	XBGR4444_Raster_A16N	bgr444le	-
11	XRGB8888_Raster_A16N	bgr0/bgra	-
12	XBGR8888_Raster_A16N	rgb0/rgba	-
40	RGB8888_Raster_A16N	bgr24	-
41	BGR8888_Raster_A16N	rgb24	-

--inputFormat	Description	FFmpeg Format	Format Defined by Microsoft
15	YUV420P10bWL_Raster_A16N (I010)	yuv420p10le	-
16	YUV420SP10bWH_Raster_A16N (P010)	p010le	P010

4 Flexible Reference Configuration

The video encoder supports specifying a reference description file for HEVC (H.265) and H.264 (AVC) for flexible reference configuration.

The reference description file contains a reference list in a required format. The test bench checks the reference list and reports errors if any. If the reference list is valid, the test bench encodes the frames specified in the list, and exits after the encoding is completed.

If a reference description file is specified, look-ahead encoding and gradual decoder refresh (GDR) are not supported.

4.1 Reference Description File Format

Each line in the reference description file must contain the following fields in order:

- **FrameN**: the frame index.
- **Type**: the slice type, which can be P, B, or IDR.
- **POC**: the display order of the frame. The display order starts from 0 at each IDR frame.
- **Qd**: the offset to be added to the QP parameter to obtain the final QP value used for the frame. The offset is either specified by the application or calculated by the rate control algorithm.
- **Qft**: the weight used during rate distortion optimization.
- **TID**: the temporal layer ID.
- **LTR**: the LTR index if the current picture as LTR frame;
- **num_ref_pics**: the number of reference frames kept for the current frame.

The reference frames can be used by either the current frame or a subsequent frame.

- **ref_pics**: a list of num_ref_pics entries in format [[^]]<poc_diff> or [[^]]L<poc> to indicate the reference frames of the current frame.
 - The optional prefix [^] indicates that the corresponding frame is used for reference by the current frame.
 - * A P-frame can use only one other frame as its reference.
 - * A B-frame can use two other frames as its reference.
 - <poc_diff> indicates the POC offset of the reference frame relative to the current frame and that the frame is used for short-term reference.
 - [[^]]L<poc> indicates the POC of the reference frame and that the frame is used for long-term reference.

For each frame, a maximum of four short-term reference frames and two long-term reference frames can be kept. The short-term reference frames must be ordered in ascending order of POC offsets, and listed before long-term reference frames.

- **used_by_cur**: a list of num_ref_pics binaries, each indicating whether the corresponding frame in the ref_pics list is used as a reference for encoding of the current frame.

4.2 Reference Description File Configurations

To specify a reference description file, you need to:

- Specify the `--flexRefs` option.
- Specify `--RPSInSliceHeader=1`.

4.3 Example of Configuring a Reference Description File

This section provides an example of encoding eight frames with a reference description file.

Command

```
./hevc_testenc -i foreman_cif.yuv -w352 -h288  
--RPSInSliceHeader=1 --flexRefs refs.lst -o flex.hevc
```

Reference Description File

The contents of the `vrrefs.lst` file in the command are shown as follows:

#	Frm	Type	POC	Qd	Qft	TID	LTR	N	ref_pics	used_by_cur
0	IDR	0	0	0.5	0	0	0	0		
4	P	4	0	0.5	0	0	0	1	^-4	1
2	B	2	0	0.5	0	0	0	2	^-2 ^2	1 1
1	B	1	0	0.5	0	0	0	3	^-1 ^1 3	1 1
3	B	3	0	0.5	0	0	0	2	^-1 ^1	1 1
8	P	8	0	0.5	0	1	1	1	^-4	1
6	B	6	0	0.5	0	0	0	2	^-2 ^L8	1 1
5	B	5	0	0.5	0	0	0	3	^-1 ^1 L8	1 1
7	B	7	0	0.5	0	0	0	2	^-1 ^L8	1 1

5 Region of Interest

The video encoder supports coding control specific to two types of region of interest (ROI):

- Rectangular ROI
- ROI map

5.1 Rectangular ROI

The video encoder supports the following rectangular ROIs:

- A maximum of two or eight ROIs with specific QP delta or absolute QP values

The encoder supports a maximum of eight ROIs if the value of `EVLHwConfig_t.ROI8Support` is 1, and a maximum of two ROIs if the value is 0.

- A maximum of eight IPCM ROIs

In IPCM ROIs, no compression is applied to achieve no data loss.

IPCM ROIs are available only if the value of `EVLHwConfig_t.IPCM8Support` is 1.

- An intra-mode ROI

Blocks inside the ROI are force encoded as intra blocks. When gradual decoder refresh (GDR) is enabled, the intra-mode ROI also serves as the GDR intra area for GDR refresh pictures.

NOTE: You can also use the cyclic intra refresh (CIR) feature to specify the MBs or CTBs to be force encoded with the intra mode. For detail, see Section

Example: Configuring Cyclic Intra Refresh.

5.1.1 Rectangular ROI Configurations

The options used to configure ROIs of each type are listed as follows:

NOTE: If two areas of the same type overlap, the ROI with a greater ID has a higher priority. For example, `Roi2Area` takes precedence over `Roi1Area`.

- QP ROI

- `roi1Area` to `roi8Area` define the eight ROIs.
- `roi1DeltaQp` to `roi8DeltaQp` define a QP delta value for each region.
You need to specify the QP delta based on the value of the slice QP.
- `roi1Qp` to `roi8Qp` define an absolute QP value for each region.
You can assign a smaller QP value for ROIs and a larger QP value for other regions.

- IPCM ROI

`ipcm1Area` to `ipcm8Area` define the IPCM ROIs.

- Intra-mode ROI

`intraArea` defines the intra-mode ROI.

5.1.1.1 Example: Configuring a QP ROI

The following example command shows how to define the area of ROI 1 and configure its QP delta value for HEVC (H.265).

```
.\video_testenc --codecFormat hevc -i foreman.yuv -w352 -h288 -b5 --gopSize=1  
--roi1Area=1:1:1:1 --roi1Qp=10 -o stream.hevc
```

The following example command shows how to define the area of ROI 1 and configure its absolute QP value for H.264 (AVC).

```
.\video_testenc --codecFormat h264 -i foreman.yuv -w352 -h288 -b5 --gopSize=1  
--roi1Area=1:1:1:1 --roi1Qp=10 -o stream.h264
```

5.1.1.2 Example: Configuring the Intra-Mode ROI

The following example commands show how to define the area of the intra-mode ROI for HEVC (H.265) and H.264 (AVC) respectively:

```
.\video_testenc --codecFormat hevc -i foreman.yuv -w352 -h288 -b5 --gopSize=1  
--intraArea=1:1:1:1 -o stream.hevc  
  
.\video_testenc --codecFormat h264 -i foreman.yuv -w352 -h288 -b5 --gopSize=1  
--intraArea=1:1:1:1 -o stream.h264
```

5.1.1.3 Example: Configuring Cyclic Intra Refresh

The following example commands show how to configure cyclic intra refresh (CIR) for HEVC (H.265) and H.264 (AVC) respectively:

```
.\video_testenc --codecFormat hevc -i foreman.yuv -w352 -h288 -b5 --gopSize=1  
--cir=1:2 -o stream.hevc  
  
.\video_testenc --codecFormat h264 -i foreman.yuv -w352 -h288 -b5 --gopSize=1  
--cir=1:2 -o stream.h264
```

In these commands, the `cir=start:interval` option indicates that CTBs or MBs with order counts ($\text{start} + \text{interval} * N$) are force encoded as intra blocks, where N is an integer. In each frame, the CTBs or MBs are counted starting from 0, from left to right and top to bottom.

5.2 ROI Map

The video encoder supports various QP/CU information formats for ROI map configurations. The support for formats varies according to the value of `EVLHwConfig_t.roiMapVersion`. For details about the formats, see *Hantro VC9000E Series Memory Buffer and Format Organization*.

You can configure an ROI map from a definition file, a text file, or a binary file.

- ROI map configuration from a definition file is supported only for formats 1 to 4.
- ROI map configuration from a text file is supported only for formats 1 to 3 and for IPCM maps.
- ROI map configuration from a binary file is supported for formats 3 to 7.

For details about how to configure an ROI map from a file, see Section *ROI Definition File* and Section *ROI Map Text or Binary File*.

5.2.1 ROI Definition File

The video encoder test bench supports input of an ROI definition file to specify ROIs and configure features for each ROI. The parser in the test bench automatically fills the ROI map with the feature data required by the selected QP/CU information format and ignores the data of other features.

ROI definition file input is supported only for HEVC (H.265) and H.264 (AVC) encoding and only for QP/CU information formats 1 to 4.

5.2.1.1 ROI Definition File Format

This section describes the format in which data in the ROI definition file must be organized.

NOTE: The ROI definition file can include QP/CU information formats 5 to 7 for forward compatibility. The support for configuration of these formats through the ROI definition file is under development.

```
# pic=xxx
rect=(left, top, width, height)

qp=(type, value)
size=32|16|8
mode=(type, partition)

intra=(L0, L1, L2, L3, Chroma)
```

```
// Picture 1 to which the following ROI configurations are applied.
// Pictures are numbered starting from 0.
// Region 1 to which the following configurations are applied.
// If the region is not aligned to the block unit specified
// in the command line, it is enlarged for alignment.
// fmt3: The QP to be forced in region 1.
// type can be "abs|delta".
// fmt4: The CU size, invalid if unspecified
// fmt4: type can be "intra|inter|skip|ipcm", invalid if unspecified.
// For the intra mode, partition can be "any|2Nx2N|NxN".
// For the inter mode, partition can be
// "any|2Nx2N|2NxN|Nx2N|2NxN|2NxN|2NxN|2NxN|2NxN|2NxN".
// For the IPCM or skip mode, partition must be "none".
// fmt5: The intra modes for the luma and chroma components.
// L0, L1, L2, and L3 indicates the intra modes for the four
// luma PUs when NxN partition mode is selected. For 2Nx2N
// partition, only L0 is valid. Each value can be an integer
// from 0 to 34.
// The valid value range of Chroma is limited, depending on
```

```

part0=(dir, ref, mvx, mvy)
part1=(dir, ref, mvx, mvy)

rect=(a,b,c,d)
...
# pic=yyy
...
final

// the setting of L0. For more information, see ITU-T Rec.
// H.265 and H.264.
// fmt6: The inter motion information for the first partition.
// dir can be "B0|B1|L0|L1".
// For B0 and B1 directions, two part0 are defined.
// dir of one part0 is B0, and that of the other is B1.
// L0 and L1 directions are for P blocks.
// ref is the reference index, which must be 0.
// mvx and mvy specify the motion vector in the first partition,
// in the unit of 1/4 pixel. The motion vector must follow the
// limits on the motion estimation range required by the SoC.
// fmt6&fmt7: The inter motion information for the second partition.
// dir must be same as that of the first partition.
// ref is the reference index, which must be 0.
// mvx and mvy specify the motion vector in the second
// partition, in the unit of 1/4 pixel. The motion vector must
// follow the limits on the motion estimation range required by
// the SoC. The difference between the motion vectors of the
// two partitions must be within 1 pixel.
//
// Region 2 to which the following configurations are applied.
// Picture N to which the following ROI configurations are applied.
// The mark of the end of file

```

5.2.1.1.1 Example of ROI Definition File

The following script shows the contents of an example ROI definition file named `vsipar_test6.roi`.

```

# pic=0
rect=(64,48,9,9)
qp=(delta, -6)
size=32
mode=(ipcm, none)
rect=(64,80,9,9)
qp=(delta, -5)
size=16
mode=(skip, none)
rect=(16,16,9,9)
qp=(delta, -5)
size=16
rect=(32,32,10,10)
qp=(abs, 5)
size=16
mode=(ipcm, none)
# pic=1
rect=(48,48,48,48)
qp=(abs,12)
size=32
mode=(skip, none)
rect=(112,48,32,32)
qp=(abs, 15)
size=32
mode=(skip, none)
rect=(0,0,17,17)
qp=(abs, 10)
size=16
rect=(304,48,31,31)
qp=(abs, 8)
size=32

```

```
mode=(skip, none)
rect=(128,64,16,16)
qp=(abs, 20)
size=16
rect=(0,104,17,17)
qp=(delta, -6)
size=32
mode=(skip, none)
# pic=2
rect=(80,80,264,136)
qp=(abs, 8)
size=16
# pic=3
rect=(80,80,264,64)
qp=(abs, 8)
size=16
# pic=4
rect=(80,80,264,24)
qp=(abs, 8)
size=16
# pic=5
rect=(80,80,64,24)
qp=(abs, 8)
size=16
# pic=6
rect=(80,80,64,136)
qp=(abs, 8)
size=16
# pic=7
rect=(80,16,32,200)
qp=(abs, 8)
size=16
# pic=8
rect=(80,80,32,64)
qp=(abs, 8)
size=16
# pic=9
rect=(80,16,8,200)
qp=(abs, 8)
size=16
# pic=10
rect=(80,80,264,7)
qp=(abs, 8)
size=16
final
```

5.2.1.2 ROI Definition File Configurations

The options used to configure an ROI definition file are listed as follows:

- `roiMapDeltaQpEnable` specifies whether to enable QP control.
- `ipcmMapEnable` specifies whether to enable IPCM mode control.
- `skipMapEnable` specifies whether to enable skip mode control.
- `roiMapConfigFile` specifies the ROI definition file.
- `roiMapDeltaQpBlockUnit` specifies the block unit size for the ROI map.
- `RoiQpDeltaVer` specifies the information format used in the ROI map.

5.2.1.2.1 Example: Configuring an ROI Map of Format 1

This section provides an example of filling and applying an ROI map of format 1 according to a definition file, in order to force the QP settings and/or control whether to enable the IPCM mode in each ROI.

Definition File

See Section *Example of ROI Definition File*.

Commands

```
# force QP only
./hevc_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
-o BQSquare_416x240_qp_block64_format1.hevc --roiMapDeltaQpBlockUnit=0
--roiMapDeltaQpEnable=1 --RoiCuCtrlVer=0 --RoiQpDeltaVer=1
--roiMapConfigFile=../cfg/roicfg/vsi_par_test6.roi

# force QP and enable IPCM mode control
./hevc_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
-o BQSquare_416x240_qp_block64_format1.hevc --roiMapDeltaQpBlockUnit=0
--roiMapDeltaQpEnable=1 --RoiCuCtrlVer=0 --RoiQpDeltaVer=1 --ipcmMapEnable=1
--roiMapConfigFile=../cfg/roicfg/vsi_par_test6.roi
```

In these commands:

- The `--roiMapDeltaQpEnable` option controls whether to force the QP settings in ROIs. To enable QP control, `--roiMapDeltaQpEnable=1` must be specified.
- If an ROI specified in the example file is not aligned to the block unit specified through `--roiMapDeltaQpBlockUnit`, the ROI is enlarged for alignment.
- The `--ipcmMapEnable` option controls whether to enable IPCM mode control. To enable IPCM mode control, `--ipcmMapEnable=1` must be specified.

NOTE: In [Figure 5.2](#), if IPCM mode control is disabled, the PUs in orange will be in another color with the IPCM mode unforced.

Results

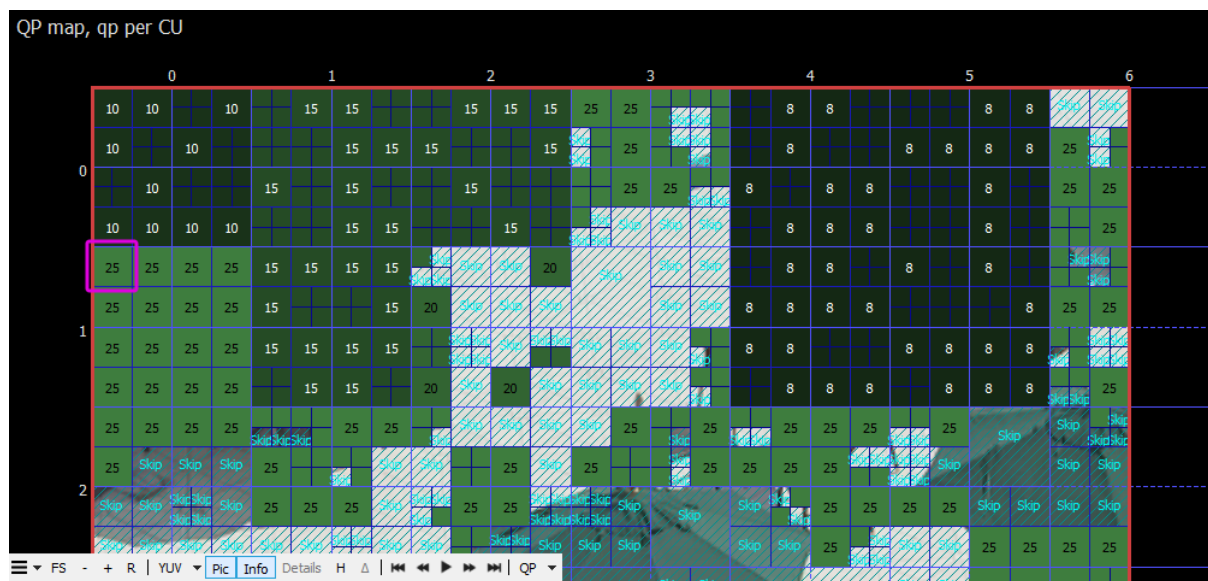


Figure 5.1 Example ROI Map of Format 1

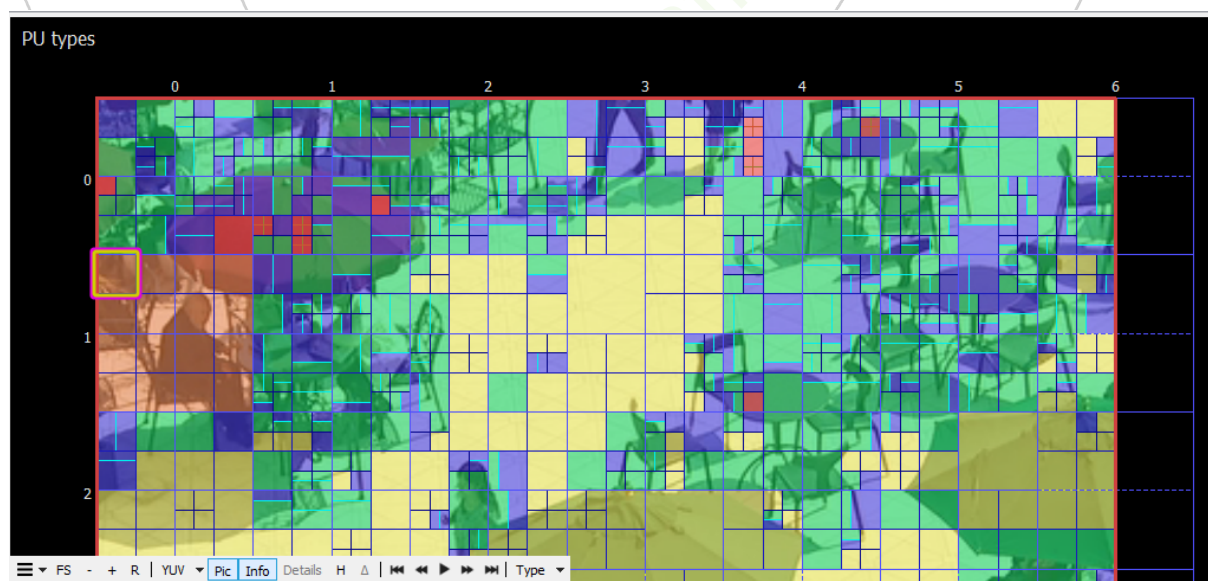


Figure 5.2 PU Types When Example ROI Map of Format 1 Is Applied with IPCM Mode Control Enabled

5.2.1.2.2 Example: Configuring an ROI Map of Format 2

This section provides an example of filling and applying an ROI map of format 2 according to a definition file, in order to force the QP settings and control whether to enable the skip mode for P- and B-frames in each ROI.

See Section *Example of ROI Definition File*.

Command

```
./hevc_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
-o BQSquare_416x240_qp_block32_format2.hevc --roiMapDeltaQpBlockUnit=1
--roiMapDeltaQpEnable=1 --RoiCuCtrlVer=0 --RoiQpDeltaVer=2 --skipMapEnable=1
--roiMapConfigFile=./cfg/roicfg/vsi_par_test6.roi
```

In this command:

- If an ROI specified in the example file is not aligned to the block unit specified through `-roiMapDeltaQpBlockUnit`, the ROI is enlarged for alignment.
- The `-roiMapDeltaQpEnable` option controls whether to force the QP settings in ROIs. To enable QP control, `-roiMapDeltaQpEnable=1` must be specified.
- The `-skipMapEnable` option controls whether to enable skip mode control. To enable skip mode control, `-skipMapEnable=1` must be specified.

Results

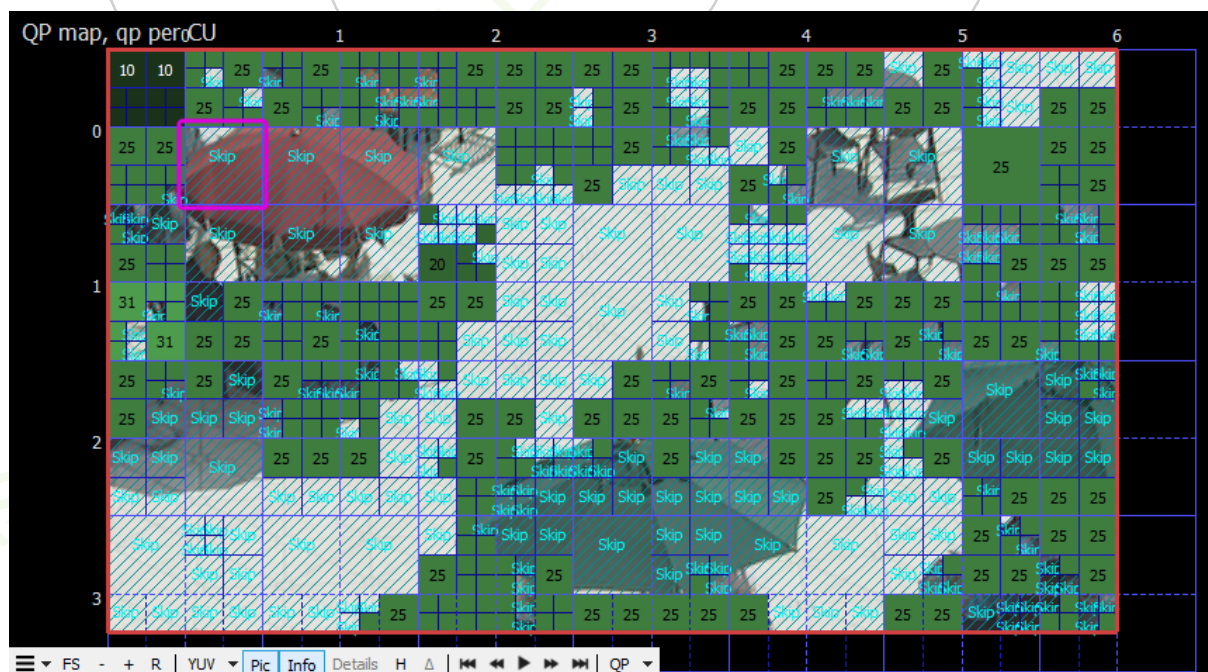


Figure 5.3 PU Types When Example ROI Map of Format 2 Is Applied with Skip Mode Control Enabled

5.2.1.2.3 Example: Configuring an ROI Map of Format 3

This section provides an example of filling and applying an ROI map of format 3 according to a definition file, in order to force the QP settings in each ROI.

Definition File

See Section [Example of ROI Definition File](#).

Command

```
./hevc_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
-o BQSquare_416x240_qp_block32_format3.hevc --roiMapDeltaQpBlockUnit=1
--roiMapDeltaQpEnable=1 --RoiCuCtrlVer=0 --RoiQpDeltaVer=3
--roiMapConfigFile=./../cfg/roicfg/vsi_par_test6.roi
```

In this command:

- If an ROI specified in the example file is not aligned to the block unit specified through `-roiMapDeltaQpBlockUnit`, the ROI is enlarged for alignment.
- The `-roiMapDeltaQpEnable` option controls whether to force the QP settings in ROIs. To enable QP control, `-roiMapDeltaQpEnable=1` must be specified.

Results

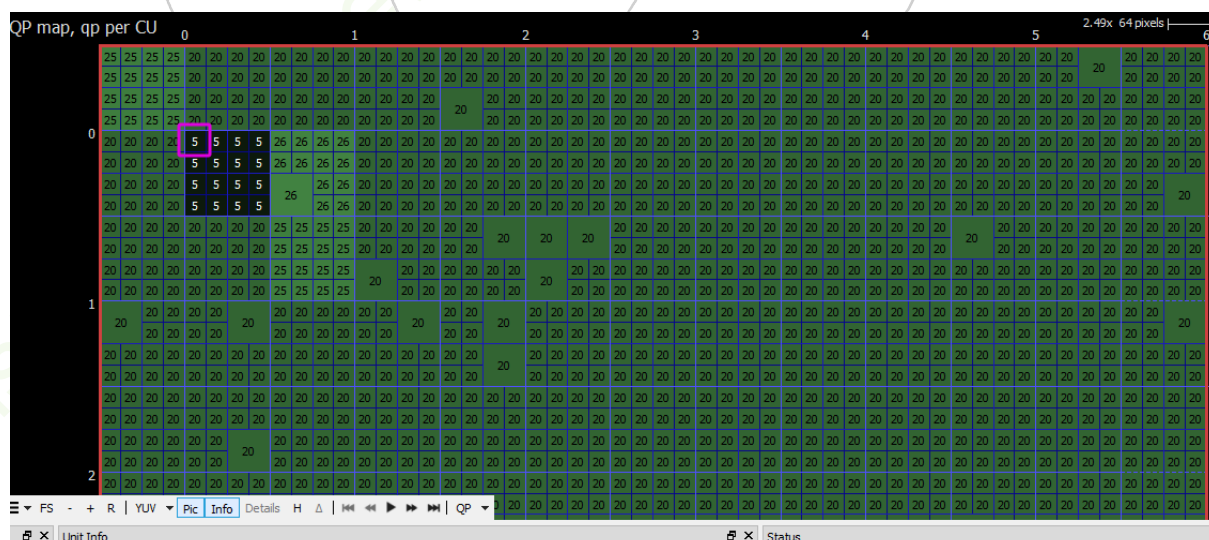


Figure 5.4 Example ROI Map of Picture 0 When Format 3 Is Used

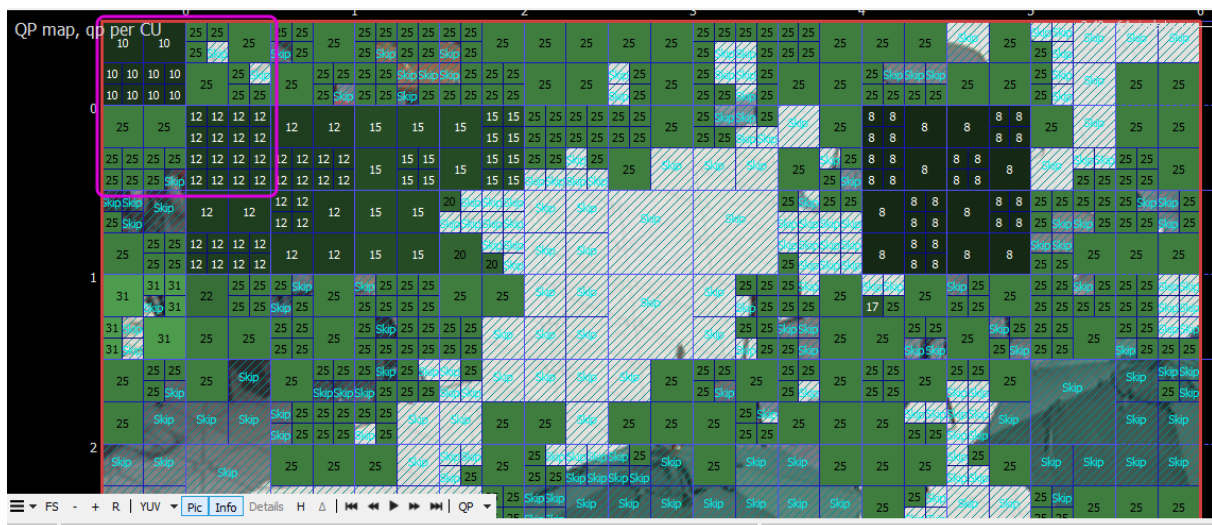


Figure 5.5 Example ROI Map of Picture 1 When Format 3 Is Used

5.2.1.2.4 Example: Configuring an ROI Map of Format 4

This section provides an example of filling and applying an ROI map of format 4 according to a definition file, in order to force the QP settings in each ROI.

NOTE: The QP/CU control information of an ROI map in format 4 consists of two bytes for each 8x8 pixel block.

Definition File

See Section [Example of ROI Definition File](#).

Command

```
./hevc_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
-o BQSquare_416x240_qp_block32_format4.hevc --roiMapDeltaQpBlockUnit=1
--roiMapDeltaQpEnable=1 --RoiCuCtrlVer=4 --RoiQpDeltaVer=3
--roiMapConfigFile=../cfg/roicfg/vsi_par_test6.roi
```

In this command:

- If an ROI specified in the example file is not aligned to the block unit specified through `--roiMapDeltaQpBlockUnit`, the ROI is enlarged for alignment.

- The `-roiMapDeltaQpEnable` option controls whether to force the QP settings in ROIs. To enable QP control, `-roiMapDeltaQpEnable=1` must be specified.
- The `-ipcmMapEnable` option controls whether to enable IPCM mode control. To enable IPCM mode control, `-ipcmMapEnable=1` must be specified.
- The `-skipMapEnable` option controls whether to enable skip mode control. To enable skip mode control, `-skipMapEnable=1` must be specified.

Results

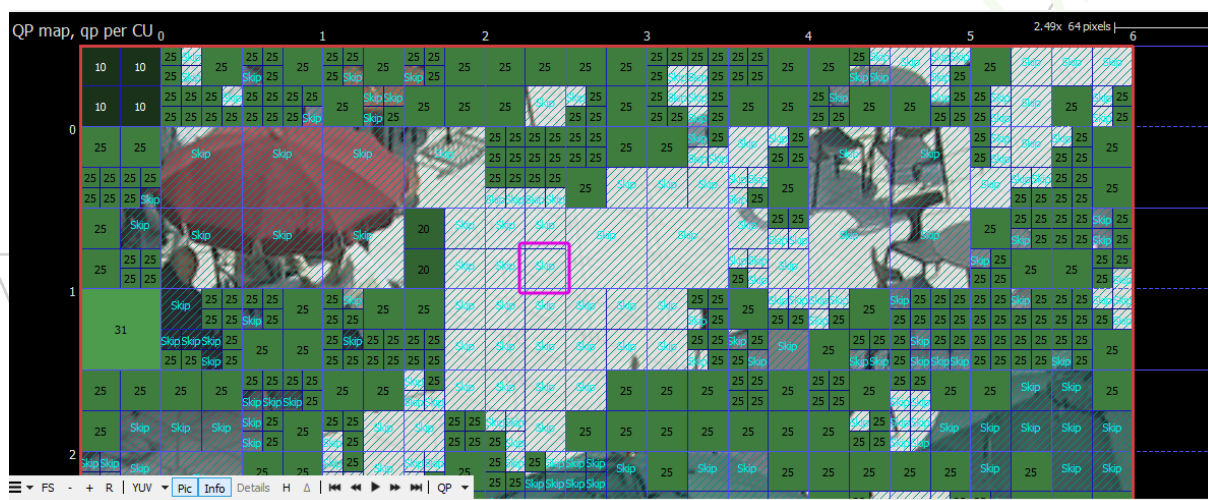


Figure 5.6 Example ROI Map of Format 4

5.2.2 ROI Map Text or Binary File

The video encoder test bench supports input of a text or binary file that contains an ROI map to specify QP/CU control information for each block unit. Seven QP/CU control information formats are supported.

- ROI maps of formats 1 to 3 are QP maps, which can be configured from text files. QP maps of format 3 can also be configured from binary files.
- ROI maps of formats 4 to 7 are CU control maps, which can be configured from a binary file.

You can also specify a text file that contains an IPCM map to control whether to enable the IPCM mode for each block unit.

For format details, see *Hantro VC9000E Series Memory Buffer and Format Organization*.

5.2.2.1 Map Text or Binary File Configurations

QP Map

The options used to configure a text or binary file for a QP map are listed as follows:

- `roiMapDeltaQpEnable` controls whether to enable the QP map.
- `roiMapDeltaQpFile` specifies a text file for the QP map.
Alternatively, if QP information format 3 is used, you can use the `roiMapInfoBinFile` option to specify a binary file for the QP map.
- `roiMapDeltaQpBlockUnit` specifies the block unit size for the QP map.
- `RoiQpDeltaVer` specifies the QP information format used in the QP map.

CU Control Map

The options used to configure a binary file for a CU control map are listed as follows:

- `RoimapCuCtrlInfoBinFile` specifies the binary map for the CU control map.
- `roiMapDeltaQpBlockUnit` specifies the block unit size for the CU control map.
- `RoiCuCtrlVer` specifies the CU information format used in the CU control map.

IPCM Map

The options used to configure a text file for an IPCM map are listed as follows:

- `RoimapCuCtrlInfoBinFile` specifies the text file for the IPCM map.
- `roiMapDeltaQpBlockUnit` specifies the block unit size for the IPCM map.

5.2.2.1.1 Example: Configuring a QP Map from a Text File

This section provides an example of configuring a QP map in QP/CU information format 1 from a text file.

Command

```
./hevc_testenc -i BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
--roiMapDeltaQpBlockUnit=2 --roiMapDeltaQpEnable=1
--roiMapDeltaQpFile=./../cfg/qpdelta_416x240_map_block16x16.cfg
--RoiCuCtrlVer=0 --RoiQpDeltaVer=1 -o stream.hevc
```

Text File

The contents of the qpdelta_416x240_map_block16x16.cfg file in the command are shown as follows:

```
-7,-4,-3,-2,-1,-5,-6,-7,-8,-9,-10,-11,-12,-13,-7,-4,-3,-2,-1,-5,-6,-7,-8,-9,-10,-11,-12,-13,
-8,0,0,0,0,0,0,0,0,0,0,0,0,0,0,-2,-8,0,0,0,0,0,0,0,0,0,0,0,0,0,-2,
-9,0,0,0,0,0,0,0,0,0,0,0,0,0,0,-3,-9,0,0,0,0,0,0,0,0,0,0,0,0,0,-3,
-10,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-4,-10,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-4,
-11,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-5,-11,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-5,
-12,0,0,0,0,0,0,0,0,0,0,0,0,0,-6,-12,0,0,0,0,0,0,0,0,0,0,0,0,0,-6,
-13,0,0,0,0,0,0,0,0,0,0,0,0,0,-7,-13,0,0,0,0,0,0,0,0,0,0,0,0,0,-7,
-14,0,0,0,0,0,0,0,0,0,-2,-2,0,-15,-14,0,0,0,0,0,0,0,0,0,-2,-2,0,-15,
-7,-4,-3,-2,-1,-5,-6,-7,-8,-9,-10,-11,-12,-13,-7,-4,-3,-2,-1,-5,-6,-7,-8,-9,-10,-11,-12,-13,
-8,0,0,0,0,0,0,0,0,0,0,0,0,-2,-8,0,0,0,0,0,0,0,0,0,0,0,0,0,-2,
-9,0,0,0,0,0,0,0,0,0,0,0,0,-3,-9,0,0,0,0,0,0,0,0,0,0,0,0,0,-3,
-10,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-4,-10,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-4,
-11,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-5,-11,0,0,0,0,-5,-5,0,0,0,0,0,0,0,-5,
-12,0,0,0,0,0,0,0,0,0,0,0,0,0,-6,-12,0,0,0,0,0,0,0,0,0,0,0,0,0,-6,
-13,0,0,0,0,0,0,0,0,0,0,0,0,0,-7,-13,0,0,0,0,0,0,0,0,0,0,0,0,0,-7,
-14,0,0,0,0,0,0,0,0,0,-2,-2,0,-15,-14,0,0,0,0,0,0,0,0,0,-2,-2,0,-15
```

5.2.2.1.2 Example: Configuring a QP Map from a Binary File

This section provides an example of configuring a QP map in QP/CU information format 3 from a binary file.

Command

```
./hevc_testenc -i BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b3
--roiMapDeltaQpBlockUnit=1 --roiMapDeltaQpEnable=1
--roiMapInfoBinFile=roi_v3/hevc/V3QpCfg_only_ver1_block32.cfg
--RoiCuCtrlVer=0 --RoiQpDeltaVer=3 -o stream.hevc
```

Binary File

V3QpCfg_only_ver1_block32.cfg in the command is a binary file dumped from the DDR.

5.2.2.1.3 Example: Configuring an IPCM Map from a Text File

This section provides an example of configuring an IPCM map for HEVC (H.265) and H.264 (AVC) from a text file.

Commands

```
./hevc_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b2
--roiMapDeltaQpBlockUnit=0 --roiMapDeltaQpEnable=0 -o BQSquare_416x240_ipcmMapFile.hevc
--ipcmMapEnable=1 --ipcmMapFile=../cfg/ipcm_416x240_map_block64x64.cfg

./h264_testenc -i /BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b2
--roiMapDeltaQpBlockUnit=0 --roiMapDeltaQpEnable=0 -o BQSquare_416x240_ipcmMapFile.h264
--ipcmMapEnable=1 --ipcmMapFile=../cfg/ipcm_416x240_map_block16x16.cfg
```

Text File

The contents of the `ipcm_416x240_map_block16x16.cfg` file in the command are shown as follows:

[illegible]

Results

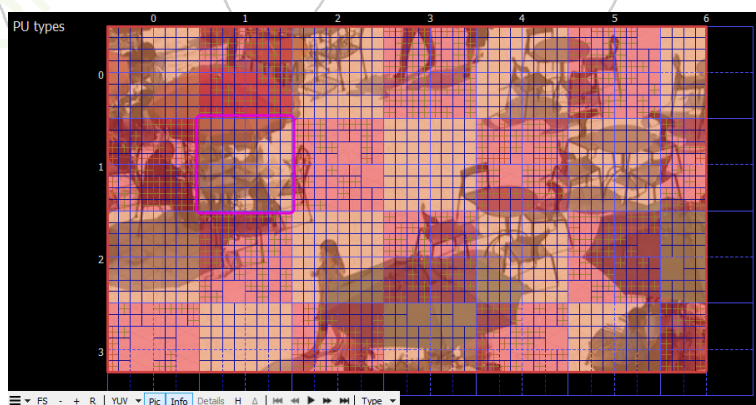


Figure 5.7 PU Types of Picture 0 When IPCM Example Command Is Applied

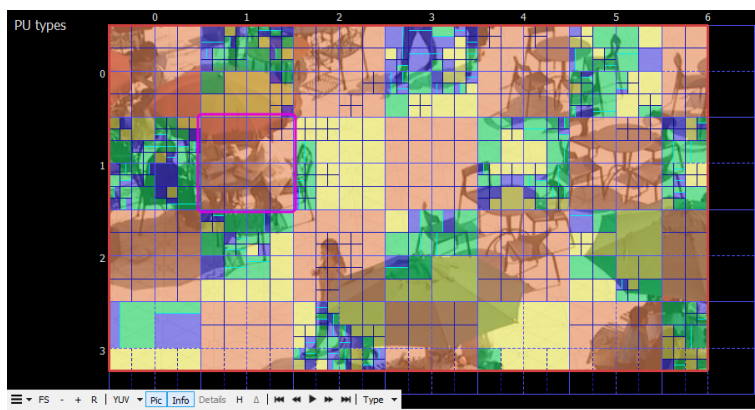


Figure 5.8 PU Types of Picture 1 When IPCM Example Command Is Applied

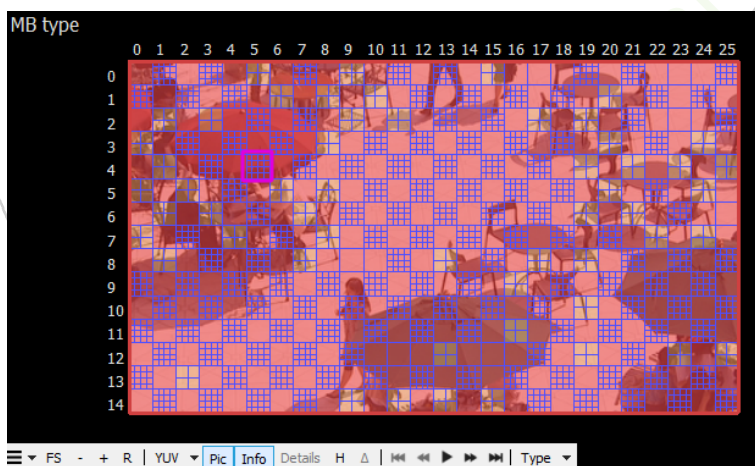


Figure 5.9 MB Types of Picture 0 When IPCM Example Command Is Applied

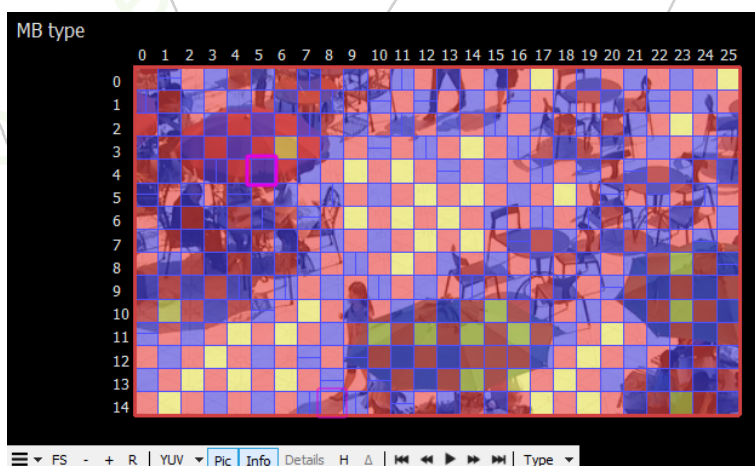


Figure 5.10 MB Types of Picture 1 When IPCM Example Command Is Applied

5.3 Generation of QP Values

The video encoder determines the final QP of a block with the following steps in the hardware pipeline:

1. Loads the following three parameters from registers:
 - Slice QP, which is used as the base QP
 - MinQP and MaxQP
2. Calculates the QP value of the block according to the ROI settings if ROI is enabled.
 - If a QP delta value is specified, the QP value is updated to the base QP minus the delta.
 - If an absolute QP value is specified, the QP value is updated to the absolute value.
3. If block-level rate control is enabled:
 - (a) Obtains a new QP delta value based on rate control settings.
 - (b) Adds the obtained delta to the QP value.
4. Clips the QP value based on the range [minQP, maxQP]. The clip result is the final QP value for the block.

NOTE:

- This procedure is a basic workflow. The actual procedure may slightly differ in some steps due to threshold operations.
- QP is valid only for blocks with residual. If a coded block has no residual (for example, when it is a skip block or the residual flag of the block is 0), the QP values are invalid though many video analyzers show the predicated QP as the final QP of the block. You can ignore these QP values.

subsection s_roifill Roimap filling process

ROI Map has different formats from 1 to 7. Each format owns different bits that correspond to Qp_info, roiAbsQq_flag, qp_valid. Qp_info affects every CU encode quality in frames. QP buffer storages specific value of CU control bits information by binary file. In the past, the binary file generated by the gen_roimap_bin.c file. After that operation, the binary file was put into hevc_testenc by -roiMapInfoBinFile option. But now, not a binary file but a script file which

involves CU control information is put into hevc_testenc. So in test_bench_utils.c, parameters from the script need to be parsed and at the same time, CU control information for ROI Buffer should be added.

Details of the roimap fill process

- First, parse the roi file parameters
- second, find the roi location, fill the data into memory

Command line of ROI:

```
./hevc_testenc -i /your_path/BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b1 -o  
BQSquare_416x240_cu_block32_roiCUFile.hevc --roiMapDeltaQpBlockUnit=2  
--roiMapDeltaQpEnable=0 --roiMapConfigFile=../../cfg/roicfg/vsi_par_test6.roi  
--RoiCuCtrlVer=4 --RoiQpDeltaVer=0 --enableOutputCuInfo=2  
  
./h264_testenc -i /your_path/BQSquare_416x240_60.yuv -w 416 -h 240 -q 25 -b1 -o  
BQSquare_416x240_cu_block32_roiCUFile.h264 --roiMapDeltaQpBlockUnit=2  
--roiMapDeltaQpEnable=0 --roiMapConfigFile=../../cfg/roicfg/vsi_par_test6.roi  
--RoiCuCtrlVer=4 --RoiQpDeltaVer=0 --enableOutputCuInfo=2
```

6 Coding Control Features

6.1 Codec Formats and Profiles

6.1.1 10-bit Encoding

The video encoder supports 10-bit encoding as a configurable feature. When 10-bit encoding is selected, the video encoder performs encoding in 10-bit color space internally and outputs 10-bit stream. The input can be 8-bit or 10-bit picture. The pre-processing module will do corresponding conversion.

The 10-bit encoding is supported only if the value of `EWLHwConfig_t.main10Eanbled` is 1.

Configuration

To enable 10-bit encoding, you need to:

- Specify `-profile=2` for HEVC (H.265) to select the Main 10 profile.
- Specify `-profile=12` for H.264 (AVC) to select the High 10 profile.
- Specify `-profile=0` for AV1 to select the Main profile.
- Specify `-profile=2` for VP9 to select the High profile.
- Specify `-bitDepthLuma=10 -bitDepthChroma=10` to enable 10-bit encoding.
- Configure the pre-processing module to enable color space conversion to 10-bit if the input picture is in a color format of different bit depth.

Examples

```
# 5 HEVC frames
./hevc_testenc -b4 -w352 -h288 -i foreman_cif.yuv --bitDepthLuma=10 --bitDepthChroma=10 --profile=2 -o
    stream.hevc
# 5 H.264 frames
./h264_testenc -b4 -w352 -h288 -i foreman_cif.yuv --bitDepthLuma=10 --bitDepthChroma=10 --profile=12 -o
    stream.h264
```

6.1.2 Monochrome Encoding

The video encoder supports the Monochrome encoding as a configurable feature. For H.265 (HEVC), When 8-bit monochrome encoding is applied, it is defined as Monochrome profile; when both 10-bit and monochrome encoding feature are selected, it is defined as Monochrome 10 profile.

In H264 (AVC), monochrome encoding is defined in High 10 Profile (both 8 bit and 10 bit).

The monochrome encoding is supported only if the value of `EVLHwConfig_t.MonoChromeSupport` is 1.

Configuration

To enable Monochrome encoding, you need to:

- Specify `--codedChromaIdc=0` to encode as Monochrome stream.
- Specify `--profile=3` for HEVC (H.265) Monochrome profile and `--profile=4` for HEVC (H.265) Monochrome 10 profile.
- Specify `--profile=12` for H.264 (AVC).

Note: Not support monochrome encoding for AV1 and VP9 now.

Examples

```
# 5 HEVC frames Monochrome Profile
./hevc_testenc -b4 -w352 -h288 -i foreman_cif.yuv --bitDepthLuma=8 --bitDepthChroma=8 \
  --profile=3 --codedChromaIdc=0 -o stream.hevc
# 5 HEVC frames Monochrome 10 Profile
./hevc_testenc -b4 -w352 -h288 -i foreman_cif.yuv --bitDepthLuma=10 --bitDepthChroma=10 \
  --profile=4 --codedChromaIdc=0 -o stream.hevc
# 5 H.264 High 10 frames
./h264_testenc -b4 -w352 -h288 -i foreman_cif.yuv --bitDepthLuma=10 --bitDepthChroma=10 \
  --profile=12 --codedChromaIdc=0 -o stream.h264
```

6.1.3 Gradual Decoder Refresh

The video encoder supports gradual decoder refresh (GDR) for HEVC (H.265) and H.264 (AVC) to provide smooth streams without bit rate spikes caused by I-frame insertion.

GDR produces an encoded stream with only one I-frame at the beginning and no following I-frames. Instead of coding more I-frames, GDR uses several P frames as a recovery point. GDR defines and refreshes an intra area in each P-frame from top to bottom, as shown in Figure 6.1.

This way, bits are distributed smoothly across the stream, which leads to a smaller network buffer size required and thus reduce network buffering delay. Figure 6.2 shows the comparison between the bit distribution with GDR and without GDR.

The GDR metadata, including location and recovery information, is defined in supplemental enhancement information (SEI) messages.

NOTE:

- Intra areas in consecutive frames are overlapped to avoid motion estimation by samples not refreshed.
- Intra-mode ROIs are not available when GDR is enabled.

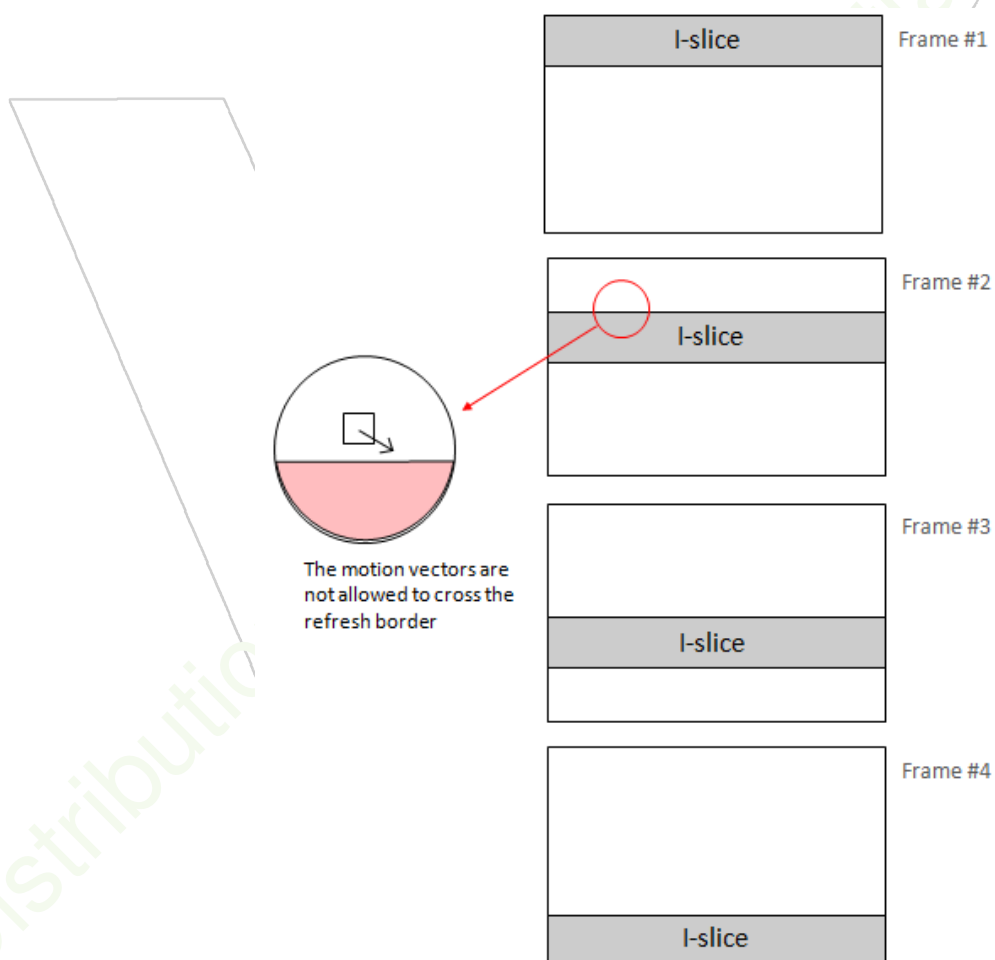


Figure 6.1 GDR Schematic Diagram

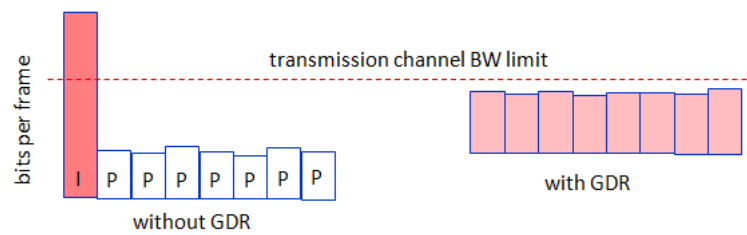


Figure 6.2 Bit Distribution Across Stream with GDR and without GDR

Configuration

The video encoder test bench provides the `--gdrDuration` option to control the GDR feature.

- If this option is set to an integer greater than 0, GDR is enabled and the integer indicates the number of P-frames constituting a recovery point.
- If this option is set to 0, GDR is disabled.

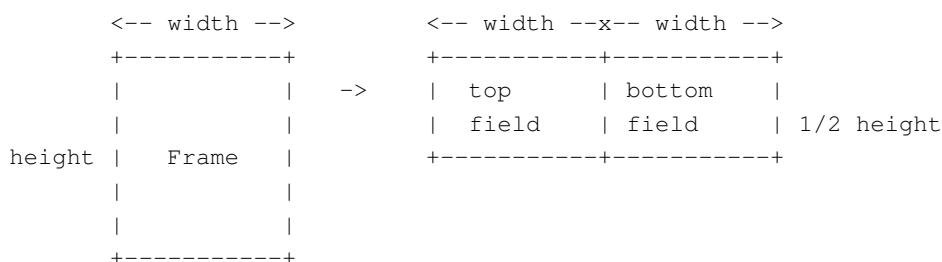
Example Command

```
# Encode 13 frames, use four P-frames as a recovery point, and encode every sixth frame with an intra area
./hevc_testenc -b12 -w352 -h288 -i foreman_cif.yuv --gopSize=1 --intraPicRate=6 --gdrDuration=4 -o
stream.hevc
./h264_testenc -b12 -w352 -h288 -i foreman_cif.yuv --gopSize=1 --intraPicRate=6 --gdrDuration=4 -o
stream.h264
```

6.1.4 Interlaced Coding for HEVC

The video encoder supports interlaced coding for HEVC (H.265) by adding proper SEI information to indicate the field structure.

The test bench assume all input will be frames. Therefore, the options described the input picture is frame-based. When interlace encoding is specified, the test bench will feed one frame twice to the encoder, with 2x original width and the 1/2 original height. The encoding picture size will keep the same size as before and the encoding picture width will be half. For the bottom field, the horizontal offset will be added to the original width. The API will treat the crop information as the valid encoding region.



Interlaced coding is supported for HEVC only if `--gopSize` is set to 1.

Configuration

To enable interlaced coding, you need to:

- Specify `--gopSize=1`.
- Specify `--interlacedFrame=1` to enable interlaced coding.
- Specify `--sei=1` to code the interlace information in picture timing SEI.
- Set `--fieldOrder` to specify the order of top and bottom fields.
 - 0: bottom field first
 - 1: top field first

Examples

```
# input 1920x1080 frames, crop each into a rectangular (left=640,top=360,width=1280,height=720),
# and encode them as two fields of 1280x360 in bottom field first order
./hevc_testenc -i BQTerrace_1920x1080_60.yuv --firstPic 40 --lastPic 50 -w 1920 -h 1080 -x 1280 -y 720
--horOffsetSrc=640 --verOffsetSrc=360 --interlacedFrame 1 --sei 1 --gopSize 1 --fieldOrder=0
-o output-bottom-first.bin

# input 1920x1080 frames, crop each into a rectangular (left=640,top=360,width=1280,height=720),
# and encode them as two fields of 1280x360 in top field first order
./hevc_testenc -i BQTerrace_1920x1080_60.yuv --firstPic 40 --lastPic 50 -w 1920 -h 1080 -x 1280 -y 720
--horOffsetSrc=640 --verOffsetSrc=360 --interlacedFrame 1 --sei 1 --gopSize 1 --fieldOrder=1
-o output-top-first.bin
```

6.1.5 HEIF Image File Support

The video encoder supports outputting encoded HEVC (H.265) or AV1 streams to files of High Efficiency Image File Format (HEIF), ISO/IEC 23008-12. Related configurations can be fed to the video encoder through an HEIF configuration file.

NOTE: To support the HEIF feature, the video encoder requires the third-party HEIF writer library for constructing the HEIF container. For information about how to build and use the library for the video encoder, see *Hantro VC9000E Video Encoder API User Manual*.

Configuration

To output an HEVC or AV1 stream to an HEIF file, you need to:

- Specify `--Heifcfg <configure_filename>` where `<configure_filename>` indicates the path to an HEIF configuration file.
- Specify `-R1 --gopSize=1` to ensure each frame is encoded as an I-frame for saving heif images.
- Specify `--gopSize=1` to avoid outputting B-Frames in the heif sequences.

HEIF Configuraiton File

The following table lists the options that can be used in an HEIF configuration file.

Table 6.1 Options in HEIF Configuration File

Option	Description
-HI	Saves an image or image collection to the HEIF output file.
-HS	Saves an image sequence to the HEIF output file.
-N [n]	Specifies the number of images (for heif sequences, this number will be setted to the number of encoded frames).
-P [n]	Specifies the sequence number of the primary image. The default value is 1, which indicates that the first image is the primary image.
-H [n]	Specifies the sequence number of the hidden image. The primary image cannot be the hidden image.
-T [file_name]	Specifies the path to the thumbnail images or sequence file.
-A --alpha [file_name] -A --depth [file_name]	Saves auxiliary images to the HEIF output file. --alpha specifies an image file for overlay input. --depth specifies a file that contains the depth information for a 3D image.

Option	Description
-C [output_width] [output_height] [horizontal_offset] [vertical_offset]	Crops the images before saving them to the HEIF file. output_width: the width of the cropped image output_height: the height of the cropped image horizontal_offset: the X offset of the clean aperture center minus (width-1)/2 vertical_offset: the Y offset of the clean aperture center minus (height-1)/2
-M [n]	Horizontally mirrors the images if the value is 0, or vertically mirrors the images if the value is 1.
-R [angle]	Rotates the images. The value can be 0, 90, 180, or 270.
-G [mode] [rows] [cols] -G [mode] [width] [height]	Specifies grid settings. mode: the gride mode. Available values include: - 0: disables the grid mode. - 1: takes each input image as a grid for composing the output image. - 2: Splits input image into multiple grids by indicated numbers of grids (rowsxcols) for encoding and storage. - 3: Splits input image into multiple grids by indicated gird image size (widthxheight) for encoding and storage. rows: the number of rows in the unit of grid (for mode 2) cols: the number of columns in the unit of grid (for mode 2) width: width of the unit grid by pixels (for mode 3) height: height of the unit grid by pixels (for mode 3)
-O [output_width][output_height] {horizontal_offset1, vertical_offset1} {horizontal_offset2, vertical_offset2} ...	Specifies overlay settings. output_width: the height of the output overlaid image output_height: the height of the output overlaid image horizontal_offset and vertical_offset: The offsets of each input overlay image.
-group [file_name]	Adds a group to the HEIF output file.

Option	Description
-o [file_name]	Specifies the path to the HEIF output file. Use extension <code>.heic</code> to HEVC stream, and <code>.avif</code> to AV1 stream.

HEIF Configuration File Examples

The following script shows the contents of an example HEIF configuration file to split each input image into multiple grids for encoding and storage and output the grid data to an HEIF file. The image will be reconstructed from the grids by an HEIF reader.

```
-HI
-N 1
-G 2 64 64
-o test.heic
```

The following script shows the contents of an example HEIF configuration file to rotate and overlay four images and output the rotated and overlaid image to an HEIF file.

```
-HI
-N 4
-R 90
-O 2560 1440 {0, 0} {416, 240} {1280, 720} {832, 480}
-o test.heic
```

6.2 Encoding Tools for Better Quality

Hantro encoder integrates some tools to achieve better quality. This chapter will introduce these tools.

6.2.1 Quality Measurement

Encoder quality can be evaluated by objective measure matrix or by subjective viewing.

For objective quality, PSNR is still the wide-accepted quality measurement. In addition, SSIM is used too. Hantro encoder is tuned for better PSNR mainly. BDR is used to evaluate the objective quality, see VCEG-M33 for detail description on what the BDR mean. In general, when measuring quality with PSNR, BDR means how many percentage of bit-rate saving is achieved to get the same PSNR. Negative percentage means better quality when we compare the proposal encoder with the anchor encoder.

[1] VCEG-M33, Calculation of average PSNR differences between RD-curves, by Bjontegaard

For subjective quality, it is hard to evaluate because different people may show different point of view for one picture. But generally, when checking some artifact, it is easy to get the agreement in most condition.

6.2.2 Anti Intra Flicker

Intra flicker is one kind of artifacts existing in video encoding system. The flicker is due to intra frame is inserted into the video sequence periodically. Since intra frame's prediction methods is different from P or B frame, such artifact is hard to remove. In hantro video encoding system, AIF(Anti Intra Flicker) feature is designed to reduce such kinds of artifact. In our design, the IDR frame is encoded twice. So when the feature is enable, the I frame will take double time to complete. Because IDR frame interval is quite large, e.g. one per 30 frames or one per 60 frames, the performance impact is small. AIF feature is useful to improve the subjective quality, especially when bit rate is low or middle and the scenario is full of slow-moving details.

To use this feature, hardware need to support tile 4x4 encoding. `EVLHwConfig_t.tile4x4FormatSupport` should be 1 to indicate it is supported.

This feature is conflicted with following features,

- 10 bit encoding: Due to hardware limitation, currently AIF cannot support 10-bit encoding.
- GDR(Gradual Decoder Refresh): GDR will change IDR frame to P frames. and current AIF only works for whole I frame.
- Parallel encoding, multi-tile encoding, lookahead: to implement AIF for parallel encoding, the two-pass encoding scheme need to adjust to avoid dependence change due to parallel jobs. Currently, it is not implemented.

To enable the feature in test bench, option "--aifEnable" should be 1. In addition, following option can control the quality of the IDR frame.

- option "--intraQpDelta": it controls the QP used in the IDR frame. generally, it should be a negative value or zero. Smaller value will provide smaller QP to encode IDR frame and give better quality for IDR frame. Suggest to use "-2" in general case.
- option "--aifQpDelta": it should be a negative value. The sum QP of previous reference frame and the aifQpDelta will be used to encode the IDR frame in first pass. So smaller delta value will get better quality but cost more bits. suggest to use "-1" or "0" in general case.

Note: when evaluate this feature in c-model, enable VCMD is recommended. or some pre-processing features will be fail to work for the IDR frame.

6.3 Metadata in Coding Streams

6.3.1 HDR10+ or Dolby Vision Information Insertion

An HDR helper option --jsonHDR is designed to help write HDR information into the T.35 SEI packets of streams. The HDR information is parsed from a .json file compatible with HDR10+ and/or Dolby Vision. HDR10+ information for one frame in the JSON data can be identified by the frame ID. Such information is repeated until new information comes.

For definitions of HDR10+ and Dolby Vision, see the following documents:

- ST 2094-40:2020 - SMPTE Standard (HDR10+)
- ST 2094-10:2016 - SMPTE Standard (Dolby Vision)

6.3.1.1 Example: Configuring a JSON File of HDR10+ Format

This section provides an example of configuring a .json file compatible with HDR10+.

Command

```
# H.264 for HDR10+
./video_testenc --codecFormat=h264 -i foreman_cif.yuv -w352 -h288 -b13
```

```
--jsonHDR=vsi_hdr10.json --vuiColordescription=9:16:9 --gopSize=4
```

JSON File

The contents of the `vsi_hdr10.json` file in the command are shown as follows:

```
{
  "Json_info": "VSI",
  "SEI_info": [
    {
      "frame_num": 1,
      "itu_t_t35_country_code": 181,
      "itu_t_t35_terminal_provider_code": 60,
      "itu_t_t35_terminal_provider_oriented_code": 1,
      "application_identifier": 4,
      "application_version": 0,
      "num_windows": 3,
      "processing_windows": [
        1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 0,
        11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 1
      ],
      "targeted_system_display_maximum_luminance": 29,
      "targeted_system_display_actual_peak_luminance": 1,
      "targeted_system_display_actual_peak_luminance": {
        "num_rows_targeted_system_display_actual_peak_luminance": 2,
        "num_cols_targeted_system_display_actual_peak_luminance": 4,
        "targeted_system_display_actual_peak_luminance": [
          15, 14, 13, 12, 11, 10, 9, 8
        ]
      },
      "distribution": {
        "maxscl": [
          1, 2, 3, 11, 12, 13, 21, 22, 23
        ],
        "average_maxrgb": [
          4, 14, 24
        ],
        "num_distribution_maxrgb_percentiles": [
          6, 3, 7
        ],
        "distribution_maxrgb_percentages": [
          1, 2, 3, 4, 5, 6, 15, 16, 17, 2, 3, 4, 5, 6, 7, 8
        ],
        "distribution_maxrgb_percentiles": [
          7, 8, 9, 10, 11, 12, 18, 19, 20, 9, 10, 11, 12, 13, 14, 15
        ],
        "fraction_bright_pixels": [
          13, 21, 21
        ]
      },
      "mastering": {
        "mastering_display_actual_peak_luminance_flag": 1,
        "num_rows_mastering_display_actual_peak_luminance": 5,
        "num_cols_mastering_display_actual_peak_luminance": 3,
        "mastering_display_actual_peak_luminance": [
          7, 6, 5, 4, 3, 2, 1, 0, 1, 2, 3, 4, 5, 6, 7
        ]
      },
      "bezier_curve": {
        "tone_mapping_flag": [
          1, 1, 0
        ],
        "knee_point_x": [
          14, 22
        ],
        "knee_point_y": [

```

```

        15,23
    ],
    "num_bezier_curve_anchors":[
        3,4
    ],
    "bezier_curve_anchors":[
        16,17,18,24,25,26,27
    ],
    "color_saturation_mapping_flag":[
        1,0,1
    ],
    "color_saturation_weight":[
        19,0,28
    ]
}
},
{
    "frame_num": 12,
    "itu_t_t35_country_code": 183,
    "itu_t_t35_terminal_provider_code": 60,
    "itu_t_t35_terminal_provider_oriented_code": 1,
    "application_identifier": 4,
    "application_version": 0,
    "num_windows": 3,
    "processing_windows":[
        1,2,3,4,5,6,7,8,9,10,0,
        11,12,13,14,15,16,17,18,19,20,1
    ],
    "targeted_system_display_maximum_luminance": 29,
    "targeted_system_display_actual_peak_luminance_flag":1,
    "targeted_system_display_actual_peak_luminance":{
        "num_rows_targeted_system_display_actual_peak_luminance":2,
        "num_cols_targeted_system_display_actual_peak_luminance":4,
        "targeted_system_display_actual_peak_luminance":[
            15,14,13,12,11,10,9,8
        ]
    },
    "distribution":{
        "maxscl":[
            1,2,3,11,12,13,21,22,23
        ],
        "average_maxrgb":[
            4,14,24
        ],
        "num_distribution_maxrgb_percentiles":[
            6,3,7
        ],
        "distribution_maxrgb_percentages":[
            1,2,3,4,5,6,15,16,17,2,3,4,5,6,7,8
        ],
        "distribution_maxrgb_percentiles":[
            7,8,9,10,11,12,18,19,20,9,10,11,12,13,14,15
        ],
        "fraction_bright_pixels":[
            13,21,21
        ]
    },
    "mastering":{
        "mastering_display_actual_peak_luminance_flag":1,
        "num_rows_mastering_display_actual_peak_luminance": 5,
        "num_cols_mastering_display_actual_peak_luminance": 3,
        "mastering_display_actual_peak_luminance":[
            7,6,5,4,3,2,1,0,1,2,3,4,5,6,7
        ]
    },
    "bezier_curve":{
        "tone_mapping_flag":[
            1,1,0
        ]
    }
}
}

```



```

    ],
    "knee_point_x": [
        14, 22
    ],
    "knee_point_y": [
        15, 23
    ],
    "num_bezier_curve_anchors": [
        3, 4
    ],
    "bezier_curve_anchors": [
        16, 17, 18, 24, 25, 26, 27
    ],
    "color_saturation_mapping_flag": [
        1, 0, 1
    ],
    "color_saturation_weight": [
        19, 0, 28
    ]
    ]
}
]
}

```

6.3.1.2 Example: JSON File of Dolby Vision Format

This section provides an example of configuring a `.json` file compatible with Dolby Vision.

Command

```

# H264 for Dolby Vision
./video_testenc --codecFormat=h264 -i foreman_cif.yuv -w352 -h288 -b13
--jsonHDR=vsi_dolby.json --vuiColordescription=9:16:9 --gopSize=4

```

JSON File

The contents of the `vsi_dolby.json` file in the command are shown as follows:

```

{
  "Json_info": "VSI",
  "HDR_TYPE": "DOLBY_VISION",
  "SEI_info": [
    {
      "frame_num": 0,
      "itu_t_t35_country_code": 181,
      "itu_t_t35_terminal_provider_code": 59,
      "itu_t_t35_terminal_provider_oriented_code": 0,
      "data_type_code": 9,
      "app_identifier": 1,
      "app_version": 0,
      "metadata_refresh_flag": 1,
      "metadata": {
        "num_ext_blocks": 4,
        "ext_dm_data_block": [
          {
            "ext_block_length": 5,
            "ext_block_level": 1,
            "min_PQ": 137,
            "max_PQ": 138,

```

```

        "avg_PQ":139
    },
    {
        "ext_block_length":11,
        "ext_block_level":2,
        "target_max_PQ":140,
        "trim_slope":141,
        "trim_offset":142,
        "trim_power":143,
        "trim_chroma_weight":144,
        "trim_saturation_gain":145,
        "ms_weight":4097
    },
    {
        "ext_block_length":11,
        "ext_block_level":2,
        "target_max_PQ":146,
        "trim_slope":147,
        "trim_offset":148,
        "trim_power":149,
        "trim_chroma_weight":150,
        "trim_saturation_gain":151,
        "ms_weight":4097
    },
    {
        "ext_block_length":7,
        "ext_block_level":5,
        "active_area_left_offset":133,
        "active_area_right_offset":134,
        "active_area_top_offset":135,
        "active_area_bottom_offset":136
    }
    ]
}
},
{
    "frame_num":1,
    "itu_t_t35_country_code": 181,
    "itu_t_t35_terminal_provider_code": 59,
    "itu_t_t35_terminal_provider_oriented_code": 0,
    "data_type_code": 9,
    "app_identifier": 1,
    "app_version": 0,
    "metadata_refresh_flag":1,
    "metadata":{
        "num_ext_blocks":1,
        "ext_dm_data_block":[
            {
                "ext_block_length":5,
                "ext_block_level":1,
                "min_PQ":204,
                "max_PQ":205,
                "avg_PQ":206
            }
        ]
    }
}
]
}
}

```

6.3.1.3 Example: JSON File of Mixed Format

This section provides an example of configuring a .json file compatible with both HDR+ and Dolby Vision.

Command

```
# HEVC for mixed format
./video_testenc --codecFormat=hevc -i foreman_cif.yuv -w352 -h288 -b13
--jsonHDR=vsi.json --vuiColorDescription=9:16:9 --vuiVideoFormat=5
--vuiVideoSignalPresent=1 --HDR10_display=0:0:3:4:5:6:2:1:100:10
--HDR10_lightlevel=50:50 --bitDepthLuma=10 --bitDepthChroma=10 --profile=2
```

JSON File

The contents of the `vsi.json` file in the command are shown as follows:

```
{
  "Json_info": "VSI",
  "SEI_info": {
    {
      "frame_num": 1,
      "itu_t_t35_country_code": 1,
      "itu_t_t35_terminal_provider_code": 60,
      "itu_t_t35_terminal_provider_oriented_code": 1,
      "application_identifier": 4,
      "application_version": 0,
      "num_windows": 3,
      "processing_windows": [
        1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 0,
        11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 1
      ],
      "targeted_system_display_maximum_luminance": 29,
      "targeted_system_display_actual_peak_luminance_flag": 1,
      "targeted_system_display_actual_peak_luminance": {
        "num_rows_targeted_system_display_actual_peak_luminance": 2,
        "num_cols_targeted_system_display_actual_peak_luminance": 4,
        "targeted_system_display_actual_peak_luminance": [
          15, 14, 13, 12, 11, 10, 9, 8
        ]
      }
    },
    "distribution": {
      "maxscl": [
        1, 2, 3, 11, 12, 13, 21, 22, 23
      ],
      "average_maxrgb": [
        4, 14, 24
      ],
      "num_distribution_maxrgb_percentiles": [
        6, 3, 7
      ],
      "distribution_maxrgb_percentages": [
        1, 2, 3, 4, 5, 6, 15, 16, 17, 2, 3, 4, 5, 6, 7, 8
      ],
      "distribution_maxrgb_percentiles": [
        7, 8, 9, 10, 11, 12, 18, 19, 20, 9, 10, 11, 12, 13, 14, 15
      ],
      "fraction_bright_pixels": [
        13, 21, 21
      ]
    },
    "mastering": {
      "mastering_display_actual_peak_luminance_flag": 1,
      "num_rows_mastering_display_actual_peak_luminance": 5,
      "num_cols_mastering_display_actual_peak_luminance": 3,
      "mastering_display_actual_peak_luminance": [
        7, 6, 5, 4, 3, 2, 1, 0, 1, 2, 3, 4, 5, 6, 7
      ]
    },
    "bezier_curve": {
      "tone_mapping_flag": [
        1, 1, 0
      ]
    }
  }
}
```

```

    ],
    "knee_point_x": [
        14, 22
    ],
    "knee_point_y": [
        15, 23
    ],
    "num_bezier_curve_anchors": [
        3, 4
    ],
    "bezier_curve_anchors": [
        16, 17, 18, 24, 25, 26, 27
    ],
    "color_saturation_mapping_flag": [
        1, 0, 1
    ],
    "color_saturation_weight": [
        19, 0, 28
    ]
}
},
{
    "frame_num": 5,
    "itu_t_t35_country_code": 2,
    "itu_t_t35_terminal_provider_code": 60,
    "itu_t_t35_terminal_provider_oriented_code": 1,
    "application_identifier": 4,
    "application_version": 0,
    "num_windows": 3,
    "processing_windows": [
        1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 0,
        11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 1
    ],
    "targeted_system_display_maximum_luminance": 29,
    "targeted_system_display_actual_peak_luminance_flag": 1,
    "targeted_system_display_actual_peak_luminance": {
        "num_rows_targeted_system_display_actual_peak_luminance": 2,
        "num_cols_targeted_system_display_actual_peak_luminance": 4,
        "targeted_system_display_actual_peak_luminance": [
            15, 14, 13, 12, 11, 10, 9, 8
        ]
    },
    "distribution": {
        "maxscl": [
            1, 2, 3, 11, 12, 13, 21, 22, 23
        ],
        "average_maxrgb": [
            4, 14, 24
        ],
        "num_distribution_maxrgb_percentiles": [
            6, 3, 7
        ],
        "distribution_maxrgb_percentages": [
            1, 2, 3, 4, 5, 6, 15, 16, 17, 2, 3, 4, 5, 6, 7, 8
        ],
        "distribution_maxrgb_percentiles": [
            7, 8, 9, 10, 11, 12, 18, 19, 20, 9, 10, 11, 12, 13, 14, 15
        ],
        "fraction_bright_pixels": [
            13, 21, 21
        ]
    },
    "mastering": {
        "mastering_display_actual_peak_luminance_flag": 1,
        "num_rows_mastering_display_actual_peak_luminance": 5,
        "num_cols_mastering_display_actual_peak_luminance": 3,
        "mastering_display_actual_peak_luminance": [
            7, 6, 5, 4, 3, 2, 1, 0, 1, 2, 3, 4, 5, 6, 7
        ]
    }
}

```

```

    ]
  },
  "bezier_curve":{
    "tone_mapping_flag":[
      1,1,0
    ],
    "knee_point_x":[
      14,22
    ],
    "knee_point_y":[
      15,23
    ],
    "num_bezier_curve_anchors":[
      3,4
    ],
    "bezier_curve_anchors":[
      16,17,18,24,25,26,27
    ],
    "color_saturation_mapping_flag":[
      1,0,1
    ],
    "color_saturation_weight":[
      19,0,28
    ]
  }
},
{
  "frame_num":7,
  "itu_t_t35_country_code": 3,
  "itu_t_t35_terminal_provider_code": 60,
  "itu_t_t35_terminal_provider_oriented_code": 1,
  "application_identifier": 4,
  "application_version": 0,
  "num_windows": 3,
  "processing_windows":[
    1,2,3,4,5,6,7,8,9,10,0,
    11,12,13,14,15,16,17,18,19,20,1
  ],
  "targeted_system_display_maximum_luminance": 29,
  "targeted_system_display_actual_peak_luminance_flag":1,
  "targeted_system_display_actual_peak_luminance":{
    "num_rows_targeted_system_display_actual_peak_luminance":2,
    "num_cols_targeted_system_display_actual_peak_luminance":4,
    "targeted_system_display_actual_peak_luminance":[
      15,14,13,12,11,10,9,8
    ]
  },
  "distribution":{
    "maxscl": [
      1,2,3,11,12,13,21,22,23
    ],
    "average_maxrgb": [
      4,14,24
    ],
    "num_distribution_maxrgb_percentiles": [
      6,3,7
    ],
    "distribution_maxrgb_percentages": [
      1,2,3,4,5,6,15,16,17,2,3,4,5,6,7,8
    ],
    "distribution_maxrgb_percentiles": [
      7,8,9,10,11,12,18,19,20,9,10,11,12,13,14,15
    ],
    "fraction_bright_pixels": [
      13,21,21
    ]
  },
  "mastering":{

```

```

"mastering_display_actual_peak_luminance_flag":1,
"num_rows_mastering_display_actual_peak_luminance": 5,
"num_cols_mastering_display_actual_peak_luminance": 3,
"mastering_display_actual_peak_luminance": [
    7, 6, 5, 4, 3, 2, 1, 0, 1, 2, 3, 4, 5, 6, 7
]
},
"bezier_curve":{
"tone_mapping_flag": [
    1, 1, 0
],
"knee_point_x": [
    14, 22
],
"knee_point_y": [
    15, 23
],
"num_bezier_curve_anchors": [
    3, 4
],
"bezier_curve_anchors": [
    16, 17, 18, 24, 25, 26, 27
],
"color_saturation_mapping_flag": [
    1, 0, 1
],
"color_saturation_weight": [
    19, 0, 28
]
}
},
{
"frame_num": 9,
"itu_t_t35_country_code": 4,
"itu_t_t35_terminal_provider_code": 60,
"itu_t_t35_terminal_provider_oriented_code": 1,
"application_identifier": 4,
"application_version": 0,
"num_windows": 3,
"processing_windows": [
    1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 0,
    11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 1
],
"targeted_system_display_maximum_luminance": 29,
"targeted_system_display_actual_peak_luminance_flag": 1,
"targeted_system_display_actual_peak_luminance": {
    "num_rows_targeted_system_display_actual_peak_luminance": 2,
    "num_cols_targeted_system_display_actual_peak_luminance": 4,
    "targeted_system_display_actual_peak_luminance": [
        15, 14, 13, 12, 11, 10, 9, 8
    ]
},
"distribution": {
"maxscl": [
    1, 2, 3, 11, 12, 13, 21, 22, 23
],
"average_maxrgb": [
    4, 14, 24
],
"num_distribution_maxrgb_percentiles": [
    6, 3, 7
],
"distribution_maxrgb_percentages": [
    1, 2, 3, 4, 5, 6, 15, 16, 17, 2, 3, 4, 5, 6, 7, 8
],
"distribution_maxrgb_percentiles": [
    7, 8, 9, 10, 11, 12, 18, 19, 20, 9, 10, 11, 12, 13, 14, 15
],

```

```

    "fraction_bright_pixels":[
        13,21,21
    ],
    },
    "mastering":{
        "mastering_display_actual_peak_luminance_flag":1,
        "num_rows_mastering_display_actual_peak_luminance": 5,
        "num_cols_mastering_display_actual_peak_luminance": 3,
        "mastering_display_actual_peak_luminance":[
            7,6,5,4,3,2,1,0,1,2,3,4,5,6,7
        ]
    },
    "bezier_curve":{
        "tone_mapping_flag":[
            1,1,0
        ],
        "knee_point_x":[
            14,22
        ],
        "knee_point_y":[
            15,23
        ],
        "num_bezier_curve_anchors":[
            3,4
        ],
        "bezier_curve_anchors":[
            16,17,18,24,25,26,27
        ],
        "color_saturation_mapping_flag":[
            1,0,1
        ],
        "color_saturation_weight":[
            19,0,28
        ]
    }
},
{
    "frame_num":11,
    "itu_t_t35_country_code": 181,
    "itu_t_t35_terminal_provider_code": 59,
    "itu_t_t35_terminal_provider_oriented_code": 0,
    "data_type_code": 9,
    "app_identifier": 1,
    "app_version": 0,
    "metadata_refresh_flag":1,
    "metadata":{
        "num_ext_blocks":1,
        "ext_dm_data_block":[
            {
                "ext_block_length":5,
                "ext_block_level":1,
                "min_PQ":204,
                "max_PQ":205,
                "avg_PQ":206
            }
        ]
    }
}
}
}

```

6.3.2 External SEI Data Input

The video encoder test bench supports input of external SEI data for each frame before encoding the frame. You can feed the test bench by specifying an SEI binary file with the `--extSEI` option.

Data in the SEI binary file must be organized in the following format:

```
<pre>
+-----+
|  frameId_0, u32  |
+-----+
|  seiCnt, u32     |
+-----+
|ExternalSEI[0]    |
+-----+
|  nalType, u8     |
+-----+
|  payloadType, u8 |
+-----+
|  payloadDataSize, u32 |
+-----+
|  pPayloadData,   |
|  u8*payloadDataSize |
+-----+
|  ...            |
+-----+
|ExternalSEI[seiCnt-1] |
+-----+
|  nalType, u8     |
+-----+
|  payloadType, u8 |
+-----+
|  payloadDataSize, u32 |
+-----+
|  pPayloadData,   |
|  u8*payloadDataSize |
+-----+
|  ...            |
|  ...            |
+-----+
|  frameId_N, u32  |
+-----+
|  seiCnt, u32     |
+-----+
|ExternalSEI[0]    |
+-----+
|  nalType, u8     |
+-----+
|  payloadType, u8 |
+-----+
|  payloadDataSize, u32 |
+-----+
|  pPayloadData,   |
+-----+
```



```

|      u8*payloadDataSize |
+-----+
|      ...                |
+-----+
|ExternalSEI[seiCnt-1] |
+-----+
|      nalType, u8       |
+-----+
|      payloadType, u8   |
+-----+
|      payloadDataSize, u32 |
+-----+
|      pPayloadData,     |
|      u8*payloadDataSize |
+-----+

```

</pre>

In the file:

- `frameId` indicates the frame sequence ID in the encoded output stream, starting from 0.
- `seiCnt` indicates the number of ExternalSEI sections for the frame.
- `ExternalSEI` indicates the SEI information of the frame.
- Before encoding loop, fetch the first `seiFrameId`

```

if (tb->extSEIFile && seiFrameId == DEFAULT)
{
    fret = fread(&seiFrameId, sizeof(u32), 1, tb->extSEIFile);
}

```

- Each time before encoding a frame, if current encode frame id equals to `seiFrameId`, test bench reads out SEI info, and fetch next `seiFrameId`:

```

if (tb->encIn.picture_cnt == seiFrameId)
{
    pEncIn->externalSEICount = ReadExternalSeiData(&pEncIn->pExternalSEI, tb->extSEIFile);
    if (!feof(tb->extSEIFile))
    {
        fret = fread(&seiFrameId, sizeof(u32), 1, tb->extSEIFile);
    }
}
else
{
    pEncIn->externalSEICount = 0;
    pEncIn->pExternalSEI = NULL;
}

```

Then do encode, finally free the allocated memory if needed:

```

if (pEncIn->externalSEICount != 0)
{
    for (int k = 0; k < pEncIn->externalSEICount; k++)
    {
        free(pEncIn->pExternalSEI[k].pPayloadData);
    }
}

```

```
}  
free(pEncIn->pExternalSEI);  
pEncIn->externalSEICount = 0;  
pEncIn->pExternalSEI = NULL;  
}
```

cml->extSEI is parsed from test bench's command line, with option like "-extSEI test_data/extSEI.dat", where test_data/extSEI.dat is the SEI binary file.
tb->extSEIFile is a FILE* opened from the SEI binary file.

6.4 Miscellaneous Coding Features

6.4.1 Multi-Stream Encoding

The video encoder provides the `--multimode` option for testing the multi-stream cases.

Configuration

To test the multi-stream feature, you need to:

- Specify `--multimode=1`.
- Specify `--streamcfg` once for each stream. Set the value of `--multimode` to the separate configuration file of the stream. The file must contain the command line used to encode the stream.

Example

The following example commands encode two streams simultaneously in multi-thread mode:

```
$ echo "-i foreman_cif.yuv -w352 -h288 -q26 -b19 -o foreman_q26_f20.h264" > 1.cfg  
$ echo "-i foreman_cif.yuv -w352 -h288 -U1 -B512000 -b29 -o foreman_512kbps_f30.h264" > 2.cfg  
$ ./h264_testenc --multimode=1 --streamcfg="1.cfg" --streamcfg="2.cfg"
```

6.4.2 Batch Mode Encoding

Batch mode encoding will aggregate multiple encoding tasks in software and trigger hardware to start the encoding tasks after the aggregation achieve some threshold. In such way, the hardware can work on the tasks one by one without interrupt until the last one. User can use this feature to reduce hardware's power because the hardware can sleep longer if the tasks are excuted in short time. At the same time, the software can reduce interrupt process for only one interrupt is processed when a batch of tasks is done.

This feature has following limitation,

- It is only valid for VCMD because software must be involved to process frame done IRQ.
- To reduce software complexity, batch mode is invalid for lookahead
- It is disable automatically if multi-instance is detected. In such condition, the hardware cannot saving power because it is hard to find proper timing to power off the codec.

To enable the feature in test bench, option `--batchEnable` should be 1. In addition, following option can control the task aggregation process.

- option `--parallelCoreNum`: it controls the maximum tasks software can aggregated.
- option `--batchCount`: it controls the threshold to trigger hardware to execute the aggregated tasks. the default and maximum number is `parallelCoreNum-1`

Note: This feature need working with VCMD.

Following command line can be used to test this feature. The threshold is set as 1 at the beginning and change to 4 when frame 1 is encoding.

```
./hevc_testenc -i foreman_cif.yuv -w352 -h288 -b20 --gopSize=1 -q26 --parallelCoreNum=10 \
--batchEnable=1 --batchCount=1 -o stream.hevc @1 --batchCount=4
```

6.4.3 Resolution Change

The video encoder supports changing the resolution to same or smaller one after initialization without memory reallocation. At the same time, the coding feature should be kept same to avoid extra buffers are required due to different coding tools.

For the requirements on resolution change, see the *Stream-Level Encoding Control* section in *Hantro VC9000E Video Encoder API User Manual*.

6.4.3.1 Resolution Change Feature Testing

The video encoder test bench provides the `--inputFileList` option to specify a file that contains a list of input pictures for testing the resolution change feature. Each line in the file must contain the following suboptions:

- `-i`: the path to an input picture
- `-w` and `-h`: the picture resolution
- `-a`: the start picture
- `-b`: the end picture
- `-o`: the path to the output file

6.4.3.2 Example of Testing the Resolution Change Feature

Input List File

The following script shows the contents of an example input list file named `input_file_list.cfg`:

```
-i yuv/soccer_30fps_4cif_704x576.yuv -w704 -h576 -a0 -b30 -o city.h264
-i yuv/forman_cif.yuv -w352 -h288 -a0 -b30 -o foreman_cif.h264
```

Commands

The following command enables encoding of the pictures listed in `input_file_list.cfg`:

```
./h264_testenc -w704 -h576 -i rush_hour_25fps_w704h576.yuv -b2 -o stream.h264
--lookaheadDepth=4 -U1 --bitPerSecond=2000000 --gopSize=4
--inputFileList=./cfg/input_file_list.cfg
```

The command with the `--inputFileList` option is equivalent to the following commands:

```
./h264_testenc -i yuv/soccer_30fps_4cif_704x576.yuv -w704 -h576 -a0 -b30 -o city.264
--lookaheadDepth=4 -U1 --bitPerSecond=2000000 --gopSize=4
./h264_testenc -i yuv/forman_cif.yuv -w352 -h288 -a0 -b30 -o foreman_cif.h264
--lookaheadDepth=4 -U1 --bitPerSecond=2000000 --gopSize=4
```

7 Frame Statistics Output

Table 7.1 lists the frame-level statistics that the video encoder can output. The statistics are output to a file named `frameInfo.txt` in a readable format.

The feature is supported only if the value of `EVLHwConfig_t.frameInfoSupport` is 1.

Table 7.1 Frame Statistics

Category	Item
CU statistics by CU type and size	N/A
Encoded bit statistics	- Total number of encoded bytes - Number of head bits, including CU/MB mode bits and MV bits - Number of residual bits, including TU CBF and residual bits
Average complexity	- Average spatial texture complexity (MAD-I) - Average temporal motion complexity (MAD-P)
SSE and PSNR	- Frame SSE and region SSE - Frame PSNR in fixed-point 16.16 format
QP statistics	- Initial QP and mean QP - QP histogram
Frame information	- POC of the frame - Number of slices in the frame - Index of the frame in the GOP (frames are counted starting from 0 for each IDR frame) - Number of the picture in the sequence

7.1 Frame Statistics Output Configurations

The options used to configure the frame statistics output feature are listed as follows:

- `enableFrameInfo` controls whether to enable the frame statistics output feature.
- `sse0Rect` to `sse7Rect` defines a maximum of eight regions for SSE statistics.
- `sse0Enable` to `sse7Enable` controls whether to enable each region.

7.2 Example: Frame Statistics Output with Frame-Level SSE Only

The following example commands show how to enable frame statistics output for HEVC (H.265) and H.264 (AVC) respectively:

```
.\video_testenc --codecFormat hevc -i foreman.yuv -w352 -h288 -b5 --gopSize=1
--enableFrameInfo 1 -o stream.hevc

.\video_testenc --codecFormat h264 -i foreman.yuv -w352 -h288 -b5 --gopSize=1
--enableFrameInfo 1 -o stream.h264
```

7.3 Example: Frame Statistics Output with Frame-Level and Region-Level SSE

The following example commands show how to enable frame statistics output with SSE of specified regions for HEVC (H.265) and H.264 (AVC) respectively:

```
.\video_testenc --codecFormat hevc -i foreman.yuv -w352 -h288 -b5 --gopSize=1
--enableFrameInfo 1 -o stream.hevc
--sse0Enable=1 --sse1Rect=0:1:2:3
--sse1Enable=1 --sse1Rect=4:4:4:4
--sse5Enable=1 --sse5Rect=8:10:9:11

.\video_testenc --codecFormat h264 -i foreman.yuv -w352 -h288 -b5 --gopSize=1
--enableFrameInfo 1 -o stream.h264
--sse0Enable=1 --sse1Rect=0:1:2:3
--sse1Enable=1 --sse1Rect=4:4:4:4
--sse5Enable=1 --sse5Rect=8:10:9:11
```

8 Miscellaneous Features

This chapter describes the following features of the video encoder test bench:

- Preset RDO Schemes
- framerate
- bw
- Force Addition of Filler Data
- Frame Re-Encoding
- Log Output Control

8.1 Preset RDO Schemes

The video encoder pre-defines several rate distortion optimization (RDO) schemes based on tradeoff between performance and compression efficiency. You can select an RDO scheme by explicitly specifying the `preset` option. A greater value of `preset` indicates higher quality but lower performance.

The following table lists the default settings of each RDO scheme and the mapping values of `preset`. When `preset` is specified, the default settings are loaded and will be overwritten by its following options.

NOTE: Performance varies according to actual stream conditions. The following table provides only approximate values in general conditions with the core frequency at 650 MHz.

Table 8.1 Default Parameter Settings for RDO Schemes

preset: H.264	preset: HEVC	lookaheadDepth	rdoLevel	enableRdoQuant	Performance
-	4	40	3	1	4K@12 fps
-	3	40	3	0	4K@18 fps
-	2	40	2	0	4K@30 fps
1	1	40	1	0	4K@60 fps
0	0	0	1	0	4K@75 fps

8.2 Force Addition of Filler Data

When HRD is enabled, filler data is added to make the average bit rate reach the target and avoid CPB underflow with a constant bit rate. When HRD is disabled, filler data is not added by default.

To avoid bit rate drop, the video encoder supports force adding filler data NAL units no matter whether HRD is enabled or disabled. This feature is useful when fixed transmission bandwidth is required.

NOTE: The filler data force addition feature is available only for HEVC (H.265) and H.264 (AVC).

Configuration

To enable force addition of filler data, you need to:

- Specify `--fillerData=1` to enable force addition of filler data even if HRD is disabled.
- Set `--bitPerSecond` to the target bit rate.
- Set `--cpbSize` to the CPB size.
- Specify `--picRc=1` to disable picture-level rate control.

Example

```
./video_testenc --codecFormat 0 -l0 -i city_30fps_w352h288.yuv -w352 -h288 -a0 -b9
--picRc=1 --fillerData=1 -B 10000000 -c 50000 --lookaheadDepth=4 -o stream.hevc
```

In this command:

- Rate control settings include:
 - `--lookaheadDepth=4` sets the look-ahead buffer size to 4 frames.
 - `--fillerData=1` enables force addition of filler data.
 - `-B 10000000` is short for `--bitPerSecond=10000000`, which sets the target bit rate for rate control to 10000000 bps.
 - `-c 50000` is short for `--cpbSize=50000`, which sets the CPB size to 50000 bits.
- Other settings include:
 - `--codecFormat 0` sets the codec format to HEVC (H.265).
 - `-l0` sets the input format to I420 (YUV420P8b_Raster_A16N).
 - `-i city_30fps_w352h288.yuv` specifies that the input file is `city_30fps_w352h288.yuv`.
 - `-w352 -h288` sets the resolution of the input video sequence to 1920x1080 pixels. If `-x` and `-y` are not specified, the width and height of the output video stream are the same as those of the input video sequence.
 - `-a0 -b9` sets the first picture to be encoded is No. 0 and the last picture to be encoded is No. 9.
A total of 10 frames are encoded.
 - `-o stream.hevc` specifies to output the encoded stream to the `stream.hevc` file.

8.3 Frame Re-Encoding

The video encoder supports frame re-encoding in exceptional scenarios such as output buffer overflow.

Without frame re-encoding, when the output stream of a frame is larger than the output stream buffer, the video encoder returns a `VCENC_OUTPUT_BUFFER_OVERFLOW` error and discards the frame. If frame re-encoding is enabled, the video encoder re-allocates a larger buffer and re-encodes the frame.

NOTE: Re-encoding causes a delay that is usually acceptable.

Configuration

To enable frame re-encoding, specify `--reEncode=1`.

Example

```
./video_testenc --codecFormat 0 -l0 -i city_30fps_w352h288.yuv -w352 -h288 -a0 -b9  
--reEncode=1 -o stream.hevc
```

In this command:

- `--codecFormat 0` sets the codec format to HEVC (H.265).
- `-l0` sets the input format to I420 (YUV420P8b_Raster_A16N).
- `-i city_30fps_w352h288.yuv` specifies that the input file is `city_30fps_w352h288.yuv`.
- `-w352 -h288` sets the resolution of the input video sequence to 1920x1080 pixels.
If `-x` and `-y` are not specified, the width and height of the output video stream are the same as those of the input video sequence.
- `-a0 -b9` sets the first picture to be encoded is No. 0 and the last picture to be encoded is No. 9.
A total of 10 frames are encoded.
- `--reEncode=1` enables frame re-encoding in exceptional scenarios such as output buffer overflow.
- `-o stream.hevc` specifies to output the encoded stream to the `stream.hevc` file.

8.4 Log Output Control

If logging is enabled by setting `LOGMSG=y` when you compile the test bench, you can configure the log output by setting the following environment variables in Linux Bash before running the test bench:

- `VCENC_LOG_OUTPUT` for specifying the target to which logs are output

Available values include:

- 0: outputs all logs to stdout.
- 1: outputs all logs to the file `vcenc_trace_p${pid}.log` where `${pid}` is the process ID.
- 2: outputs logs of each thread to a separate log file `vcenc_trace_p${pid}_t${tid}.log` where `${pid}` is the process ID and `${tid}` is the thread ID.

By default, all logs are output to stdout.

NOTE: Alternatively, you can use the `--logOutDir` option.

- `VCENC_LOG_LEVEL` for specifying the log levels to be output

Available values include:

- 0: does not output logs.
- 1: outputs logs of the fatal level.
- 2: outputs logs of the error level and above.
- 3: outputs logs of the warning level and above.
- 4: outputs logs of the information level and above.
- 5: outputs logs of the debugging level and above.
- 6: outputs all logs.

By default, logs of the error level and above are output.

NOTE: Alternatively, you can use the `--logOutLevel` option.

- `VCENC_LOG_TRACE` for selecting the trace log types to be output

Available options include:

- Bit 0: encoder API call log
- Bit 1: register configuration log
- Bit 2: EWL APL call log
- Bit 3: memory usage log
- Bit 4: rate control status log
- Bit 5: command line log
- Bit 6: performance log (under development)

By default, all trace logs are output.

NOTE: Alternatively, you can use the `--logTraceMap` option.

- (Reserved) `VCENC_LOG_CHECK` for selecting the check log types to be output

Available options include:

- Bit 0: reconstructed YUV data
- Bit 1: PSNR/SSIM for each frame
- Bit 2: VBV information for rate control checking
- Bit 3: feature information for coverage checking

NOTE: Alternatively, you can use the `--logCheckMap` option.

8.4.1 Example of Setting Log Output

The following example commands enable encoder API call logs and register configuration logs of the information level and above to be output.

```
export VCENC_LOG_LEVEL=4    # information level
export VCENC_LOG_TRACE=3    # encoder API call log and register configuration log
```

9 Command Line Examples

9.1 Example: Low-Delay P with GOP Structure of Size 1 and Fixed QP

The following example command applies the Low-Delay P configuration and configures a GOP structure of size 1 and fixed QP:

```
./hevc_testenc -R0 -A0 --gopSize=1 --gopConfig=../vc9000e_gop_config_lowdelay_P.cfg  
-w1920 -h1080 -i BBD_1920x1080_50.yuv -b52 -q22 -o 0.hevc --outReconFrame=0
```

In this command:

- **-R0** sets the intra frame rate to 0.
No frame is force encoded as an intra frame.
- **-A0** sets the intra QP delta to 0.
The intra QP equals the target QP.
- **--gopSize=1** sets the GOP structure size to 1.
- **--gopConfig=../vc9000e_gop_config_lowdelay_P.cfg** specifies the GOP configuration file.
For the contents of the file, see [Example File: vc9000e_gop_config_lowdelay_P.cfg](#).
- **-w1920 -h1080** sets the input frame size to 1920x1080 pixels.
- **-i BBD_1920x1080_50.yuv** specifies that the input file is BBD_1920x1080_50.yuv.
- **-b52** specifies that the last frame to be encoded is No. 52 in the input file.
A total of 53 frames are encoded because frames are counted starting from 0.
- **-q22** sets the default target QP to 22.
- **-o 0.hevc** specifies to output the encoded stream to the 0.hevc file.
- **--outReconFrame=0** disables the output of reconstructed frames.

9.1.1 Example File: vc9000e_gop_config_lowdelay_P.cfg

The following script shows the contents of the example file vc9000e_gop_config_lowdelay_P.cfg:

```
*          Type POC QPoffset QPfactor TemporalId num_ref_pics ref_pics used_by_cur
* Frame1: P    1   3         0.4624   0           1           -1           1
```

9.2 Example: Subjective Rate Control

The following example command enables subjective rate control targeting 1080P@30 fps and 500 Kbps:

```
./hevc_testenc -R300 -w1920 -h1080 -U1 -u1 -g30 -b239 -A0 -q-1 --bitPerSecond=500000
-i BBD_1920x1080_50.yuv -o BBD_1920x1080_50.bin --gopSize=1
```

In this command:

- **-R300** sets the intra frame rate to 300.
The video encoder forces one frame to be encoded as an intra frame every 300 frames.
- **-w1920 -h1080** sets the input frame size to 1920x1080 pixels.
- **-U1** enables picture-level rate control.
- **-u1** enables block-level rate control to improve subjective quality.
- **-g30** sets the bit rate window to 30 frames.
The rate control algorithm allocates bits for each 30-frame window and tries to match the target bit rate at the end of the window.
- **-b239** specifies that the last frame to be encoded is No. 239 in the input file.
A total of 240 frames are encoded because frames are counted starting from 0.
- **-A0** sets the intra QP delta to 0.
The intra QP equals the target QP.
- **-q-1** enables the video encoder to calculate the initial target QP.
- **--bitPerSecond=500000** sets the target bit rate for rate control to 500000 bps.

- `-i BBD_1920x1080_50.yuv` specifies that the input file is `BBD_1920x1080_50.yuv`.
- `-o BBD_1920x1080_50.bin` specifies to output the encoded stream to the `BBD_1920x1080_50.bin` file.
- `--gopSize=1` sets the GOP structure size to 1.

9.3 Example: GOP Structure of Size 4 with Long-Term Reference

The following example command configures a GOP structure of size 4 with long-term reference (LTR) enabled:

```
./hevc_testenc -A-2 -R300 --gopSize=4 --gopConfig=../vc9000e_gop_config_lowdelay_ltr.cfg
-w1920 -h1080 -i ParkScene_1920x1080_24.yuv -b49 -q22 -o 0.hevc --outReconFrame=0
```

In this command:

- `-A-2` sets the intra QP delta to -2.
The intra QP equals the intra frame QP minus 2.
- `-R300` sets the intra frame rate to 300.
The video encoder forces one frame to be encoded as an intra frame every 300 frames.
- `--gopSize=4` sets the GOP structure size to 4.
- `--gopConfig=../vc9000e_gop_config_lowdelay_ltr.cfg` specifies the GOP configuration file.
For the contents of the file, see [Example File: vc9000e_gop_config_lowdelay_ltr.cfg](#).
- `-w1920 -h1080` sets the input frame size to 1920x1080 pixels.
- `-i ParkScene_1920x1080_24.yuv` specifies that the input file is `ParkScene_1920x1080_24.yuv`.
- `-b49` specifies that the last frame to be encoded is No. 49 in the input file.
A total of 50 frames are encoded because frames are counted starting from 0.
- `-q22` sets the default target QP to 22.
- `-o 0.hevc` specifies to output the encoded stream to the `0.hevc` file.
- `--outReconFrame=0` disables the output of reconstructed frames.

9.3.1 Example File: vc9000e_gop_config_lowdelay_ltr.cfg

The following script shows the contents of the example file vc9000e_gop_config_lowdelay_ltr.cfg:

```
* # normal frame configure
* #      Type POC  QPoffset QPfactor TemporalId num_ref_pics ref_pics  used_by_cur
* Frame1: B    1    0      0.4624    0          2          -1 L1      1 1
* Frame2: B    2    0      0.4624    0          2          -1 L1      1 1
* Frame3: B    3    0      0.4624    0          2          -1 L1      1 1
* Frame4: B    4    0      0.578     0          2          -1 L1      1 1
*
* # long-term reference and special frame configure
* #      Type  QPoffset QPfactor TemporalId num_ref_pics ref_pics used_by_cur LTR Offset Interval
* #default -255 -255 -255 -255
* Frame0: -255 -255 -255 -255      2          -1 L1      1      1 1 0      24
```

9.4 Example: Quality Test

To measure video quality, you can either enable picture-level rate control with a target bit rate or use a fixed QP value with rate control disabled. When picture-level rate control is enabled, you can also enable block-level rate control for better objective quality.

This section provides example commands for encoding in these two cases respectively.

NOTE: Enabling block-level rate control may decrease the objective quality (for example, in terms of PSNR) compared with the best case. Therefore, if you want to measure the quality by PSNR only, it is recommended that you disable block-level rate control.

Picture-Level and Block-Level Rate Control Enabled

The following example command enables both picture-level and block-level rate control:

```
./hevc_testenc -w 832 -h 480 -b 100 -B1000000 -U1 -ul -i BQM_832x480_60.yuv
-o out.bin --gopSize=1
```

In this command:

- `-w 832 -h 480` sets the input frame size to 832x480 pixels.

- `-b 100` specifies that the last frame to be encoded is No. 100 in the input file.
A total of 101 frames are encoded because frames are counted starting from 0.
- `-B1000000` sets the target bit rate for rate control to 1000000 bps.
- `-U1` enables picture-level rate control.
- `-u1` enables block-level rate control.
- `-i BQM_832x480_60.yuv` specifies that the input file is `BQM_832x480_60.yuv`.
- `-o out.bin` specifies to output the encoded stream to the `out.bin` file.
- `--gopSize=1` sets the GOP structure size to 1.

Fixed QP with Rate Control Disabled

The following example command configures a fixed QP value and disables both picture-level and block-level rate control:

```
./hevc_testenc -w 832 -h 480 -b 100 -q 22 -i BQMall_832x480_60.yuv -o out.bin  
--gopSize=4 --gopLowdelay=1
```

In this command:

- `-w 832 -h 480` sets the input frame size to 832x480 pixels.
- `-b 100` specifies that the last frame to be encoded is No. 100 in the input file.
A total of 101 frames are encoded because frames are counted starting from 0.
- `-q 22` sets the default target QP to 22.
- `-i BQMall_832x480_60.yuv` specifies that the input file is `BQMall_832x480_60.yuv`.
- `-o out.bin` specifies to output the encoded stream to the `out.bin` file.
- `--gopSize=4` sets the GOP structure size to 4.
- `--gopLowdelay=1` enables the video encoder to use the default low-delay GOP configurations.

9.5 Example: Picture Mirroring and Rotation

The following example command mirrors and rotates the input picture before encoding. If both picture mirroring and rotation are enabled, the input picture is mirrored before being rotated.

```
./hevc_testenc --mirror=1 --rotation=3 -i BQSquare_416x240_60.yuv -w416 -h240  
-b 10 -g 60 -R 30 -u 0 -q 20 --intraQpDelta=-2 -o stream.hevc
```

In this command:

- Mirroring and rotation settings include:
 - `--mirror=1` horizontally mirrors the input picture.
 - `--rotation=3` rotates the input picture by 180 degrees.
- Other settings include:
 - `-i BQSquare_416x240_60.yuv` specifies that the input file is `BQSquare_416x240_60.yuv`.
 - `-w416 -h240` sets the resolution of the input video sequence to 416x240 pixels.
 - `-b 10` specifies that the last frame to be encoded is No. 10 in the input file.
A total of 11 frames are encoded because frames are counted starting from 0.
 - `-g 60` sets the bit rate window to 60 frames.
The rate control algorithm allocates bits for each 60-frame window and tries to match the target bit rate at the end of the window.
A smaller bit rate window ensures a more accurate moving bit rate. However, the more accurate the moving bit rate, the worse the video quality. A 60-frame bit rate window is recommended.
The first frame in each bit rate window is not required to be an intra frame, but typical bit rate windows start with intra frames.
 - `-R150` sets the intra frame rate to 30.
The video encoder forces one frame to be encoded as an intra frame every 30 frames.
 - `-u 0` disables block-level rate control.
 - `-q 20` sets the default target QP to 20.
 - `--intraQpDelta=-2` sets the intra QP delta to -2.
The intra QP equals the intra frame QP minus 2.
 - `-o stream.hevc` specifies to output the encoded stream to the `stream.hevc` file.

9.6 Example: Low-Delay B with Variable Bit Rate

The following example command applies the Low-Delay B configuration and selects the variable bit rate (VBR) mode for rate control. This example assumes the input is a locally stored video and the VBR mode is used to provide a higher tolerance on the variation of the moving bit rate.

```
./hevc_testenc -R150 --gopSize=1 -U1 -g150 --gopLowdelay=1 -A-5 -w1920 -h1080  
-i /mnt/sequences/jctvc/ParkScene_1920x1080_24.yuv  
-b 239 -q26 -f24 -F1 -j24 -J1 -B 6780054 --vbr=1 --qpMin=10 -o 0.hevc  
--outReconFrame=0
```

In this command:

- Rate control settings include:

- `-U1` enables picture-level rate control.

When rate control is enabled, this option must be set to 1 unless a fixed QP value is used.

- `-g150` sets the bit rate window to 150 frames.

The rate control algorithm allocates bits for each 150-frame window and tries to match the target bit rate at the end of the window.

A smaller bit rate window ensures a more accurate moving bit rate. However, the more accurate the moving bit rate, the worse the video quality. A 150-frame bit rate window is recommended.

The first frame in each bit rate window is not required to be an intra frame, but typical bit rate windows start with intra frames.

- `-A-5` sets the intra QP delta to -5.

The intra QP equals the intra frame QP minus 5.

For VBR, intra QP delta in the range from -3 to -5 is recommended to ensure better objective video quality.

- `-q26` sets the initial target QP to 26.

An accurate initial QP is strongly recommended. Alternatively, you can specify `-q-1` to enable the video encoder to calculate the initial target QP. However, the initial QP automatically calculated may be inaccurate in some cases.

- `-B 6780054` sets the target bit rate for rate control to 6780054 bps.

- `--vbr=1` enables the variable bit rate control based on the minimum QP allowed.

- `--qpMin=10` sets the minimum QP allowed to 10.
- Other settings include:
 - `-R150` sets the intra frame rate to 150.
The video encoder forces one frame to be encoded as an intra frame every 150 frames.
 - `--gopSize=1` sets the GOP structure size to 1.
 - `--gopLowdelay=1` enables the video encoder to use the default low-delay GOP configurations.
 - `-w1920 -h1080` sets the resolution of the input video sequence to 1920x1080 pixels.
If `-x` and `-y` are not specified, the width and height of the output video stream are the same as those of the input video sequence.
 - `-i /mnt/sequences/jctvc/ParkScene_1920x1080_24.yuv` specifies that the input file is `ParkScene_1920x1080_24.yuv`.
 - `-b 239` specifies that the last frame to be encoded is No. 239 in the input file.
A total of 240 frames are encoded because frames are counted starting from 0.
 - `-f24 -F1` sets the output frame rate to $24/1 = 24$ fps.
 - `-j24 -J1` sets the input frame rate to $24/1 = 24$ fps.
 - `-o 0.hevc` specifies to output the encoded stream to the `0.hevc` file.
 - `--outReconFrame=0` disables the output of reconstructed frames.

9.7 Example: Constant Bit Rate

The following example command selects and configures the constant bit rate (CBR) mode for rate control. This example assumes the input is a locally stored video.

```
./hevc_testenc -g120 -R120 --gopSize=1 -A-2 --rdoLevel=3 -w1920 -h1080
-i input.yuv -b 99 -q-1 -U 1 -B 1000000 --cpbMaxRate=100000 --cpbSize=2000000
-o 0.hevc --outReconFrame=0
```

In this command:

- Rate control settings include:

- `-g120` sets the bit rate window to 120 frames.

The rate control algorithm allocates bits for each 120-frame window and tries to match the target bit rate at the end of the window.

- `-U1` enables picture-level rate control.

- `-A-2` sets the intra QP delta to -2.

The intra QP equals the intra frame QP minus 2.

- `-B 1000000` sets the target bit rate for rate control to 1000000 bps.

- `--cpbMaxRate=1000000` sets the maximum local bit rate to 1000000 bits per second.

The value of `--cpbMaxRate` is equal to that of `--bitPerSecond (-B)`, the rate control algorithm works in CBR mode.

- `--cpbSize=2000000` sets the coded picture buffer (CPB) size to 2000000 bits.

For CBR mode, set this option to two time the target bit rate.

- Other settings include:

- `-R120` sets the intra frame rate to 120.

The video encoder forces one frame to be encoded as an intra frame every 120 frames.

- `--gopSize=1` sets the GOP structure size to 1.

- `--rdoLevel=3` sets the RDO level to 3, which provides the best video quality but the worst performance (encoding speed) among all RDO levels.

- `-w1920 -h1080` sets the resolution of the input video sequence to 1920x1080 pixels.

- `-i input.yuv` specifies that the input file is `input.yuv`.

- `-b 99` specifies that the last frame to be encoded is No. 99 in the input file.

A total of 100 frames are encoded because frames are counted starting from 0.

- `-q-1` enables the video encoder to calculate the initial target QP.

- `-o 0.hevc` specifies to output the encoded stream to the `0.hevc` file.

- `--outReconFrame=0` disables the output of reconstructed frames.

9.8 Example: Rate Distortion Optimization

The following example command configures rate distortion optimization (RDO).

```
./hevc_testenc -g 120 -R 120 -u 0 --enableRdoQuant=1 --gopSize=0 --rdoLevel=3  
-w832 -h480 -i input.yuv -b99 -B 1000000 -q-1 -o 0.bin --outReconFrame=0
```

In this command:

- RDO settings include:
 - `--enableRdoQuant=1` enables rate-distortion optimized quantization (RDOQ).
 - `--rdoLevel=3` sets the RDO level to 3, which provides the best video quality but the worst performance (encoding speed) among all RDO levels.
- Other settings include:
 - `-g 120` sets the bit rate window to 120 frames.
The rate control algorithm allocates bits for each 120-frame window and tries to match the target bit rate at the end of the window.
 - `-R 120` sets the intra frame rate to 120.
The video encoder forces one frame to be encoded as an intra frame every 120 frames.
 - `-u 0` disables block-level rate control to improve the PSNR.
 - `--gopSize=0` enables the video encoder to adaptively decide the GOP structure size.
 - `-w832 -h480` sets the input frame size to 832x480 pixels.
 - `-i input.yuv` specifies that the input file is `input.yuv`.
 - `-b99` specifies that the last frame to be encoded is No. 99 in the input file.
A total of 100 frames are encoded because frames are counted starting from 0.
 - `-B 1000000` sets the target bit rate for rate control to 1000000 bps.
 - `-q-1` enables the video encoder to calculate the initial target QP.
 - `-o 0.bin` specifies to output the encoded stream to the `0.bin` file.
 - `--outReconFrame=0` disables the output of reconstructed frames.

9.9 Example: Look-Ahead Encoding

The following example command enables look-ahead encoding:

```
./hevc_testenc -A-2 -g 60 -R 60 -f 30 -j 30 --gopSize 0 --lookaheadDepth=40 -q-1 -u 0 -U 1  
--rdoLevel 3 --enableRdoQuant 1 -w1920 -h1080 -i BasketballDrive_1920x1080_50.yuv -b239  
-B 6230000 -o 0.hevc --outReconFrame=0
```

In this command:

- Rate control settings:

- -A-2 sets the intra QP delta to -2.

The intra QP equals the intra frame QP minus 2.

- -g 60 sets the bit rate window to 60 frames.

The rate control algorithm allocates bits for each 60-frame window and tries to match the target bit rate at the end of the window.

A smaller bit rate window ensures a more accurate moving bit rate. However, the more accurate the moving bit rate, the worse the video quality.

The first frame in each bit rate window is not required to be an intra frame, but typical bit rate windows start with intra frames.

- -R60 sets the intra frame rate to 60.

The video encoder forces one frame to be encoded as an intra frame every 60 frames.

- --lookaheadDepth=40 sets the look-ahead buffer size to 40 frames.

- -q-1 enables the video encoder to calculate the initial target QP.

- -u 0 disables block-level rate control.

- -U 1 enables picture-level rate control.

- --rdoLevel 3 sets the RDO level to 3.

- --enableRdoQuant 1 enables rate-distortion optimized quantization (RDOQ) to improve video quality while the performance drops to about 80% compared with when RDOQ is disabled.

- -B 6230000 sets the target bit rate for rate control to 6230000 bps.

- General settings:

- -f 30 sets the output frame rate to $30/1 = 30$ fps.

- -j 30 sets the input frame rate to $30/1 = 30$ fps.

- --gopSize=0 enables the video encoder to adaptively decide the GOP structure size.

- `-w1920 -h1080` sets the resolution of the input video sequence to 1920x1080 pixels.
If `-x` and `-y` are not specified, the width and height of the output video stream are the same as those of the input video sequence.
- `-i BasketballDrive_1920x1080_50.yuv` specifies that the input file is `BasketballDrive_1920x1080_50.yuv`.
- `-b239` specifies that the last frame to be encoded is No. 239 in the input file.
A total of 240 frames are encoded because frames are counted starting from 0.
- `-o 0.hevc` specifies to output the encoded stream to the `0.hevc` file.
- `--outReconFrame=0` disables the output of reconstructed frames.

9.10 Example: Temporal Scalable Video Coding

The following example command configures temporal scalable video coding (SVC-T):

```
./hevc_testenc -A-2 -w352 -h288 -b49 -g 60 -R 30 -u 0 -q 22 -qpHdr=-1 --rdoLevel=1 --gopSize=2  
--gopConfig h2_gop_config_lowdelay_2TLayer.cfg -i input.yuv -o stream.hevc
```

In this command:

- `-A-2` sets the intra QP delta to -2.
The intra QP equals the intra frame QP minus 2.
- `-R30` sets the intra frame rate to 30.
The video encoder forces one frame to be encoded as an intra frame every 30 frames.
- `--rdoLevel 3` sets the RDO level to 1.
- `--gopSize=1` sets the GOP structure size to 1.
- `--gopConfig=./h2_gop_config_lowdelay_2TLayer.cfg` specifies the GOP configuration file.
For the contents of the file, see `cmdLine_ex11_file`.
- `-w352 -h288` sets the input frame size to 352x288 pixels.
- `-i input.yuv` specifies that the input file is `input.yuv`.

- `-b49` specifies that the last frame to be encoded is No. 49 in the input file.
A total of 50 frames are encoded because frames are counted starting from 0.
- `-u 0` disables block-level rate control.
- `-q 22` sets the default target QP to 22.
- `-o stream.hevc` specifies to output the encoded stream to the `stream.hevc` file.
- `--outReconFrame=0` disables the output of reconstructed frames.

9.11 Example: Saving Heif and Avif Files

9.11.1 Saving Heif Images

The following example command saves images of heif (hevc stream inside):

```
./video_testenc --codecFormat=hevc -i foreman_cif.yuv -w352 -h288 -R1 --gopSize=1 -b4  
-q26 --Heifcfg=heif_overlay_image.cfg -o stream.hevc
```

In this command:

- `-R1 --gopSize=1` must be specified to output I frames only.
- `--Heifcfg=heif_overlay_image.cfg` specifies the heif configurations.

Example files: `heif_overlay_image.cfg`

```
-HI  
-N 4  
-O 832 480 {0, 0} {176, 144} {352, 288} {512, 352}  
-o test.heic
```

This configure file specifies four overlaid images on a canvas of 832x480.

The following example command saves images of avif (heif files of av1 stream inside):

```
./video_testenc --codecFormat=av1 -i foreman_cif.yuv -w352 -h288 -R1 --gopSize=1 -b4  
-q26 --Heifcfg=avif_gird_image.cfg -o stream.av1
```

In this command:

- `-R1 --gopSize=1` must be specified to output I frames only.
- `--Heifcfg=avif_gird_image.cfg` specifies the heif configurations.

Example files: `avif_gird_image.cfg`

```
-HI  
-N 1  
-G 3 176 144  
-o test.avif
```

This configure file specifies to split the input image to grids of 176x144 and saving as an avif file.

9.11.2 Saving Heif Sequence

The following example command saves sequence of heif (hevc stream inside):

```
./video_testenc --codecFormat=hevc -i foreman_cif.yuv -w352 -h288 -b4 -q26  
--Heifcfg=heif_simple_image.cfg -o stream.hevc
```

In this command:

- `--Heifcfg=heif_simple_image.cfg` specifies the heif configurations.

Example files: `heif_simple_image.cfg`

```
-HS  
-N 5  
-o test.heic
```

This configure file specifies a 5 frames heif sequence, note that if this number does not equal to which specified in command line, it will be set to the number specified in command line (5 frames here).

The following example command saves sequence of avif (av1 stream inside):

```
./video_testenc --codecFormat=av1 -i foreman_cif.yuv -w352 -h288 -b4 -q26  
--gopSize=1 --Heifcfg=avif_thumbnail_seq.cfg -o stream.av1
```

In this command:

- `--gopSize=1` must be specified to avoid B frames.
- `--Heifcfg=avif_thumbnail_seq.cfg` specifies the heif configurations

Example files: `avif_thumbnail_seq.cfg`

```
-HS  
-N 5  
-T thumbnail_seq.av1  
-o test.avif
```

This configure file specifies a 5 frames avif sequence, associates with a thumbnail sequence which has been encoded as an av1 stream outside.

Appendix: Glossary

The following table describes the terms used in this document. For additional HEVC acronyms, see <http://www.hevcbook.de/acronyms/>.

Term	Description
1080p	high-definition resolution of 1920 x 1080 progressive video
720p	high-definition resolution of 1280 x 720 progressive video
ABR	average bit rate
AIoT	artificial intelligence of things
APB	Advanced Peripheral Bus
API	application programming interface
ARIB	Association of Radio Industries and Businesses
ASIC	application-specific integrated circuit
AV1	AOMedia Video 1
AVC	Advanced Video Coding (H.264)
AXI	Advanced eXtensible Interface
bps	bits per second
CABAC	context-adaptive binary arithmetic coding
CAVLC	content-adaptive variable-length coding
CB	coding block
CBR	constant bit rate
CIF	Common Interchange Format (352 x 288 pixels)
CIR	cyclic intra refresh
codec	compressor/decompressor
CPB	coded picture buffer
CPU	central processing unit
CQP	constant quantization parameter
CRF	constant rate factor
CSC	color space conversion
CTB	coding tree block

Term	Description
CTU	coding tree unit
CU	coding unit
CVBR	constrained variable bit rate
DDR	double data rate
DRM	digital rights management
EFBC	external frame buffer compression
EOS	end of sequence
EWL	encoder wrapper layer
FPS	frames per second
GDR	gradual decoder refresh
GOP	group of pictures
HD	high definition
HDTV	high-definition television
HEIF	High Efficiency Image File Format
HDR	high dynamic range
HEVC	High Efficiency Video Coding, a video coding standard (H.265) developed by JCT-VC
HRD	hypothetical reference decoder
IDR	Instantaneous Decoder Refresh
IPCM	intra pulse code modulation
ITU	International Telecommunication Union
ITU-T	ITU Telecommunication Standardization Sector
ITU-R	ITU Radiocommunication Sector
Kbps	kilobits per second
LCU	largest CU
LTR	long-term reference
MAD	mean absolute difference
MB	macroblock
MMU	memory management unit
MPEG-4	Moving Picture Experts Group 4
MV	motion vector
NAL	network abstraction layer
NALU	NAL unit, the smallest decodable unit of a stream
OBU	open bitstream unit
OSD	on-screen display
PAL	Phase Alternate Line (720 x 576 pixels resolution)
POC	picture order count
PPS	picture parameter set

Term	Description
PSNR	peak signal to noise ratio
PU	prediction unit
QCIF	Quarter CIF (176 x 144 pixels)
QVGA	Quarter Video Graphics Array (320 x 240 pixels)
QP	quantization parameter
RC	rate control
RDO	rate-distortion optimization
RDOQ	rate-distortion optimized quantization
REE	rich execution environment
RFC	reference frame compression
RGB	red, green, blue color space representation
ROI	region of interest
RPS	reference picture set
SAO	sample adaptive offset
SBI	segment buffer interface
SEI	supplemental enhancement information
SMPTE	Society of Motion Picture and Television Engineers
SPS	sequence parameter set
SSIM	structural similarity index measure
STR	short-term reference
sub-QCIF	Sub Quarter CIF (128 x 96 pixels resolution)
SVC-T	scalable video coding-Temporal
SXGA	Super Extended Graphics Array (1280 x 1024 pixels resolution)
TU	transformation unit
TV	television
VBR	variable bit rate
VBV	video buffer verifier
VGA	Video Graphics Array (640 x 480 pixels resolution)
VLC	variable-length code
VPS	video parameter set
VUI	video usability information
YCbCr	color space with one luma and two chroma components, used for digital encoding
YUV	color space with one luma and two chroma component, used for either digital or analog encoding

Document Revision History

This section describes top level differences in the versions of this document.

Note

This document is not necessarily updated for each patch or minor revision. The information in this document tends to be stable across a revision (nnn) series.

Document Revision	Date	Compatible Core	Description
1.11	2024-01-02	VC9000E	Updated the document name and template.
1.10	2023-08-21	VC9000E	- Added support for multi-instance multi-slice encoding in VCMD mode. - Added support for UFBC. - Added input color formats: RGBX8888_Raster_A16N, BGRX8888_Raster_A16N, RGBX1010102_Raster_A16N, BGRX1010102_Raster_A16N, YVU420SP10bWH_Raster_A16N, and Y8b_Raster_A16N. - Added support for the OSD map. - Added minIProp, maxIprop, and changePos to control QP for each intra picture. - Updated the default input and reference frame buffer alignments. - Added anti intra flicker (AIF) for single-core one-pass encoding. - Added support for High Efficiency Image File Format (HEIF).
1.00	2021-03-15	VC9000E	Initial release.